Determining Shear-Wave Velocity Profiles in Langley using Constrained Microtremor Horizontal-to-Vertical Spectral Ratio Inversions

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Key Points

1. Metro Vancouver is situated in a high-risk seismic zone and has a subsurface that contains a variety of geologic material requiring detailed hazard assessment
2. Constrained MHVSR inversions performed on sites in the township of Langley, BC can produce ground profiles giving the depth and shear wave velocity layers in the subsurface
3. Shear wave velocity profiles can be used to build accurate hazard maps to enhance emergency preparedness in the region

Abstract

Metro Vancouver is located in a high-risk seismic zone and is threatened by earthquake related hazards resulting from activity in the Cascadia Subduction Zone. The Cascadia plate boundary is over 1000 kilometers and accumulates significant amounts of strain. This accumulated strain will eventually be released and has the potential to create a magnitude 9.0 earthquake. In order to predict ground shaking intensity of an earthquake, the region's complex geology must be understood. The subsurface includes a deep basin filled with high-velocity Tertiary bedrock, thick Ice Age sediments, and soft Holocene Salish deposits. Large seismic impedance contrasts between these layers can trap seismic waves, significantly increasing the amplitude and duration of surface shaking based on the principle of conservation of energy. To improve emergency preparedness, the Metro Vancouver Seismic Microzonation Mapping Project creates region specific hazard maps. As part of this project, 209 Microtremor Horizontal-to-Vertical Spectral Ratio (MHVSR) measurements have been collected in the region of Langley. This data is typically used only to estimate layer depths and is not inverted to model shear wave velocity (v_s). This research project suggests that we can invert existing MHVSR data using priori subsurface information to constrain inverted shear-wave velocity models using Geopsy's

DINVER software. To do this, we use established depth and velocity values as parameters to guide the inversions. This will generate one-dimensional v_s profiles that are consistent with the subsurface geology. The data produced will be visualized in cross section view to understand the widespread layering in the area. V_{s30} values will also be calculated for each of the sites inverted to better evaluate potential ground motion amplification across the study area.

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Introduction

Predicting the shaking effects during an earthquake in the seismically active area of Vancouver and its surrounding greater area is uniquely difficult due to the region's complex geology. To address this, the Metro Vancouver Seismic Microzonation Mapping Project was created to produce region specific seismic hazard maps. This project is in collaboration with the British Columbia Ministry of Emergency Management and Climate Readiness and the Institute of Catastrophic Loss Reduction (Molnar et al., 2023). For nearly ten years, members of this team have focused on determining local site effects such as shaking amplification, liquefaction, and landslides (Molnar et al., 2023). The hazard maps produced from these investigations are important tools as they provide the information needed to improve developments and enhance emergency preparedness in the densely populated region.

The primary survey method utilized in this research project is the single-station microtremor horizontal-to-vertical spectral ratio (MHVSR). MHVSR is a highly efficient, non-invasive technique that records ambient background noise such as wind, traffic or ocean waves using a single three component seismometer to identify a site's resonance frequency. This method requires minimal personnel and equipment compared to traditional seismic array methods and allows for rapid data collection over large urban areas. Figure 1 shows the 209 MHVSR sites collected in the region of Langley as part of two summer field campaigns in 2024 and 2025.

The Metro Vancouver Seismic Microzonation Project (MVSMMP) has compiled a database of over 500 MHVSR measurements in Metro Vancouver. This data is traditionally used to determine depths of subsurface layers. The objective of this research is to utilize the MHVSR

data collected and derive subsurface shear wave velocity profiles for each of the sites through the process of inversions that are constrained based on predetermined parameters. This approach involves restricting the possible mathematical solutions to those that match the known local geology. The working hypothesis of this study is that by using these specific regional parameters, we can successfully invert our MHVSR data to develop a new set of v_s profiles. This will increase the subsurface information available for Eastern Metro Vancouver, providing a foundation for more reliable seismic hazard models.

Background

Seismic Activity on the West Coast

The Cascadia Subduction Zone is a massive active tectonic plate boundary that stretches over 1000 km from Northern California up to central Vancouver Island (Clague, 1997). The boundary, seen on the map in Figure 2, involves the Explorer, the Juan de Fuca and the Gorda plates which are all are currently subducting beneath the North American continent at a rate of approximately four centimeters per year (Clague, 1997). The subducting plates are considered to be 'locked'. This means that the plates are not smoothly moving past each other. As the plates move, the crust undergoes significant elastic bending and bucking (Clague, 1997). This process creates a significant amount of strain.

This strain exists as stored energy and can be eventually released as catastrophic megathrust earthquakes. Geologic evidence and oral histories indicate that the last megathrust event was a Magnitude 9.0 earthquake on January 26, 1700, which ruptured the entire length of the subduction zone (NRCAN, 2011). This event was extraordinarily devastating. Oral histories from Vancouver Island First Nations describe a massive tsunami that destroyed a village at Pachena Bay, leaving no survivors. The tsunami was so powerful it traveled across the Pacific,

with written records from Japan documenting the precise time of the waves' arrival. Other effects of this event involved coastal forests and marshes suddenly subsiding by 0.5 to 2 meters, causing them to be buried by tidal mud and tsunami sand (NRCAN, 2011).

While these magnitude 9.0 megathrust events are the most catastrophic, the Metro Vancouver area is also subject to much frequent, smaller seismic events. Approximately 400 earthquakes occur each year in the region stretching from northern Vancouver Island to Seattle (NRCAN, 2011). Majority of these earthquakes are too small to be felt and have a magnitude of less than 3.0. Earthquakes in the region originate both within the overriding North American crust and deep within the subducting Juan de Fuca plate. Even moderate earthquakes, with magnitudes between 5.0 and 6.0, present significant hazards. Research suggests they occur frequently enough to trigger local effects like submarine landslides in Saanich Inlet roughly once every century (Clague, 1997). According to National Resources Canada, an earthquake capable of causing structural damage strikes the region on average once every decade. These events serve as a constant reminder of the active tectonic environment and the associated threats.

Geologic History and Lithology

The Metro Vancouver region and the surrounding Lower Fraser Valley is geographically bordered by the Coast Mountains to the North and the Cascade Mountains and Chuckanut Hills in the south (Turner et al., 1997). The area consists of mountain peaks, gently rolling uplands, and lowlands. Beneath the surface, the region is characterized by a deep geological basin sitting above granitic rock. This basin is filled with a variety of layered materials. This subsurface structure includes a layer of sandstone bedrock that reaches thicknesses over 4 kilometers. Overlying this bedrock is a significant accumulation of Ice Age sediments, which can be up to 1 kilometer thick. A relatively thin layer of modern sediments overlies the bedrock (Turner et al.,

1997). The variety of geological material present exhibit a variety of physical properties that influence the region's response to seismic hazards.

Subsurface Layers

Bedrock

The oldest geologic units in the area are sedimentary and volcanic rocks exposed in the Cascade Mountains and as small hills in the eastern Fraser Valley (Turner et al., 1997). These rocks, dating back up to approximately 185 million years, underwent significant metamorphism. This metamorphism resulted in foliation of the rocks. Following this, granitic rocks were formed between 165 and 95 million years ago. These medium to coarse-grained igneous rocks such as granite and granodiorite, are the rocks seen in The Coast Mountains and are highly resistant to erosion and make up the mountain slopes. During the Tertiary period, between 85 and 37 million years ago, a massive sequence of sandstones, siltstones, and conglomerates were deposited. While these rocks are mostly hidden at depth throughout the Fraser Valley, they appear as scattered outcrops in Vancouver and Burnaby, such as at Burnaby Mountain and Stanley Park, where southward-tilted sandstone layers form prominent ridges. The youngest bedrock units are volcanic rocks, specifically basalt and andesite, formed between 35 and 17 million years ago. These rocks form many of the valley's hills, such as Queen Elizabeth Park and Sentinel Hill. The bedrock present also occusr as thick tabular sills and subvertical dykes that cut through older rock layers (Turner et al., 1997).

Ice Age Sediments

Glacial activity during The Pleistocene Epoch, which occurred from 2 million to 11,000 years ago, had significant geologic influence in the region, particularly during the last glaciation and the subsequent retreat of the ice (Turner et al., 1997). These Ice Age sediments are part of

the gently rolling uplands of the Fraser Valley, which range from 15 to 250 meters in elevation. The most widespread surface sediments in these uplands are thick marine silts and clays. These fine-grained materials were deposited when the sea covered the area during the period of glacial retreat. The bearing capacity of these sediments varies significantly based on their history. East of Aldergrove, these deposits were compressed by massive glaciers, which significantly increased their bearing strength. However, their fine-grained nature results in poor water infiltration and drainage, and they are notoriously prone to landsliding when situated on slopes steeper than 20 degrees. Another foundational unit present in this region is till. In the Vancouver upland, till is often the dominant near-surface material, while in the Langley and Surrey areas, it is frequently buried beneath thick sequences of marine silt and clay. Till is generally compact and concrete-like, offering high bearing capacity and serving as an excellent foundation material. For the Langley and Abbotsford regions, Ice Age gravel and sand deposits are of important economic and hydrogeologic significance. These units can reach thicknesses of 40 meters and are useful sources of aggregate for construction. Furthermore, they host major aquifers, such as the Brookswood and Abbotsford aquifers, which provide drinking water. Older Ice Age sediments are exposed in steep escarpments along the edges of the uplands, where they are susceptible to undercutting by waves or streams, frequently leading to localized landslides (Turner et al., 1997).

Modern Sediments

The youngest geological units in the region are modern sediments, which are less than 10,000 years old and primarily occupy the lowlands along the Fraser River (Turner et al., 1997). These loose materials underlie about half of the Fraser Valley, including the flat, flood-prone lands of the Fraser delta. The lithology of these lowlands varies by throughout the region. Gravel and sand dominate the Fraser River floodplain between Hope and Mission, while sand and silt

become the primary deposits farther downstream and throughout the Nicomekl-Serpentine valley. These deposits often exist as shallow aquifers and offer moderate to high bearing capacity, yet they pose a significant seismic risk as they are highly susceptible to liquefaction during a strong earthquake. In poorly drained depressions and parts of the Fraser delta, peat has accumulated in thicknesses of up to 5 meters. Peat is an exceptionally poor foundation material due to its high compressibility. Additionally, modern marine silts and clays are common in the Nicomekl-Serpentine flats, where they were formed by fine river sediment settling in the sea. These sediments generally make poor foundations due to low bearing capacity. Finally, human activity has introduced landfill along shorelines like False Creek and Burrard Inlet. These deposits are highly heterogeneous and can be problematic for structural foundations and are prone to settling or liquefaction during seismic events. Together, these modern sediments constitute the most geologically volatile areas of the Lower Mainland (Turner et al., 1997).

Site Effects

In Metro Vancouver, the intensity of ground shaking is determined by the stiffness of the soil, which is measured by shear-wave velocity. Figure 3 displays the large variety of geological material that can respond differently to seismic waves in Metro Vancouver. Soft Holocene sediments have velocities in the range of 200-300 m/s, which leads to significant amplification of the travelling seismic wave (Assaf et al., 2019). The Pleistocene sediments beneath however reach velocities around 500 m/s, and the bedrock beneath can transport shear waves at velocities of 1500m/s (Assaf et al., 2019). These large contrasts can trap the seismic waves, and the waves are forced to reflect within the soft upper layers (Şafak, 2001). This reflection prevents the energy from dissipating, which significantly increases the amplitude and duration of shaking at the surface. To standardize these geologic variations for engineering practice, the National

Building Code of Canada (NBCC) utilizes National Earthquake Hazards Reduction Program (NEHRP) Site Classes based on the time-averaged shear-wave velocity in the upper 30 meters (Vs_{30}) (Table 1). Under this system, the region's high-velocity bedrock is categorized as Site Class A or B, while the dense Older Pleistocene uplands typically fall into Site Class C. In contrast, the thick, soft silts and clays of the Fraser River delta and the Nicomekl-Serpentine valley are designated as Site Class E, representing the highest potential for ground motion amplification (Taylor et al., 2006).

The physical basis for this amplification is described by the concept of seismic impedance. Seismic impedance is defined as the product of a material's mass density (ρ) and its wave propagation velocity (V). Based on the conservation of energy principles and the seismic energy flux (E) along a ray of seismic waves can be expressed as Equation 1.

$$E(t) = \left(\frac{1}{2}\right)(\rho V)x^2(t) \quad (1)$$

In equation 1, $x(t)$ represents the velocity of seismic shaking. The energy flux must remain constant as the wave travels. Therefore, changes in the impedance must be compensated for by an increase in the shaking velocity (Şafak, 2001). The magnitude of this amplification (A), Equation 2, depends on how abruptly the impedance changes. The amplification is determined by the square root of the impedance ratio.

$$A = \frac{x_2(t)_{max}}{x_1(t)_{max}} = \sqrt{\frac{(pV)_1}{(pV)_2}} \quad (2)$$

Beyond impedance contrasts, several other factors influence site amplification in the region. As described in Şafak (2001) in Figure 4, these other factors include the focusing of waves caused by subsurface topography (such as the shape of the sedimentary basin), and the conversion of body waves into surface waves. Additionally, high water content and the inherent randomness in the geological medium can further complicate how seismic energy is distributed,

often leading to the extreme increased amplification.

The importance of understanding geology's role in site amplification is illustrated by the aftermath of an Earthquake on December 13th, 1990 in Sicily, Italy. Although considered to be a moderate event with a magnitude of 5.6, it caused significant structural failure, 20 casualties, and left 15,000 people homeless (Panzera et al., 2018). In the cities of Lentini and Carlentini, Panzera et al (2018) observed that the distribution of damage was not uniform. The destruction observed was heavily influenced by the underlying lithology. Buildings on soft silty-clay alluvial deposits and Early Pleistocene marly clays suffered significantly higher damages, and structures built on rigid calcarenite or volcanic rock experienced much less structural distress. To quantify these variations, the seismic response of the area was categorized into four distinct patterns based on subsurface impedance contrasts, shown in Figure 5. Category (a), represented by the red lines in Figure 5b, identifies sharp, low-frequency peaks (< 3.0 Hz). These curves indicate thick sequences of soft marly clays and marine sands (> 40 m) overlying bedrock, a type of layering that is capable of trapping large amounts of seismic energy. Category (b), shown by the orange lines, displays high-frequency peaks (3.0 Hz) and represent thinner alluvial deposits. Category (c), the green lines in Figure 5c, identifies less pronounced peaks (3.0–4.0 Hz) that indicate a moderate velocity contrast, such as calcarenites overlying basaltic lavas. Finally, Category (d), the blue lines, captures "broadband" behavior where no single strong contrast exists to trap energy, typically occurring in areas with many alternating layers of varied velocities.

These variations in subsurface responses controlled the level of destruction in the area. One dimensional numerical modeling revealed that peak ground acceleration was nearly three times higher on the shallow alluvial deposits compared to the firmer Carlentini plateau. These observations emphasize that local site effects, determined by both the stiffness and thickness of

the geological layers are a primary factor in structural survival and earthquake devastation.

Hazard Maps

Seismic hazard maps visualize the hazards associated with earthquakes across different areas by integrating historical earthquake and fault information, the behavior of seismic waves as they travel through the crust, and local subsurface geology. According to the U.S. Geological Survey, these maps show the relative hazards in different regions to assist in disaster mitigation and the development of building codes. While technical experts use these products to follow regulated engineering rules, they also serve as critical tools for non-technical users to make informed decisions regarding earthquake insurance or the structural rehabilitation of their homes.

While many hazard maps exist for a variety of natural disasters at a multitude of scales, the Metro Vancouver Seismic Microzonation Project aims to produce seismic microzonation maps that subdivide the region into zones of similar risk. A suite of 29 maps has been published for Western Metro Vancouver. These maps were constructed using extensive seismological simulations. The maps produced are called seismic hazard susceptibility maps. Susceptibility maps, display characteristics such as V_{s30}, site period (T_0), and sediment thickness (H_{pg}), landslide susceptibility and are designed to characterize the ground conditions in a given area (Molnar, 2024). Probabilistic maps depict specific hazards like shaking amplification, liquefaction potential, or landslide displacement at 2% or 10% probability of exceedance in 50 years. Research is currently expanding to the Eastern region to fill existing data gaps and provide a more comprehensive assessment of seismic risk for the entire Metro Vancouver area.

Shaking amplification maps, which display the factor by which seismic waves are altered as they interact with subsurface sediments, are important maps for the goals of this project. These maps show amplification factors relative to a reference site condition, which for Metro

Vancouver is defined as a time-averaged shear-wave velocity in the top 30 meters (V_{s30}) of 760 m/s. Factors greater than 1 indicate increased shaking, while values less than 1 represent deamplification. The project distinguishes between 1D site effects, where waves become trapped and resonant within a local sediment column, and 3D basin effects, where amplitude and duration change as waves interact with the geometry of the deep sedimentary basin.

As illustrated by figure 6, amplification hazards are often mapped for specific spectral periods to reflect the risks posed to different structure types. Figure 6 shows long-period (2.0 seconds) amplification hazard for a 2475-year return period. Shaking at these longer periods is particularly relevant for taller buildings and is significantly increased by the presence of the Georgia sedimentary basin. Regional observations documented in these maps show that the soft Holocene sediments of the Fraser River delta amplify shaking by factors of 3 to 6 relative to stiffer glaciated sediments, with the highest amplification occurring near the delta's northern edge. By providing this region-specific data, these maps are an improvement of general models and accurately reflect the unique geological nature of the Lower Mainland.

Objectives

The primary goal of this research project is to determine and interpret the subsurface seismic properties of Langley to contribute to microzonation hazard mapping efforts. To achieve this, the following objectives were defined:

1. To perform seismic inversions on MHVSR field data collected from summer field campaigns to Langley to generate one-dimensional shear-wave velocity profiles. These inversions are performed on MHVSR data (amplification curves) and are constrained by existing regionally appropriate models.

2. To calculate V_{s30} values (the average shear-wave velocity in the top 30 meters) for each site, providing a standardized measurement for site specific subsurface behaviour.
3. To integrate newly acquired data subsurface data into the existing Metro Vancouver Seismic Microzonation Project database to be included in the site amplification hazard maps

Methodology

The MHVSR Method

The seismic data that will be inverted for this study was collected using the single-station microtremor horizontal-to-vertical spectral ratio (MHVSR) method. This is non-invasive technique that estimates the characteristics of subsurface layers by measuring ambient seismic noise, or microtremors, with a single three-component seismometer. These background vibrations consist of a mixture of various seismic waves, typically dominated by surface waves (Rayleigh and Love waves) at large distances from sources. This method identifies subsurface characteristics by calculating the ratio between the horizontal and vertical Fourier amplitude spectra (FAS) recorded at the surface (Molnar, 2022).

Yutaka Nakamura (1989) was one of the first researchers to pioneer this method for site characterization. In his 1989 paper, Nakamura proposed that the MHVSR could reliably estimate the fundamental resonance frequency (f_0) and the amplification factor of a site without requiring an actual earthquake. This technique defines the spectral ratio (R_s) as equation 3.

$$R_s = \frac{S_{HS}}{S_{VS}} \tag{3}$$

S_{HS} and S_{VS} represent the horizontal and vertical Fourier amplitude spectra of the vibrations recorded at the surface, respectively. His theory assumes that while horizontal microtremors are

significantly amplified by soft surface layers, the vertical component undergoes negligible amplification. By further assuming that the spectral ratio in the underlying bedrock substrate is approximately unity, the ratio at the surface can be treated as a "quasi-transfer function" for the soil column. Nakamura validated this method by demonstrating that the microtremor-derived closely resembles the actual transfer function (S_{TT}) obtained from earthquake recordings (Figure 7). As shown in Figure 7, these two functions share nearly identical peak positions and magnitudes, confirming the method's reliability for identifying site resonance. While the full composition of the wavefield remains a subject of academic debate, Nakamura's assertion that the lowest frequency peak approximates a site's natural resonance remains a standard foundation for seismic microzonation.

Data Collection

The data used for this project was collected over two field campaigns in the summers of 2024 and 2025 and was accomplished by members of the Metro Vancouver Seismic Microzonation Mapping Project team. During these campaigns, single-station measurements were recorded at 433 locations across Eastern Metro Vancouver, with 227 sites specifically located within the City and Township of Langley. This data was collected using instruments called TROMINO units, which are portable, high-sensitivity three-component seismometers manufactured by MoHo s.r.l (Figure 8). At each measurement site, a seismometer was placed on the ground surface to record ambient vibrations for approximately 30 minutes. To ensure a comprehensive assessment of the region's subsurface variability, the data was collected with a variable grid resolution of 600–1000 meters. Standard data was documented for each site, including the latitude and longitude, and the specific geological material on which the sensor was placed. To protect the sensitive equipment from wind disturbances or light rain during the

recording window, a bucket was typically placed over the TROMINO unit (Motuzas, 2025).

Data Processing

The determined resonance frequency identified from these recordings is used to interpret the thickness of layers in the subsurface. Data processing involves dividing the recorded time-series into overlapping windows of 60 seconds in length and applying a Fourier transform to calculate the spectra for the vertical and horizontal (Molnar et al., 2022). The spectra are then smoothed using the Konno and Ohmachi function to reduce variance. The Konno and Ohmachi function is a logarithmic frequency that works to avoid bias in the peak amplification regardless of the frequency. The horizontal components of the readings are typically combined using the geometric mean method to define a representative horizontal spectrum. The amplitude of the resulting spectral peak is proportional to the impedance contrast between the soft overburden and the underlying stiffer glaciated sediments or bedrock (Molnar et al., 2022). The MHVSR data collected and used for this project are in Appendix A.

Inversions

To invert the H/V peaks of each MHVSR curve, this study will use the DINVER software, which is part of the open-source geophysical data analysis package called Geopsy. DINVER utilizes a neighborhood algorithm to search for 1D layered earth models, specifically shear-wave (v_s) and compressional-wave (v_p) velocity profiles, whose theoretical elliptical curves best match the observed H/V peak (Arimuko et al., 2020). The algorithm runs thousands of times, iteratively refining subsurface parameters with the intention of finding the best-fit model that minimizes the misfit function between experimental and calculated data. MHVSR inversion is a nonlinear problem with no unique solution for different combinations of layer thicknesses and velocities and can often produce nearly identical spectral ratios (Arimuko et al.,

2020).

Our inversion attempts rely on the parameters we set to constrain this non-uniqueness. Traditionally, MHVSR data is used only to estimate subsurface layer depths, as the frequency curves cannot be inverted in isolation to produce reliable velocity profiles without a priori information. However, because there exists an established baseline for local geologic layering and velocity profiles in Langley from previous geophysical surveys, we can use these velocity models as parameters to guide the software. As detailed in Table 2, the inversions are constrained by specific depth and velocity ranges for each layer, reflecting the regional lithology where speed values generally increase in proportion to depth. Different 'options' were used on different sites in different areas based on what is already known about the subsurface conditions. After the first round of inversions, parameters for sites were adjusted and options 4 through 10 were added (Table 3). More on this to be discussed in the results section.

For this project, the study area is divided into sub-regions F2, H, I, J, K, (Figure 9). In sub-regions F2 and H, two options for parameters were initially used. Sites with narrow, high-amplitude (> 8) peaks around 0.4 Hz are inverted using Option 1 boundaries. These parameter boundaries assume softer surface materials and deeper layer interfaces (reaching depths up to 500 m). Sites in these same regions with broader, moderate-amplitude peaks at 0.7 Hz or higher indicate stiffer and shallower conditions, utilizing Option 2. For the I, J, K, and N sub-regions, which primarily exhibit moderate amplification and broader peaks, a consistent four-layer boundary parameterization model (Option 3) was initially applied. This option uses established ranges, such as 350–800 m/s for intermediate glaciated layers and 1500–2500 m/s for the bedrock half-space, to ensure the resulting 1D profiles are geologically consistent. For each MHVSR site, three DINVER inversions are performed, and the final model is selected by

identifying the profile that achieves both the lowest numerical misfit and the best visual agreement between the observed spectral data and the theoretical Rayleigh wave ellipticity curve. As shown in the provided ellipticity plots in Appendix B, this visual inspection ensures that the theoretical model (indicated by the red curve) accurately captures the position of the experimental peak (black data points), providing a qualitative check on the numerical results produced by the software.

Results

For this project, inversions on 86 sites across the six regions (F2, H, I, J and K) were performed. Of these 86 inversions, 70 sites produced accurate models where the theoretical Rayleigh wave ellipticity curve effectively matched the observed H/V data. Figure 10 displays the ground profiles for each of the subregions. In region F2, 14 sites were inverted, H had 13 inversions, I and J saw 15 inversions, K had 14 inversions, and N had 15 inverted sites. Figure 11 shows the locations of the inverted sites in their respective subregions. The ground profiles in Figure 10 allows for an understanding of the differences in geologic material across Langley. In subregions F2 and H (Figures 10A and 10B), the depth profiles display relatively high lateral variability. The estimated bedrock interface which is characterized by the sharp velocity step where shear-wave velocities exceed 1500 m/s is split into two distinct depth ranges. Several sites resolve bedrock at relatively shallow depths between 150 and 300 meters, while others place it much deeper at 450 to 550 meters. These differences reflect the lateral geological transition between the stiff upland till sequences and the deeper, softer sedimentary channels in the western part of the study area. The profiles in subregions I, J, K, and N (Figures 10C, 10D, 10E, and 10F) show a more consistent ground profile structure. Across these four subregions, the subsurface layers are more uniform, with the bedrock interface mapped between 450 and 550 meters depth.

This lack of variation among dozens of independent site models confirms that the thick, glaciated sedimentary basin is widespread, flat-lying, and continuous across these central and eastern portions of Langley

A total of 16 sites were identified to produce unreasonable inversion results. This is likely because proper parameterization bounds could not be established to achieve an accurate fit with the H/V curve. It is also possible that these sites represent areas where there was an issue with data collection and/or processing. A detailed breakdown of which sites produced or did not produce results can be seen in Appendix D. Appendix D also provides the coordinates, subsurface layering for each MHVSR site.

The neighbourhood algorithm within the DINVER software was executed thousands of times for every station to search for the best-fit 1D shear-wave velocity model. Following these runs, the misfits and ellipticity curves of three separate runs were compared for each site. After performing each inversion, the misfits and ellipticity curves of runs 1, 2 and 3. The final model was chosen by identifying the run that achieved the best visual agreement between the theoretical red ellipticity curve and the experimental black H/V data points. Figure 12, an example of an inversion for site LAN079, shows the ellipticity and ground profile plots for the three respective runs. For this inversion, run 1 was selected as it was the best fit. The best fit runs selected for each initial inversion can be seen in Appendix B. LAN079 is an example of a site where the initial parameters used for the inversion produced reasonable results. After first attempts at inversion, approximately 42 sites produced inaccurate results.

Successful Reinversions

For these 42 inaccurate inversions, reinverting the MHVSR curve with the hope of producing more accurate results was performed. A variety of problems were addressed through

this reinversion process. The first step in hoping to successfully reinvert these sites was to analyze the inaccurate curve. This involved observing the behavior of the model relative to the initial constraints. Most of the sites required adjusting the parameterization boundaries. The additional parameterization boundaries used to reinvert these 42 sites are listed in Table 3. Since both three and four layer models are assumed in this region, a common reversion method practiced was to switch the number of layers used in the model parameterization bounds. The inversion and reinversion of LAN081 is an example where switching from a three layer to a four layer model produced a more accurate inversion (Figure 13).

Another example of an inaccurate result and two reinversion attempts can be seen in LAN076 (Figure 14). In the case of this site, adjustments were necessary to correct the issue of overfitting. This problem was caused by the mathematical algorithm creating "phantom" peaks (Figure 14A) not supported by the actual microtremor recordings. The first approach to try and correct this was to switch to a three layer model (Figure 14B). For site LAN076, this approach was not successful as the model missed the first peak and incorrectly fit the second peak. The inversion of the site was eventually resolved by returning to a 4-layer model with deeper and faster parameter boundaries (Option 5). These parameters provided a significantly more accurate fit to the observed H/V curve.

Knowing the location and understanding the geologic conditions of the sites was an important piece of the reinversion process. In the region of Langley, there are both uplands and lowlands. When the initial inversion for LAN050B did not produce accurate results (Figure 15A), the analysis shifted to how the specific location of the site might require different parameters. LAN050B is situated in the uplands. The initial inversion failed because the model assumed the existence of a soft surface layer that in reality, did not exist. For the reinversion, the

upper low velocity layer was removed, and the upper layer was forced to be thicker and faster with the new model parameters. The initial and modified inversion for LAN050B is displayed in Figure 15.

LAN102 is a site in the lowlands that falls in a clay channel in the region. This site required specific adjustments to resolve the inaccurate model initially produced (Figure 16A). This site could be fit by either a "soft clay" or "moderate glacial" model, but the soft clay model was selected as it matched the known surficial geology.

Cutting the MHVSR curve was another approach taken when it came to reinverting the inaccurate curves. Often times the MHVSR curves have another peak in the higher frequency. For sites like LAN199 (Figure 17), cutting the MHVSR curve at 20 Hz prevented the algorithm from wasting low-velocity solutions on high-frequency ambient noise, resulting in a more accurate model of the actual resonance peak.

Unsuccessful Reinversions

Despite the reinversion attempts discussed, some sites were unable to produce results that gave rise to reasonable models. LAN208 (Figure 18) is an example of a site where the parameters were adjusted multiple times, and the ellipticity curve always produced an extra peak. This site is situated in the deep lowlands. Given this geological context, the parameters were adjusted to allow for very deep low velocities. Even with these new model parameterization boundaries, the theoretical ellipticity curve consistently produced an extra peak that did not match the recorded data.

Other unsuccessful efforts, such as site LAN092B (Figure 19), were due to the software's inability to properly identify the front or lead-in of the resonance peak despite parameter adjustments and curve cutting. These failures are often attributed to high levels of transient noise

or a lack of a clear impedance contrast at the site.

Depth Profiles and Vs30 Data

The targeted primary output of the successful inversions is a compilation of 1D shear-wave velocity (V_s) depth profiles. These profiles characterize the subsurface layering and provide data needed to evaluate ground shaking response. The initial depth profiles are provided in Appendix B, while the revised final depth profiles are detailed in Appendix C. From these velocity profiles, V_{s30} values were calculated for every site. V_{s30} values represent the time-averaged shear-wave velocity in the top 30 meters. These values provide a standardized metric for subsurface stiffness used in modern building codes. The calculated values for each site are listed in the master spreadsheet in Appendix D, and their spatial distribution is visualized in the gradient map (Figure 20). The MATLAB code used to generate this visualization is provided in Appendix E.

Discussion

Subsurface Transects

To evaluate the consistency of the 1D velocity models, subsurface transects were constructed across the region of Langley. These transects allow for a comparison of generated depth profiles with known regional geologic trends and help understand the nature and trends of the geologic area. Figures 21 and 22 display transects spanning West to East across the study area. These figures demonstrate that the subsurface layering is relatively consistent across large portions of Langley. These subsurface visuals highlight a sequence of stiff Older Pleistocene glaciated sediments, typically compact till and dense marine silts and clays, overlying high-velocity Tertiary bedrock. In the eastern portions of these transects, the Pleistocene layers represented by the higher velocities maintain a consistent thickness, providing a stable

geotechnical foundation compared to the softer lowland units.

Figures 23 and 24 illustrate the transition going from North to South in the region. This area spans from the flat lowlands of the Fraser River in the north to the rolling central uplands and the southern low-lying flats. These transects highlight the variations in the thickness of soft Holocene Salish sediments. These modern sediments, including peat, organic silts, and deltaic sands, exhibit low shear-wave velocities (200–300 m/s) and are prone to trapping seismic waves, which can significantly increase the duration and amplitude of shaking. As the transects move from the lowlands into the central uplands, these soft Salish layers thin out or disappear, and the stiffer Pleistocene glaciomarine and till units become the dominant near-surface materials.

V_{s30} Values

The V_{s30} map (Figure 20) aided in identifying regional trends across Langley. While majority of the region exhibits moderate velocities consistent with the stiff glaciated deposits of the rolling uplands, a low-velocity region is present between 49.110°N to 49.132°N and -122.604°W to -122.548°W. This area corresponds to a buried clay channel where soft Holocene deposits are significantly thicker than the surrounding areas. These low V_{s30} values, often falling into Site Class E, indicate a higher potential for ground motion amplification during a seismic event. This effect occurs because seismic energy flux must remain constant. As waves move from high-impedance bedrock into low-impedance soft clays, the shaking velocity must increase to compensate.

Expanding to Maple Ridge and Pitt Meadows

While these sites in Langley were being inverted, BSc student Ari Chote was performing the same process of inversion to sites in the Northern part of Eastern Metro Vancouver in areas called Maple Ridge and Pitt Meadows. His Maple Ridge and Pitt Meadows V_{s30} values have

been integrated into the Langley dataset and have been put into a combined Vs30 map (Figure 25). The values across most of the Langley region primarily fall within a moderate range of 150-300 m/s, which is consistent with the stiff Pleistocene glaciomarine and till sequences that dominate the central uplands. In contrast, the data for the Maple Ridge and Pitt Meadows areas to the north exhibit a broader range of velocities, including several sites with significantly higher values exceeding 450–600 m/s. These higher values visually indicate the transition toward the granitic rocks of the Coast Mountains, where the thick basin sediments begin to pinch out.

Despite these regional differences, both datasets show consistent patterns where low-velocity anomalies, often dropping below 150 m/s, align with specific geographic features. In both Langley and the regions to the north, these lower values correlate with the presence of soft Holocene Salish sediments, such as peat, organic silts, and channel-fill deposits found in low-lying areas and abandoned river valleys. By mapping these values together, the project provides a visual representation of near-surface stiffness across the entire eastern study area, highlighting that while Langley's geology is relatively widespread and consistent, the northern regions experience more dramatic lateral changes in soil properties as they approach the mountain front. A detailed list containing the coordinates and V_{s30} values for the Maple Ridge and Pitt Meadows is provided in Appendix F.

Conclusion

This project focused on determining the subsurface seismic properties of the City and Township of Langley to contribute to ongoing regional hazard mapping efforts. Ambient seismic noise recordings from 86 sites were processed using the MHVSR method to identify fundamental resonance peaks. These spectral curves were then inverted using Geopsy's DINVER software to produce 1D shear-wave velocity depth profiles. To address the non-uniqueness of the

inversion problem, the models were constrained by regional geologic parameters, and final selection was based on a combination of numerical misfit and visual agreement with theoretical Rayleigh wave ellipticity curves. This process resulted in accurate velocity profiles used to calculate V_{s30} values, providing a standardized value for near-surface ground conditions across the study area.

The data generated through these inversions will be integrated into the Metro Vancouver Seismic Microzonation Mapping Project (MVSMMP) database. This database supports the creation of Level 3 hazard maps, which is the most advanced category of microzonation. Level 3 maps utilize a large collection of data points and complex 1D and 3D simulations to provide the hazard assessment. The maps produced by the Metro Vancouver Microzonation Mapping Project identify specific amplification factors. Long-period (2.0 s) shaking is predicted to be much more intense in areas with deep basin sediments, which is important information for the design of tall buildings. By replacing general models with this region-specific, data-driven information, the project provides a more accurate assessment of seismic risk for the Lower Mainland. These maps are intended for use in regional land-use planning and disaster mitigation.

Data and Resources

The microtremor data used in this project was collected during the 2024 and 2025 field campaigns as part of the Metro Vancouver Microzonation Mapping Project. This data was recorded using TROMINO three-component seismometers manufactured by MoHo s.r.l. This data, along with the rest of the data for the Eastern phase of the mapping project has not been released to the public and currently belongs to the Metro Vancouver Microzonation Mapping Project. Upon completion of the project, the data and resulting hazard maps will be made available through the project's public website at metrovanmicromap.ca. Information regarding

the Western Metro Vancouver phase of the project, including the initial suite of 29 hazard maps, are currently accessible on that site. All inversion processing was completed using the Geopsy software package, which is available as an open-source resource. The coding software MATLAB was used for data analysis and figure construction.

Acknowledgments

Thank you to Dr. Sheri Molnar for taking me on as a student and for all of your guidance with this project. Her and the rest of the Good Vibrations and Excitations lab group are an incredibly supportive and hardworking group of scientists, and I am very happy to have had the pleasure of working with them.

Thank you to Dr. Robert Shcherbakov for taking the time to review this project and for your support as a professor.

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Tables

Table 1: Geologic Site Classes for National Earthquake Hazards Reduction Program. Retrieved from (Taylor et al., 2006)

Site Class	General Description	Definition (V_{s30}=average shear-wave velocity in upper 30 m, m/sec; N_{30}=average N in the upper 30 m)
A	Hard rock	$V_{s30}>1500$
B	Rock	$760<V_{s30}<1500$
C	Very dense soil and soft rock	$360<V_{s30}<760$; or $N_{30}>50$; or >3 m of soil over bedrock, where $V_{s30}>760$ m/sec
D	Stiff soils	$180<V_{s30}<360$; $15<N_{30}<50$
E	Soft soils, or soil profile with >3 m soft silt and clay	$V_{s30}<180$; or $N_{30}<15$; or >3 m silt and clay with plasticity index >20, moisture content >40%, and undrained shear strength <25 kPa)

Table 2: Initial Depth and v_s boundaries of each subsurface layer for the MHVSR inversions

Initial Model Parameterization Bounds		
Option 1		
Layer Number	Depth (m)	Vs (m/s)
1	1-100	50-250
2	30-100	200-400
3	75-500	400-1000
4	half space	1000-2500
Option 2		
Layer Number	Depth (m)	Vs (m/s)
1	1-50	50-400
2	10-500	400-800
3	half space	400-1000
Option 3		
Layer Number	Depth (m)	Vs (m/s)
1	1-50	50-250
2	30-100	350-800
3	100-600	800-1500
4	half space	1500-2500

Table 3: Additional Depth and Vs boundaries of each subsurface layer for the MHVSR inversions

Additional Model Parameterization Bounds						
Option 4				Option 8		
Layer Number	**Depth (m)**	**Vs (m/s)**		Layer Number	**Depth (m)**	**Vs (m/s)**
1	1-500	100-500		1	1-200	10-200
2	halfspace	500-1000		2	50-600	100-500
				3	half space	200-1000
Option 5						
Layer Number	**Depth (m)**	**Vs (m/s)**		Option 9		
1	1-300	100 - 400		Layer Number	**Depth (m)**	**Vs (m/s)**
2	30 - 500	200 - 700			1-200	50-200
3	75 - 600	500 - 1500			30-400	200-700
4	half space	1500 - 2500			100-800	600-1500
					half space	1500-2500
Option 6						
Layer Number	**Depth (m)**	**Vs (m/s)**		Option 10		
1	50-400	100-400		Layer Number	**Depth (m)**	**Vs (m/s)**
2	100-600	200-1700			1-200	10-200
3	half space	1700-2500			30-600	100-700
						600-1500
Option 7						
Layer Number	**Depth (m)**	**Vs (m/s)**				
1	50-400	100-300				
2	75-500	200-600				
3	100-600	500-1700				
4	half space	1700-2500				

Figures

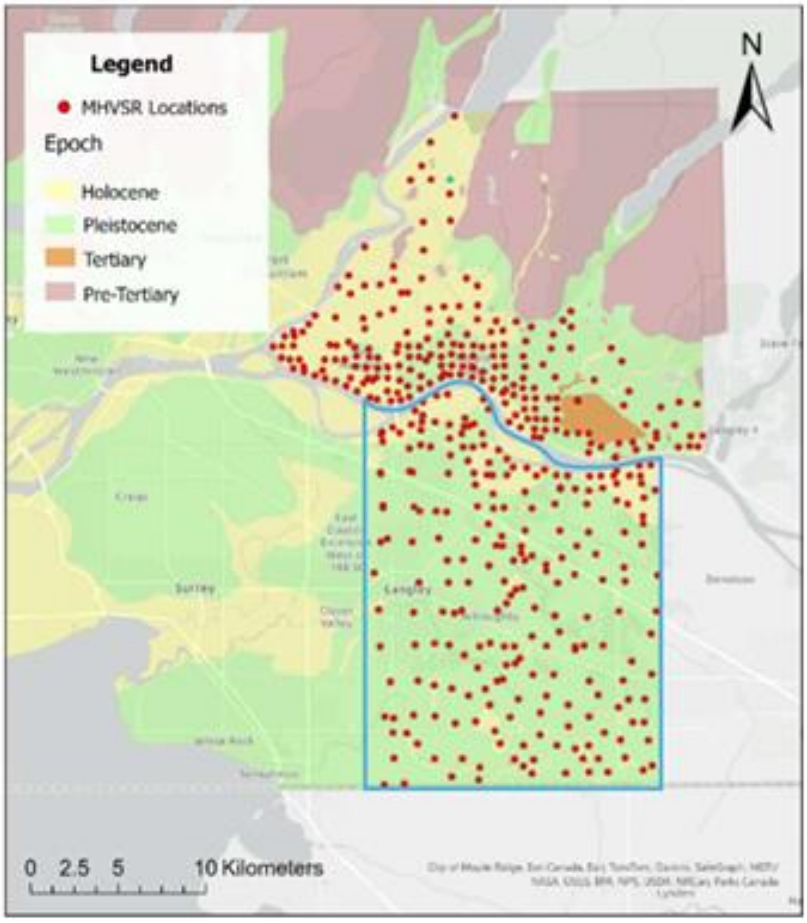


Figure 1: Map of MHVSR Sites and Geology of Eastern Vancouver. The Blue box contains the city of Langley. Retrieved from Motuzas (2025).

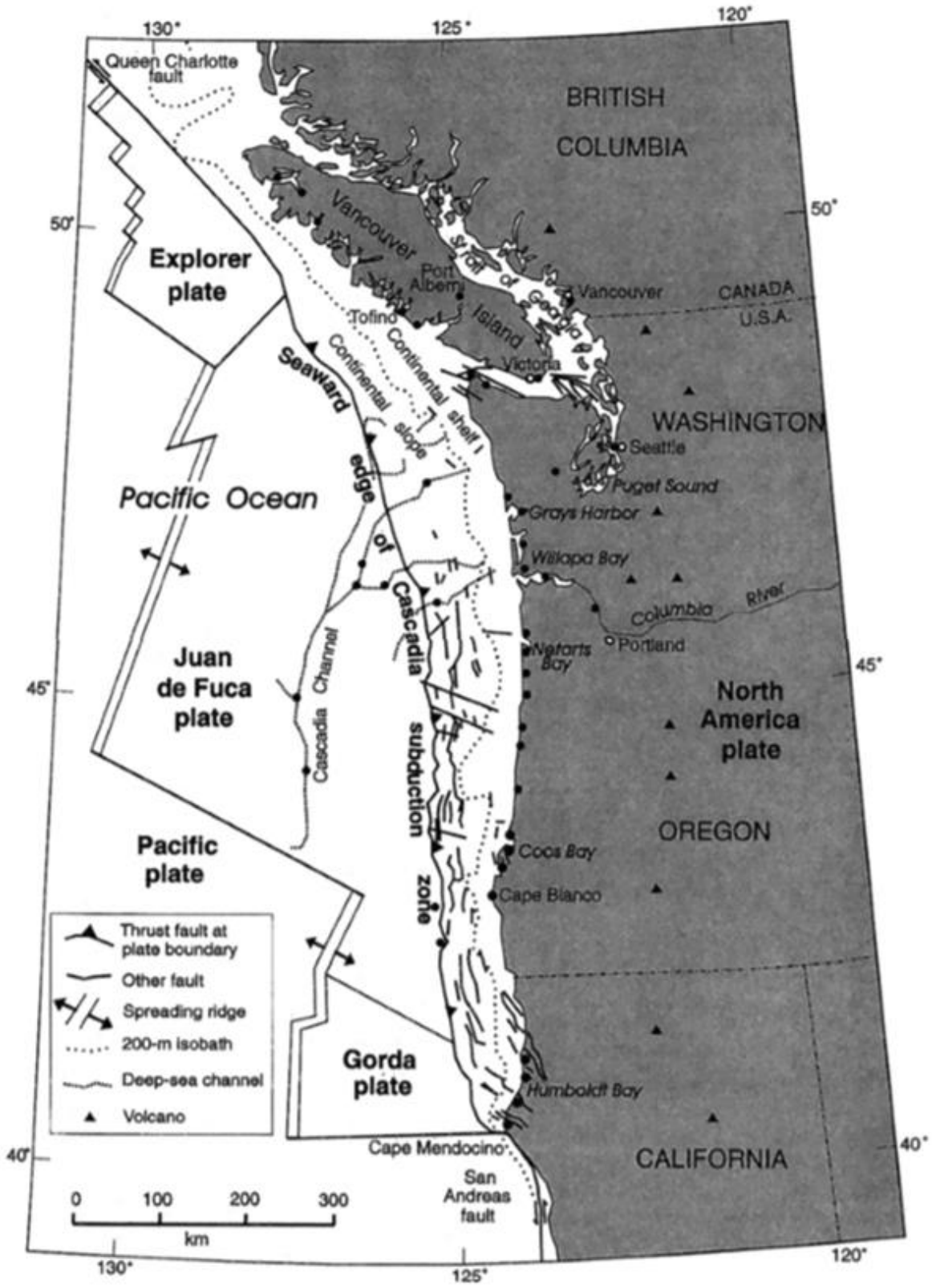


Figure 2: Map of the plates, faults, ridges, volcanos, islands and continent involved in the Cascadia Subduction Zone. Retrieved from Clague (1997).

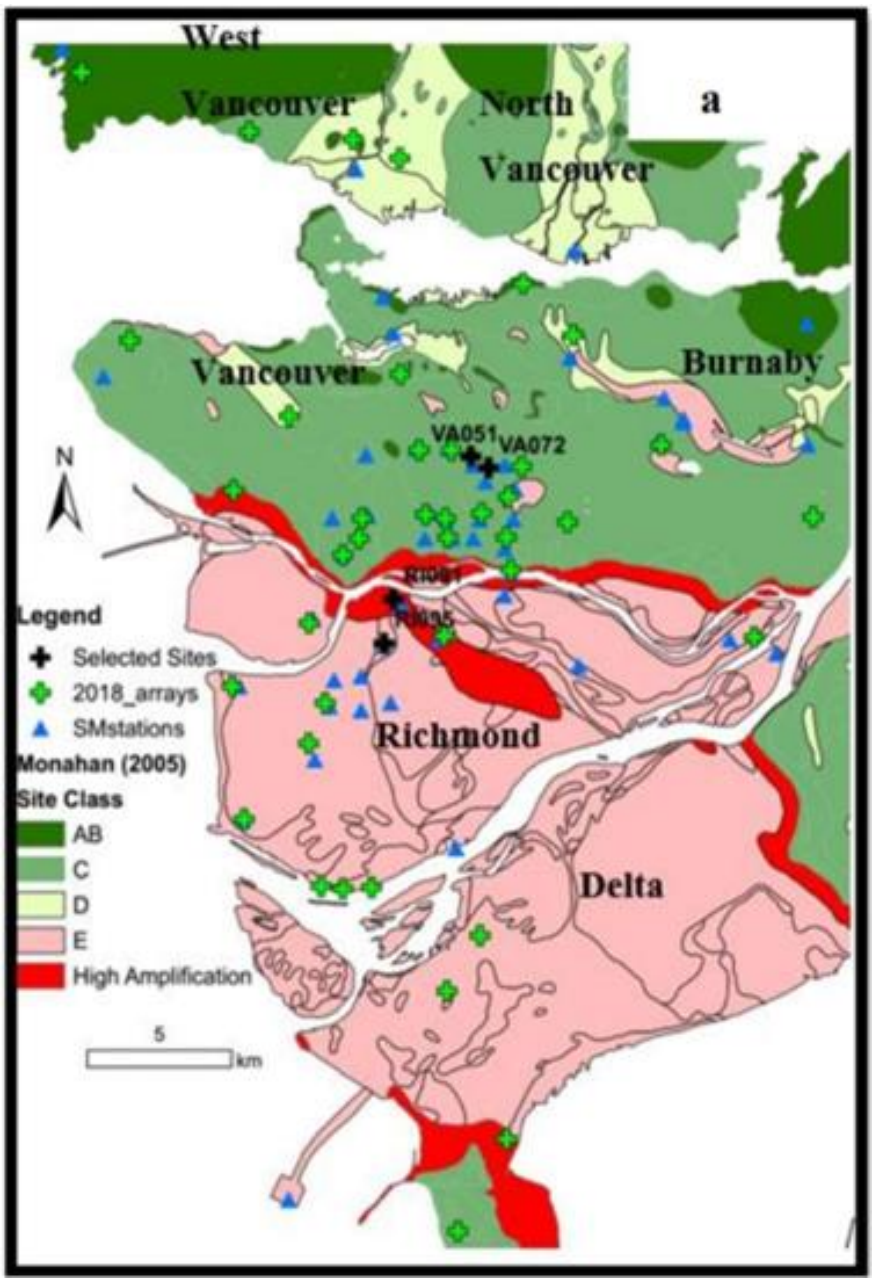


Figure 3: Greater Vancouver and its seismic site classes. Retrieved from Assaf et al (2019)

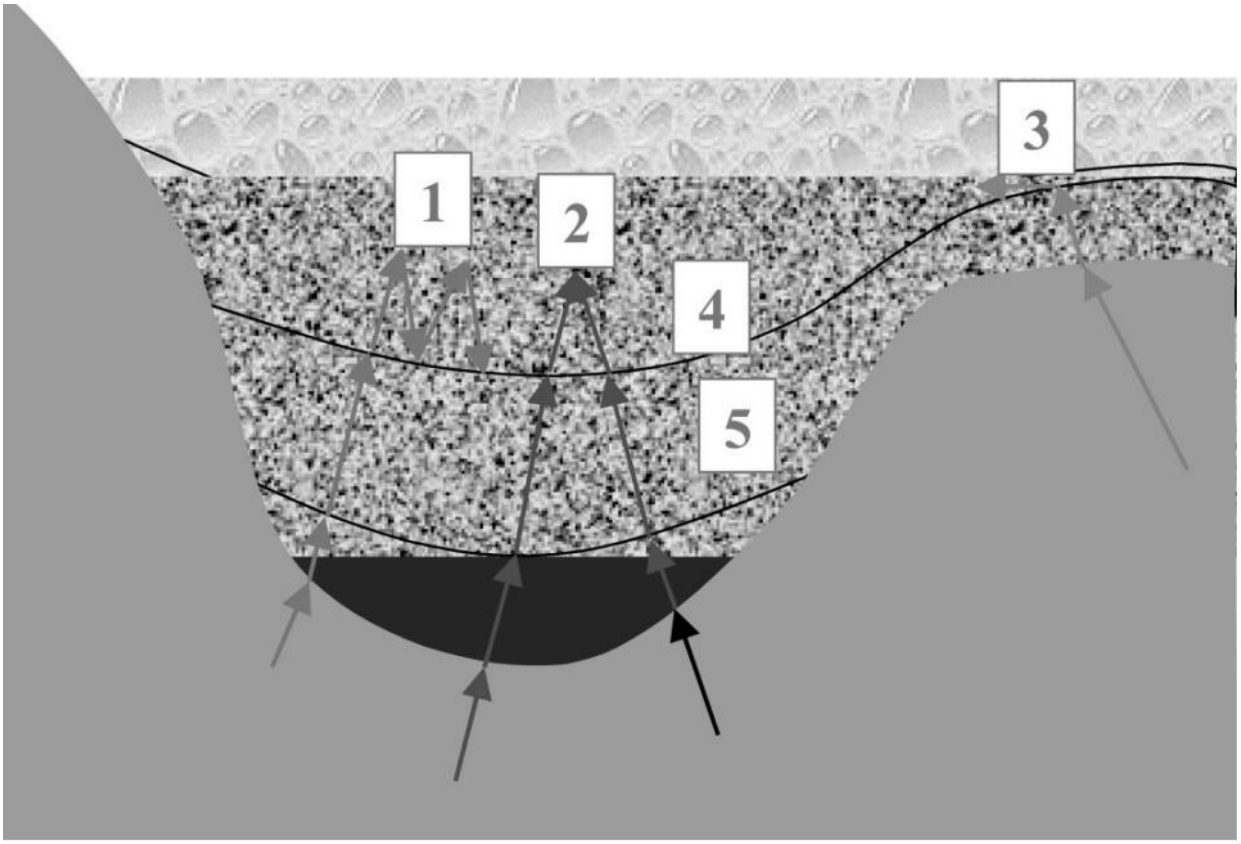


Figure 4: Visual of factors such as (1) impedance contrasts, (2) topography focusing, (3) converted surface waves, (4) water content, and (5) randomness that influence seismic amplification. Retrieved from Şafak (2001).

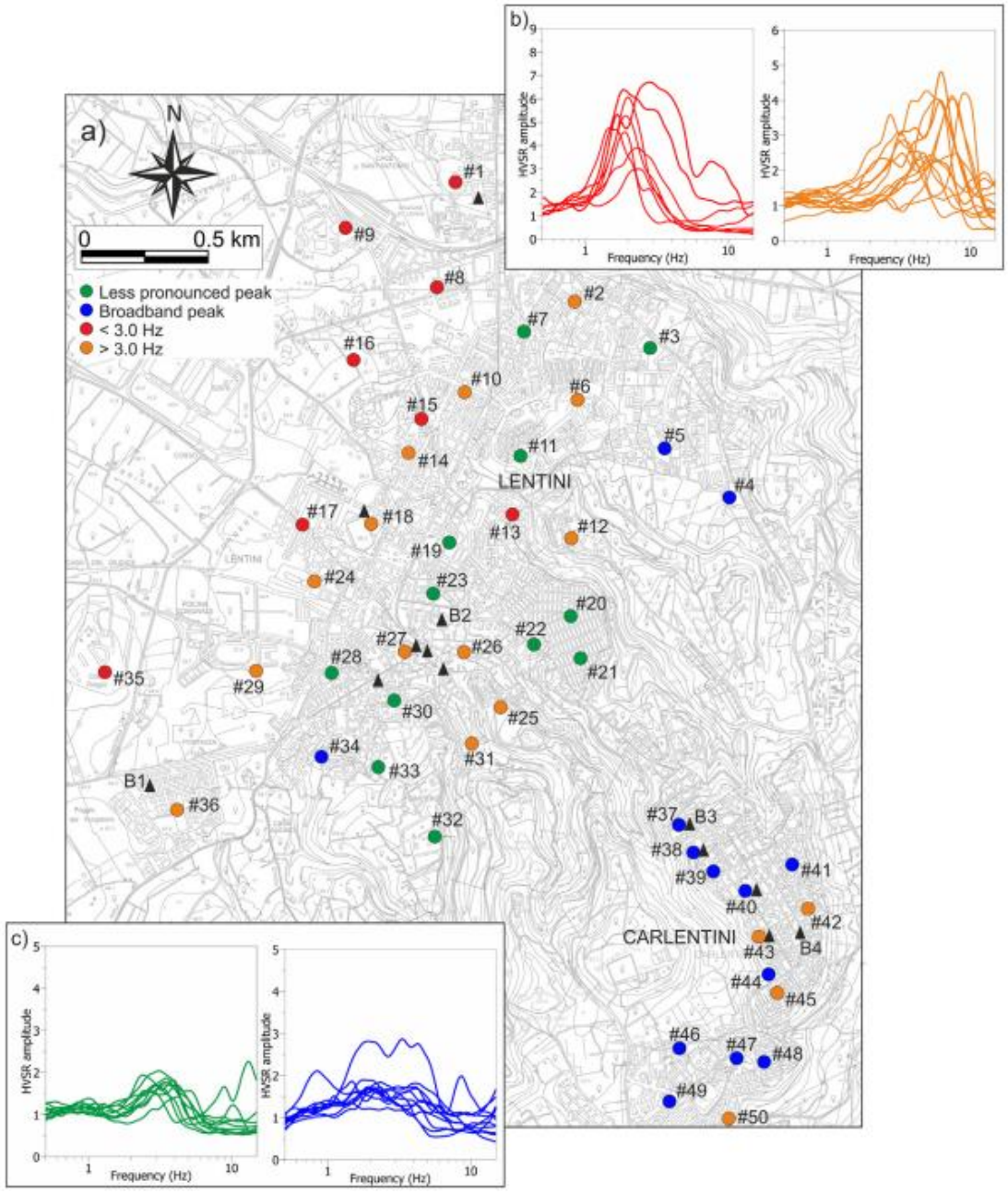


Figure 5: Distribution of resonance frequencies and HVSR patterns in the Lentini and Carlentini area. Retrieved from Panzera et al. (2018).

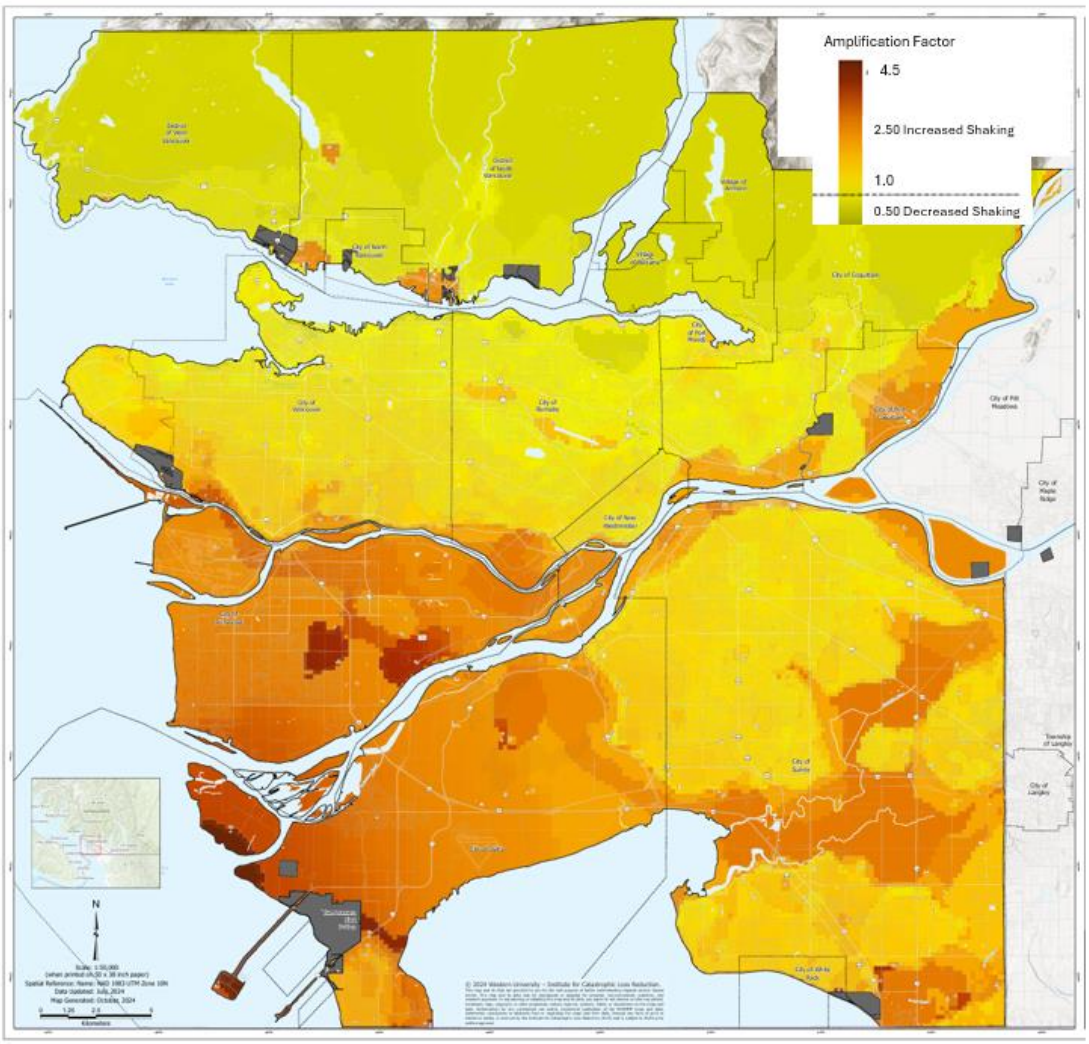


Figure 6: Western Metro Vancouver Amplification hazard Map. Modified from Assaf et al. (2024)

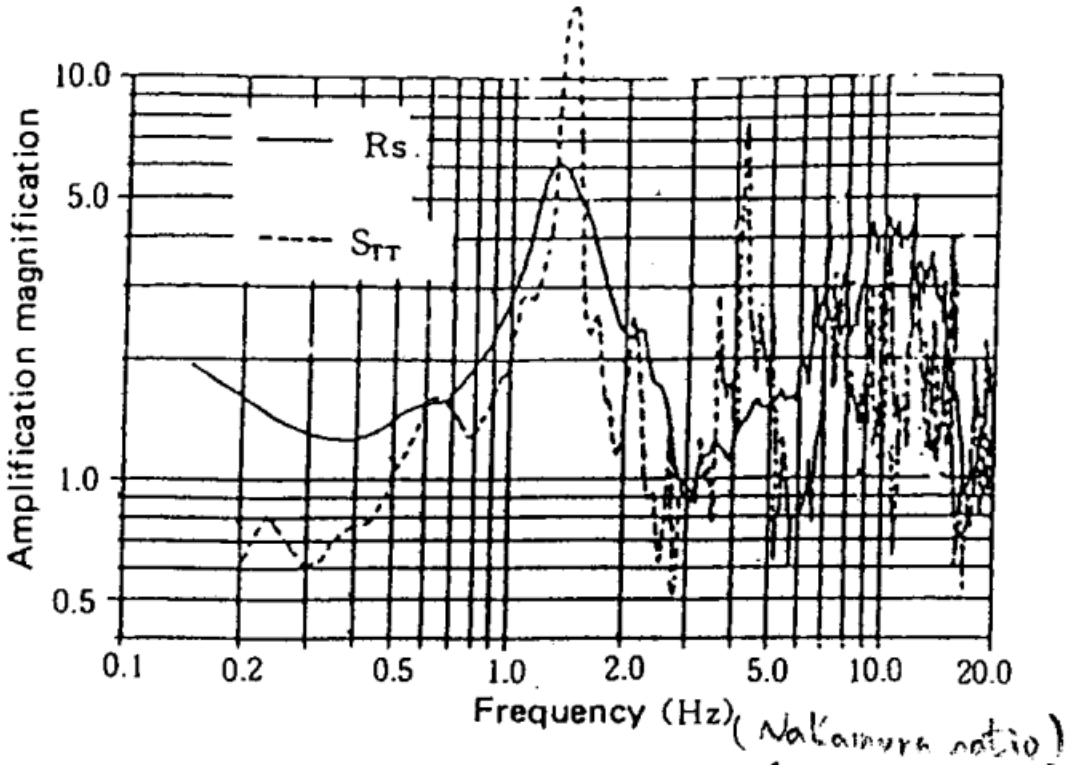


Figure 7: Comparison of the microtremor spectral ratio (R_{-s}) and earthquake transfer function (S_{TT}). Retrieved from Nakamura (1989).



Figure 8: Image of TROMINO for MHVSR measurement in Langley. Retrieved from Motuzas (2025)

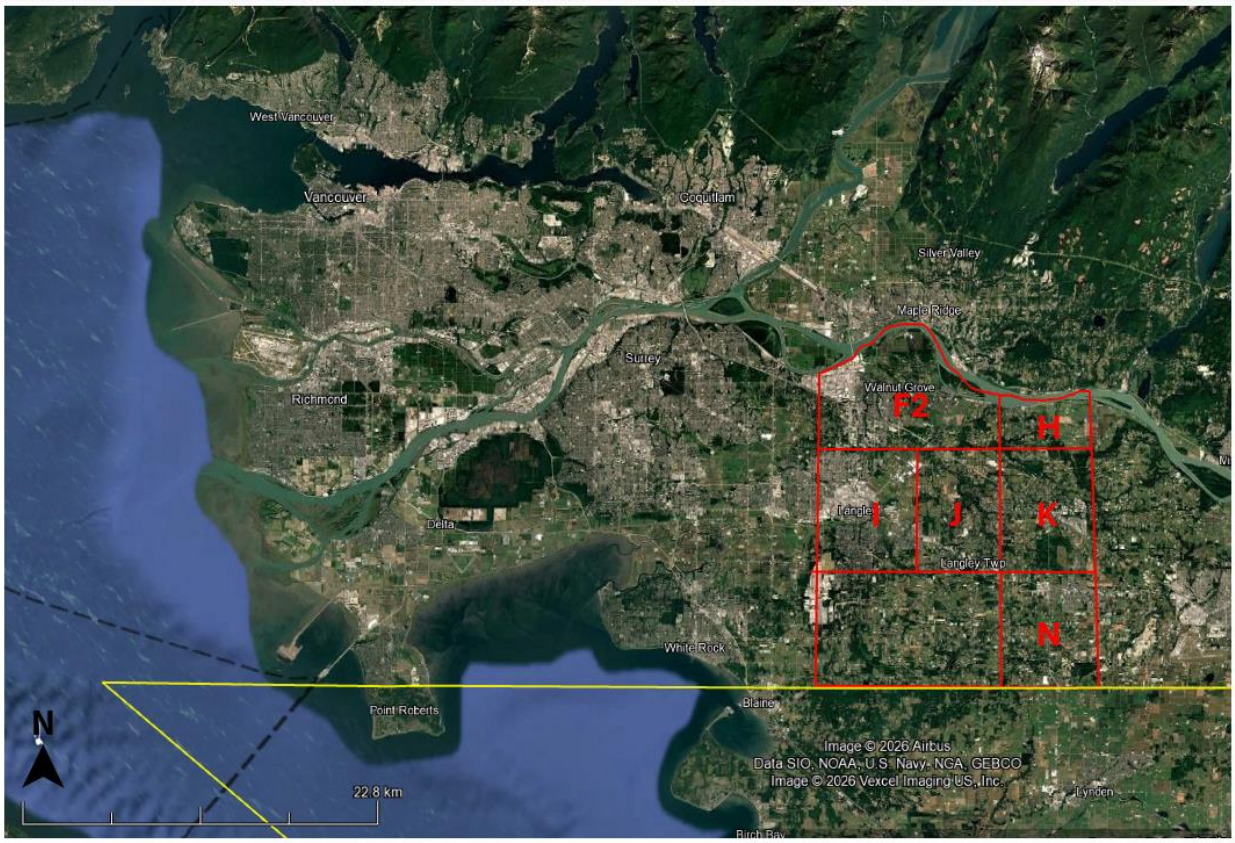


Figure 9: project area and regional subsections for MHVSR sites

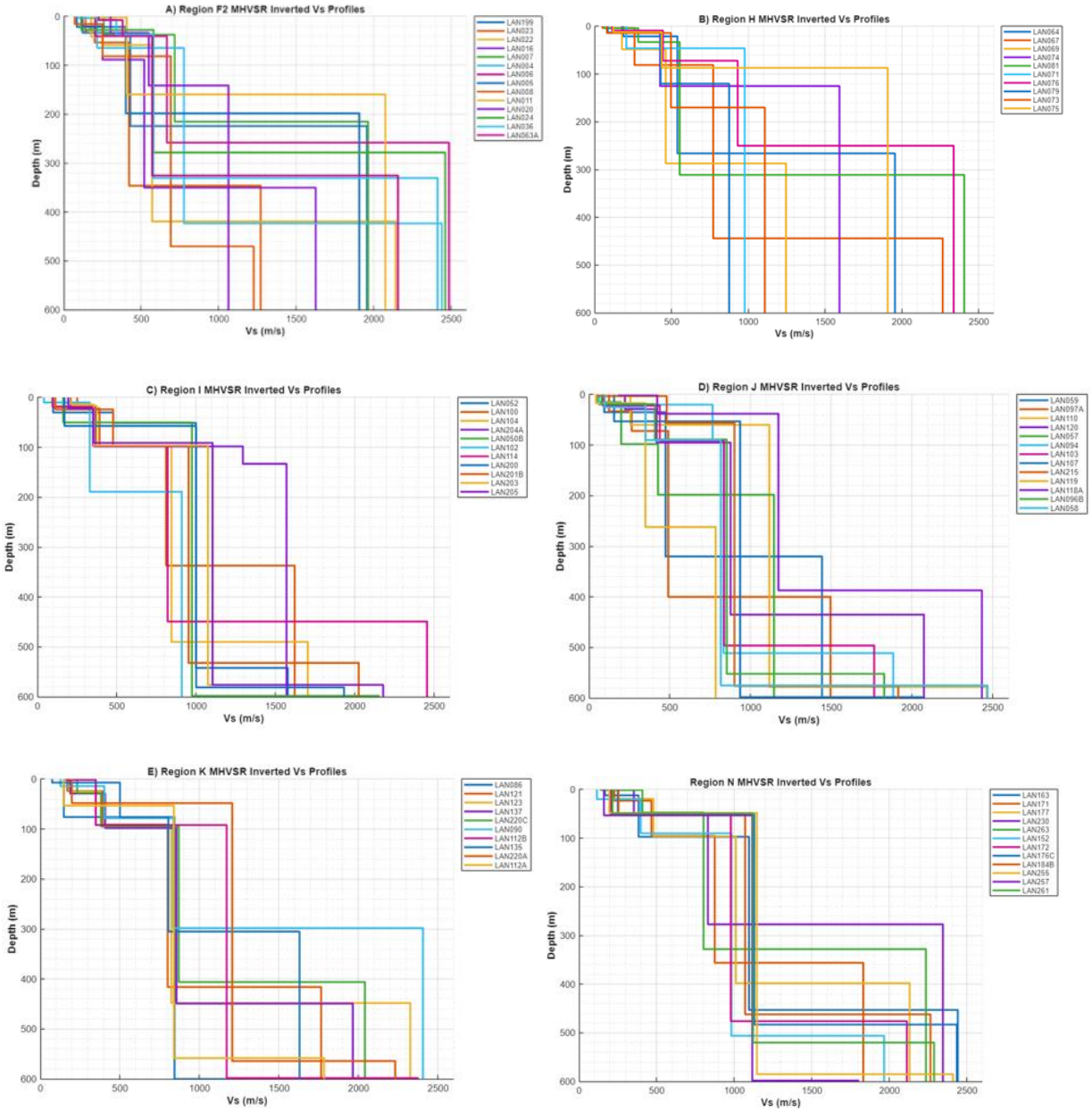


Figure 10: Ground profiles for final inversions from **A)** Region F2 **B)** Region H **C)** Region I **D)** Region J **E)** Region K **F)** Region N

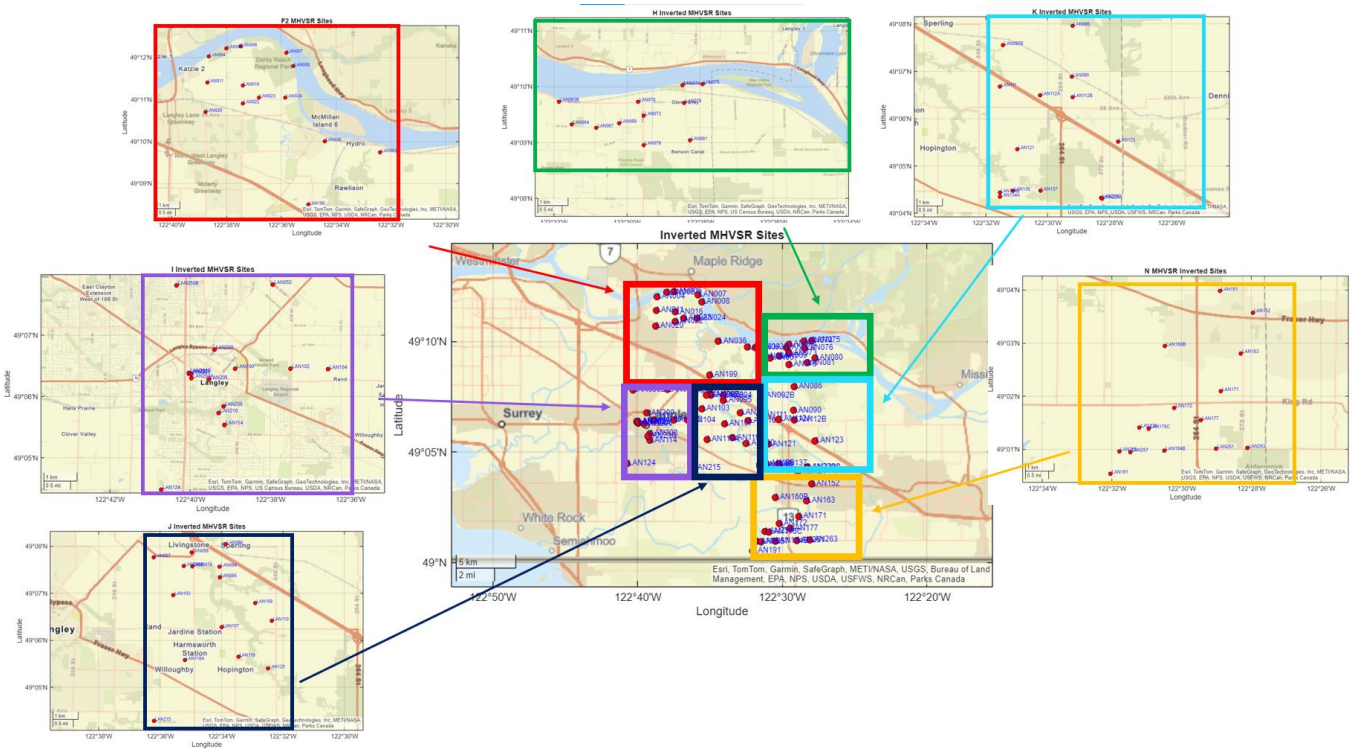


Figure 11: Maps of the Sites used for MHVSR Inversions

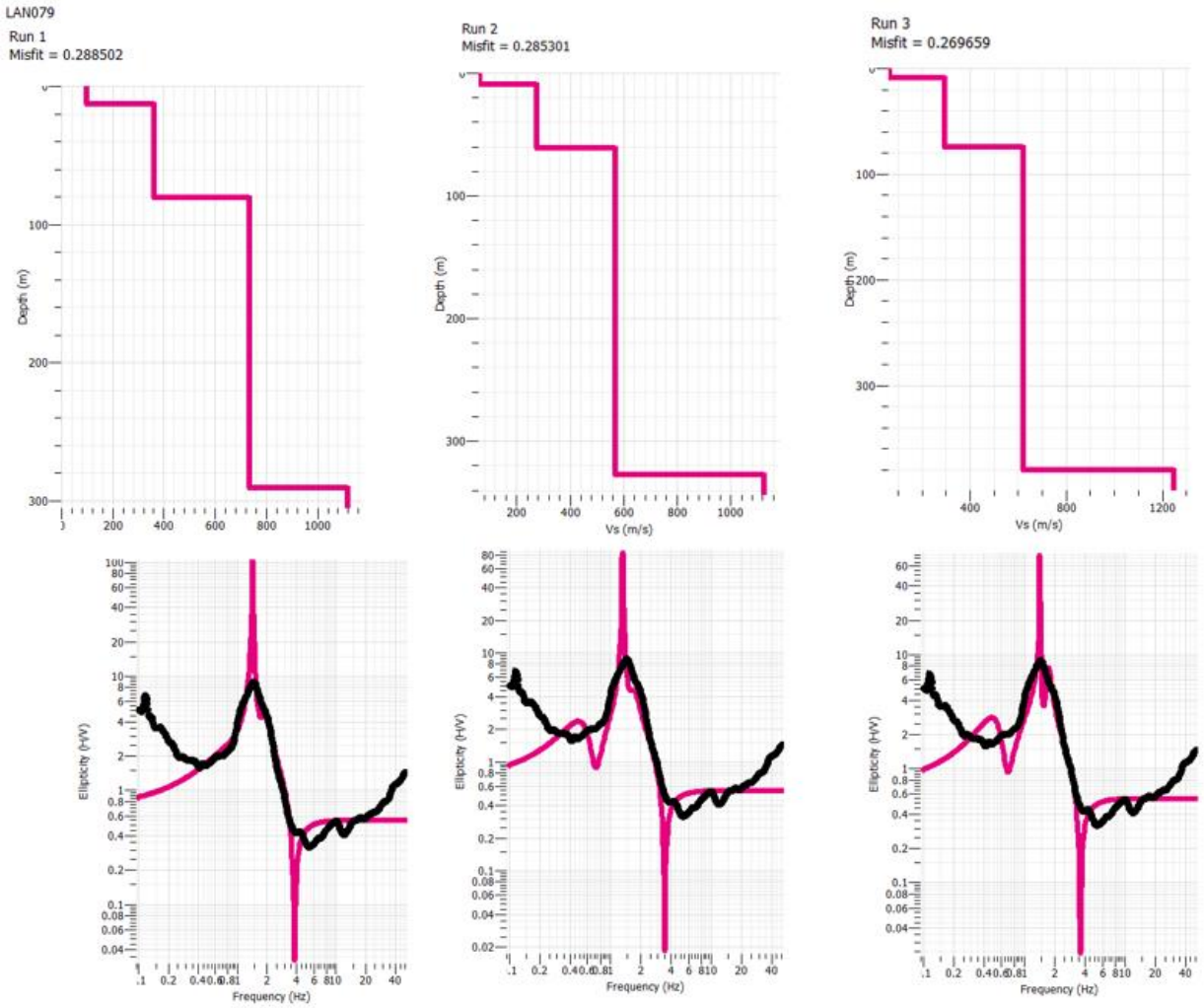
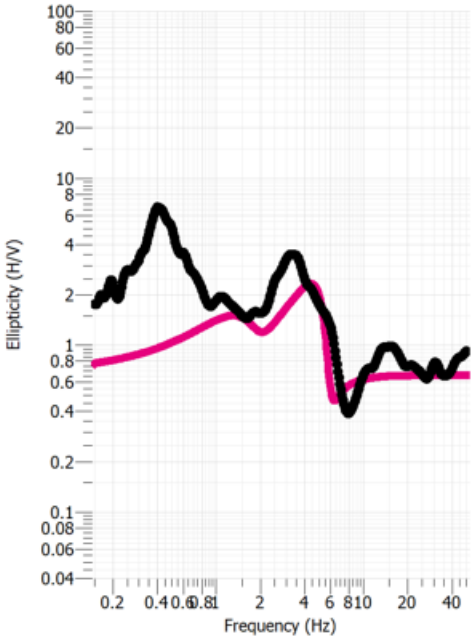


Figure 12: Example of results produced for the MHVSR Inversions.

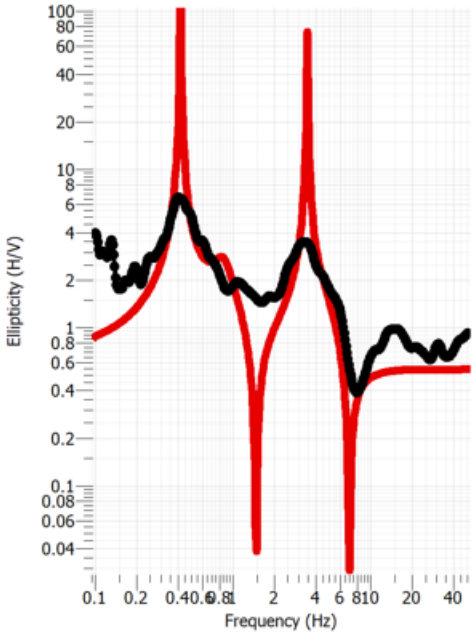
LAN081

A)



Option 2		
Layer Number	Depth (m)	Vs (m/s)
1	1-50	50-400
2	10-500	400-800
3	half space	400-1000

B)



Option 1		
Layer Number	Depth (m)	Vs (m/s)
1	1-100	50-250
2	30-100	200-400
3	75-500	400-1000
4	half space	1000-2500

Figure 13: A) Frequency/Ellipticity curve produced through the inversion of site LAN081 using option 2 parameters **B)** Frequency/Ellipticity curve produced through the inversion of site LAN076 using option 1 parameters

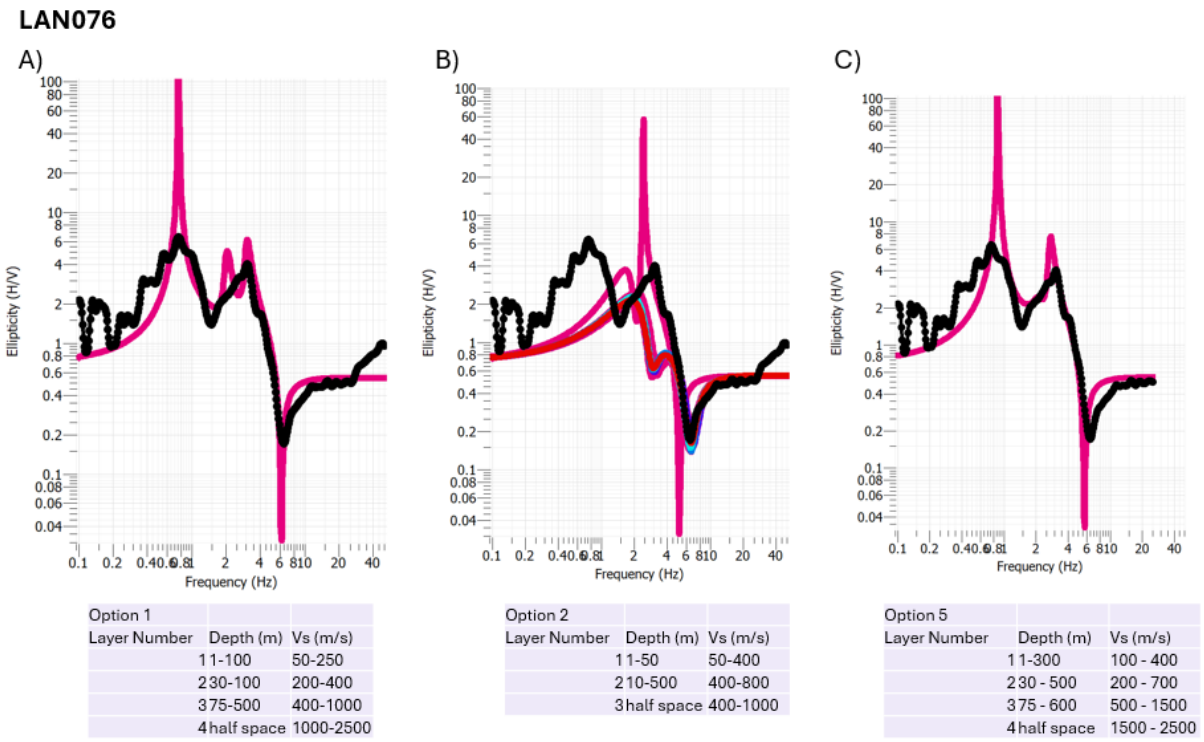
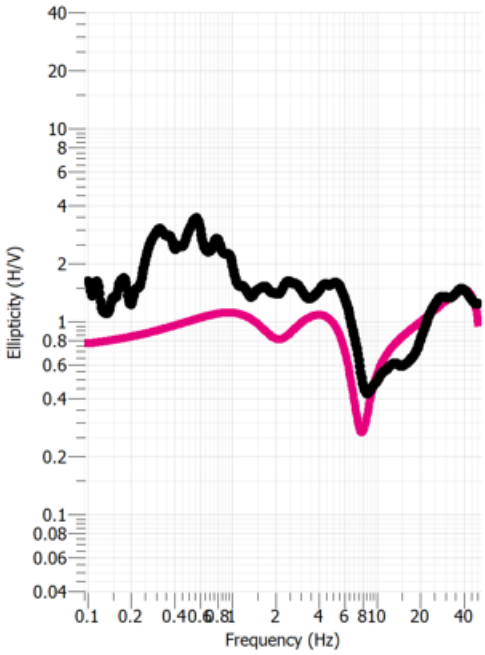


Figure 14: A) Frequency/Ellipticity curve produced through the inversion of site LAN076 using option 1 parameters **B)** Frequency/Ellipticity curve produced through the inversion of site LAN076 using option 2 parameters **C)** Frequency/Ellipticity curve produced through the inversion of site LAN076 using option 5 parameters

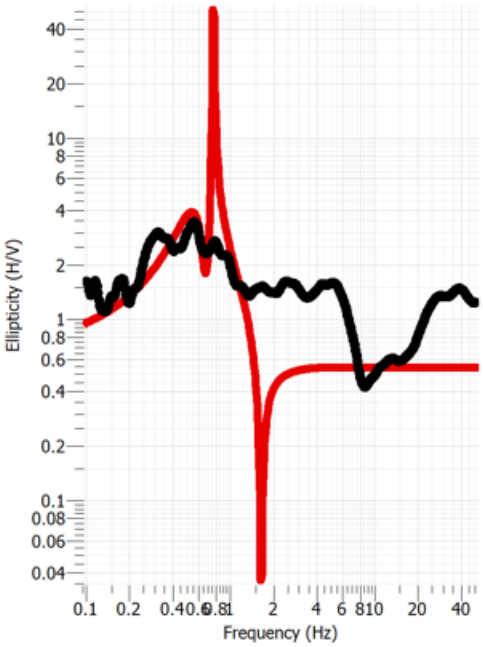
LAN050B

A)



Option 1		
Layer Number	Depth (m)	Vs (m/s)
1	1-100	50-250
2	30-100	200-400
3	75-500	400-1000
4	half space	1000-2500

B)

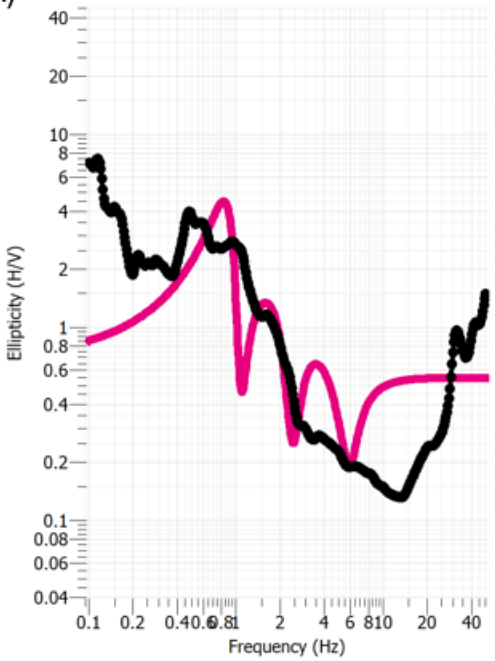


Option 6		
Layer Number	Depth (m)	Vs (m/s)
1	50-400	100-400
2	100-600	200-1700
3	half space	1700-2500

Figure 15: A) Frequency/Ellipticity curve produced through the inversion of site LAN050B using option 1 parameters **B)** Frequency/Ellipticity curve produced through the inversion of site LAN050B using option 6 parameters

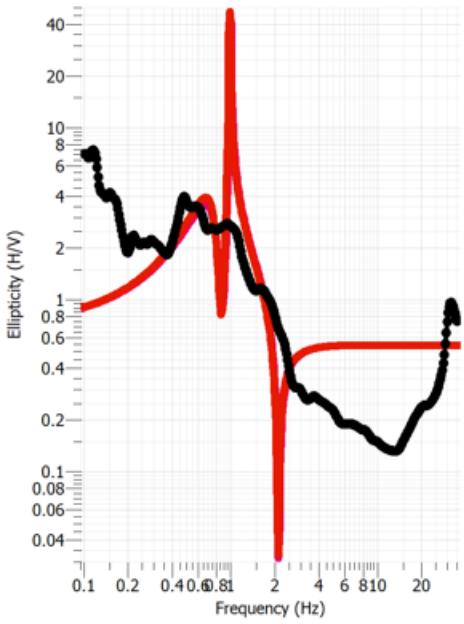
LAN102

A)



Option 1		
Layer Number	Depth (m)	Vs (m/s)
1	1-100	50-250
2	30-100	200-400
3	75-500	400-1000
4	half space	1000-2500

B)



Option 6		
Layer Number	Depth (m)	Vs (m/s)
1	50-400	100-400
2	100-600	200-1700
3	half space	1700-250

Figure 16: A) Frequency/Ellipticity curve produced through the inversion of site LAN102 using option 1 parameters **B)** Frequency/Ellipticity curve produced through the inversion of site LAN102 using option 6 parameters

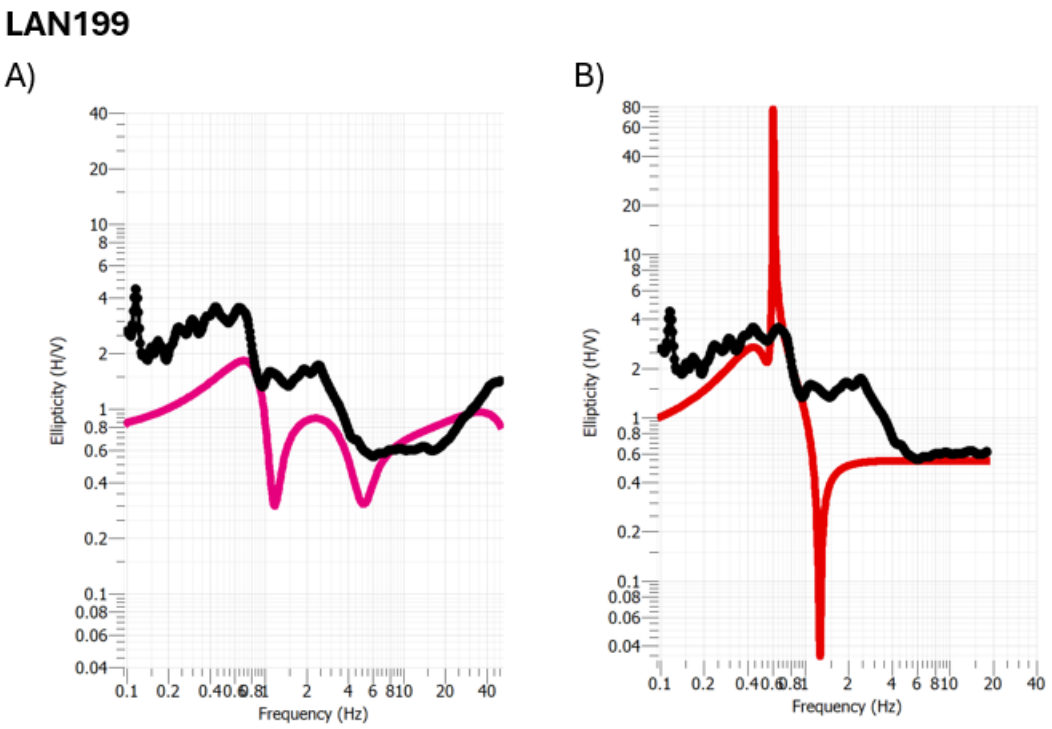


Figure 17: A) Frequency/Ellipticity curve produced through the inversion of site LAN199 with no cutting of the MHVSR curve **B)** Frequency/Ellipticity curve produced through the inversion of site LAN199 after cutting off the MHVSR curve at 20 Hz

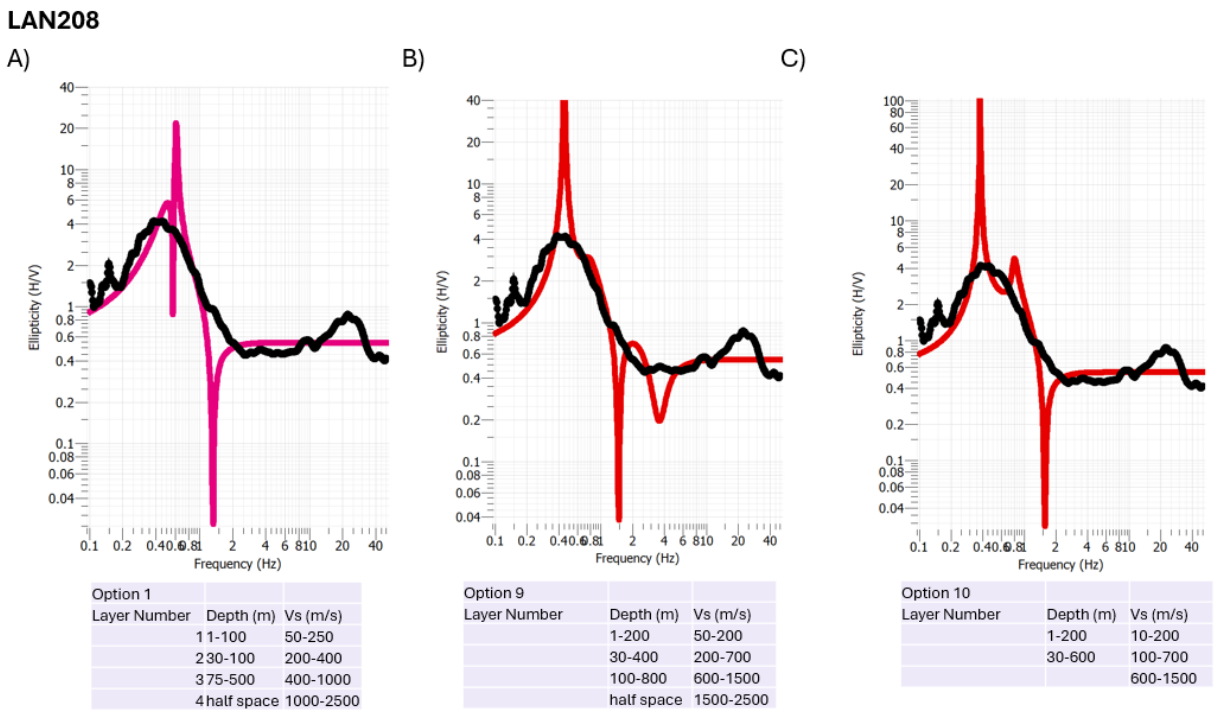
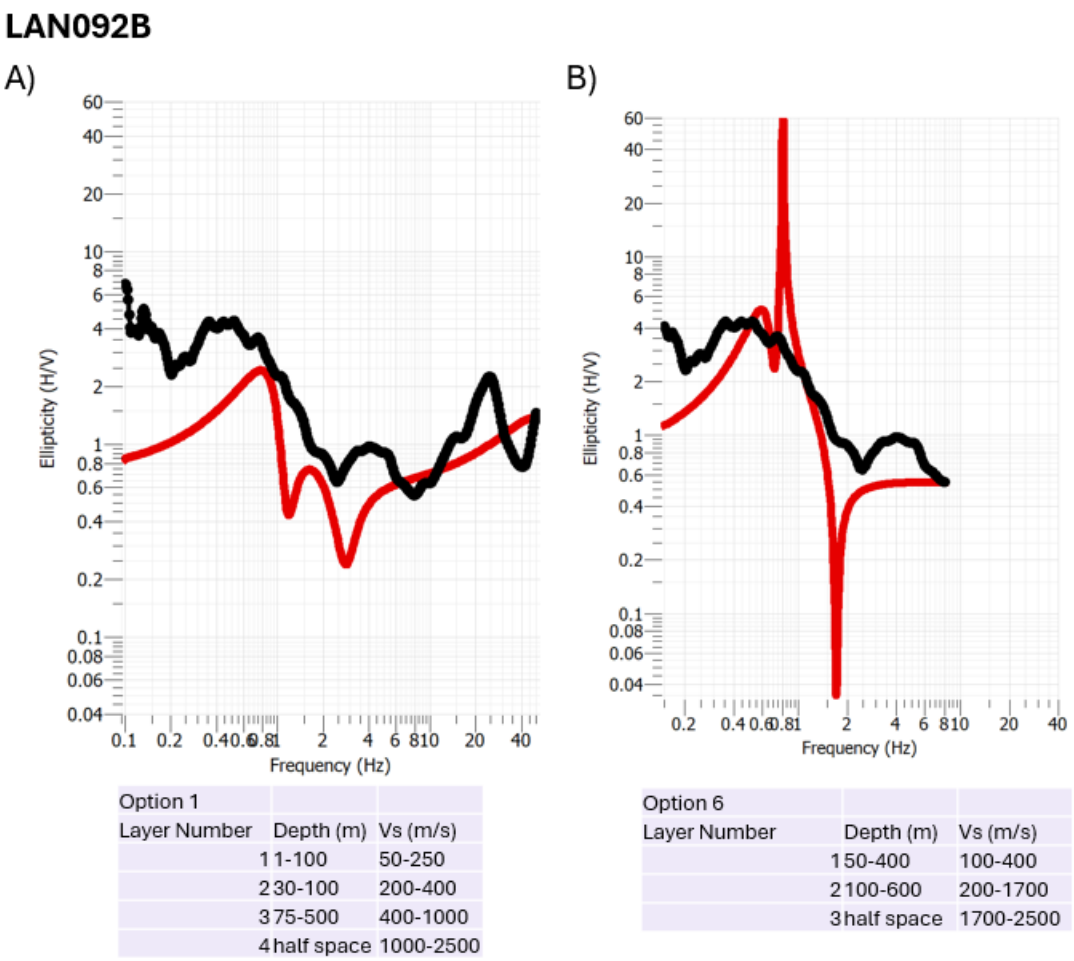


Figure 18: A) Frequency/Ellipticity curve produced through the inversion of site LAN208 using option 1 parameters **B)** Frequency/Ellipticity curve produced through the inversion of site LAN208 using option 9 parameters **C)** Frequency/Ellipticity curve produced through the inversion of site LAN208 using option 10 parameters



Option 1		
Layer Number	Depth (m)	Vs (m/s)
1	1-100	50-250
2	30-100	200-400
3	75-500	400-1000
4	half space	1000-2500

Option 6		
Layer Number	Depth (m)	Vs (m/s)
1	50-400	100-400
2	100-600	200-1700
3	half space	1700-2500

Figure 19: A) Frequency/Ellipticity curve produced through the inversion of site LAN092B with option 1 parameters and no cutting of the MHVSR curve **B)** Frequency/Ellipticity curve produced through the inversion of site LAN199 using option 6 parameters after cutting off the MHVSR curve at 10 Hz

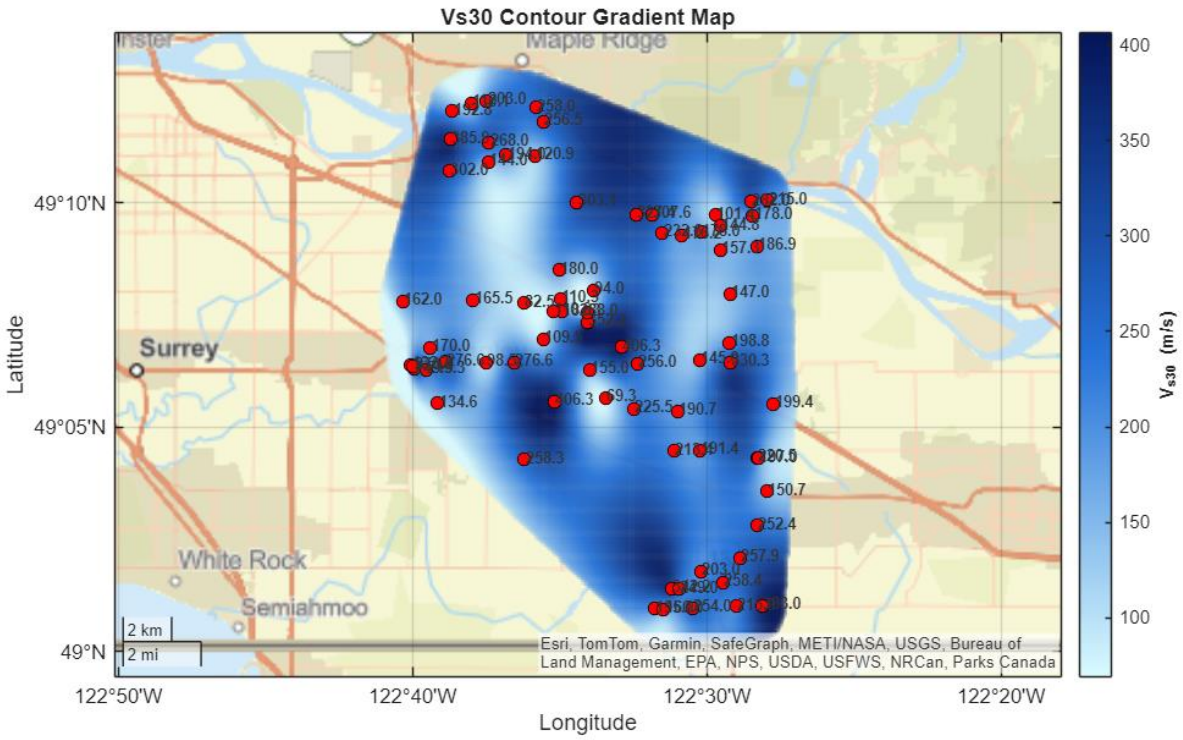


Figure 20: Vs_{30} Gradient Map

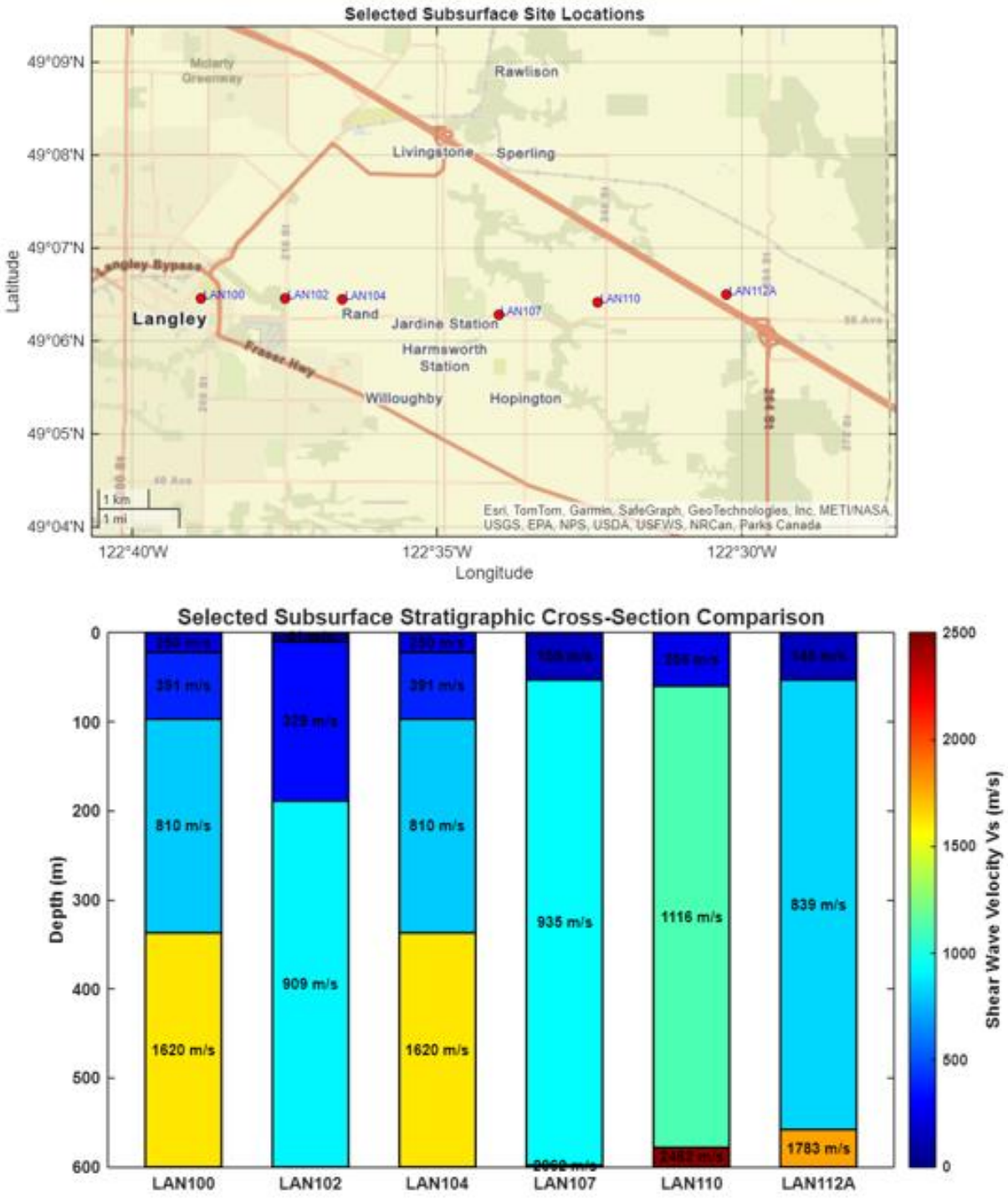


Figure 21: East-West transect showing subsurface layering for specified sites

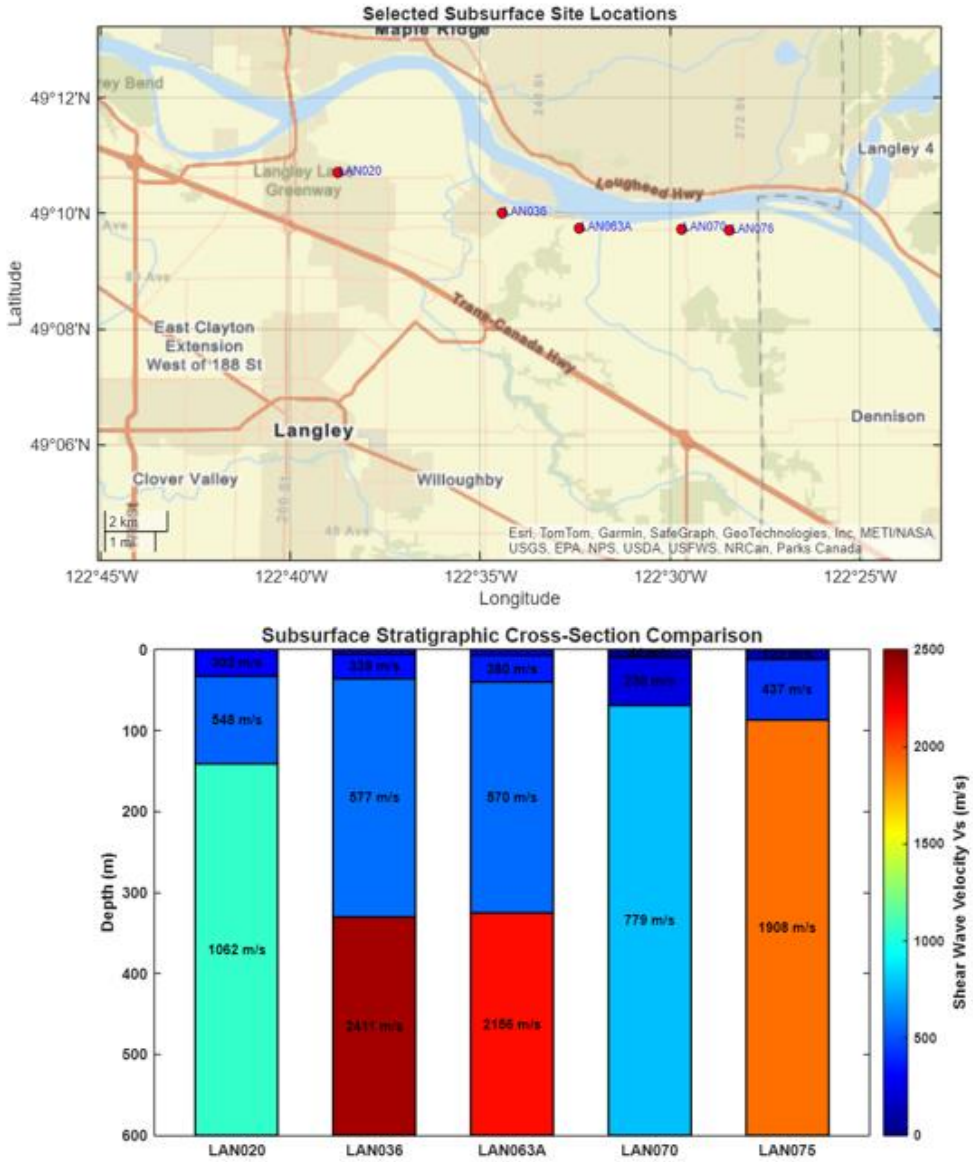


Figure 22: East-West transect showing subsurface layering for specified sites

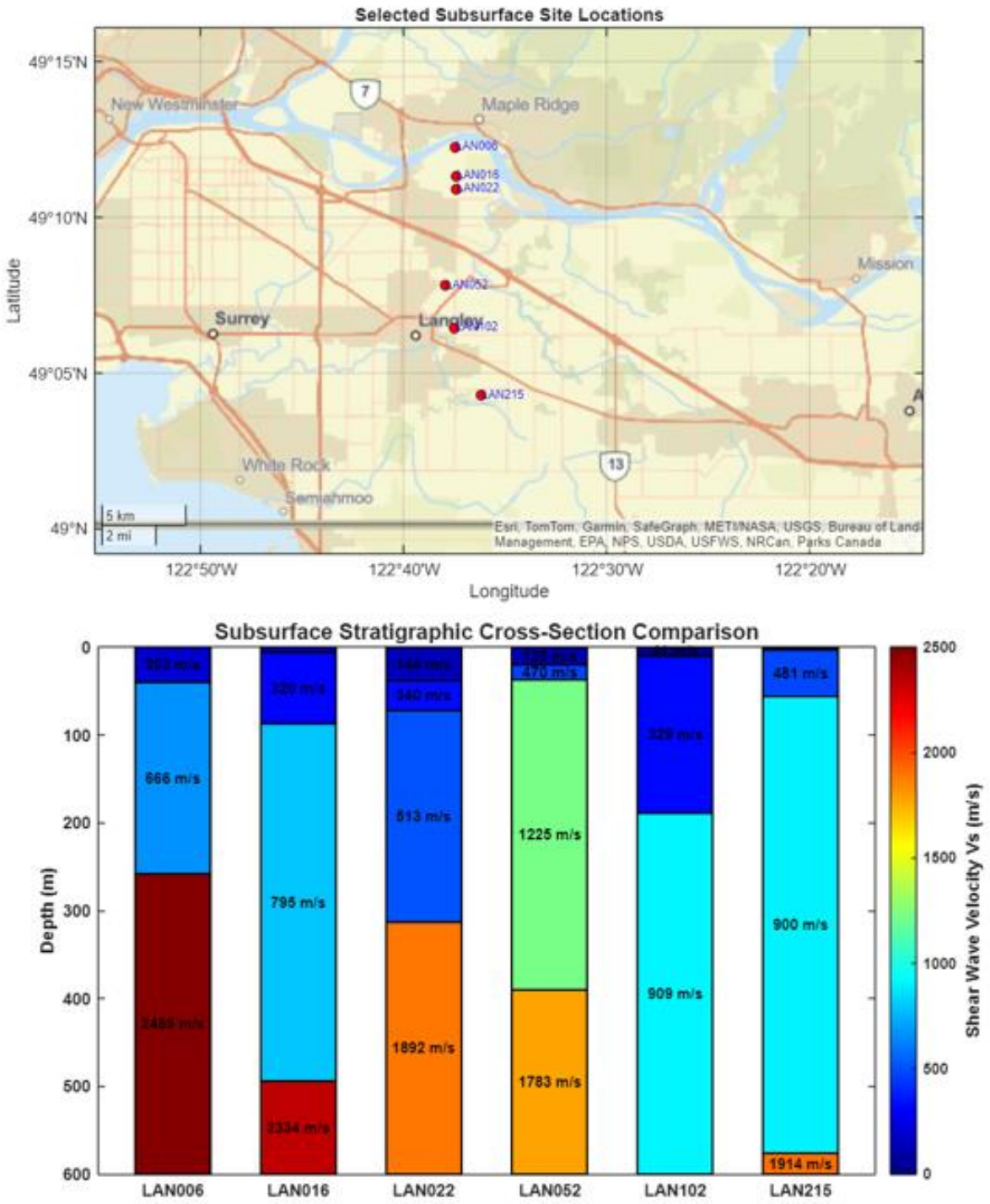


Figure 23: North-South transect showing subsurface layering for specified sites

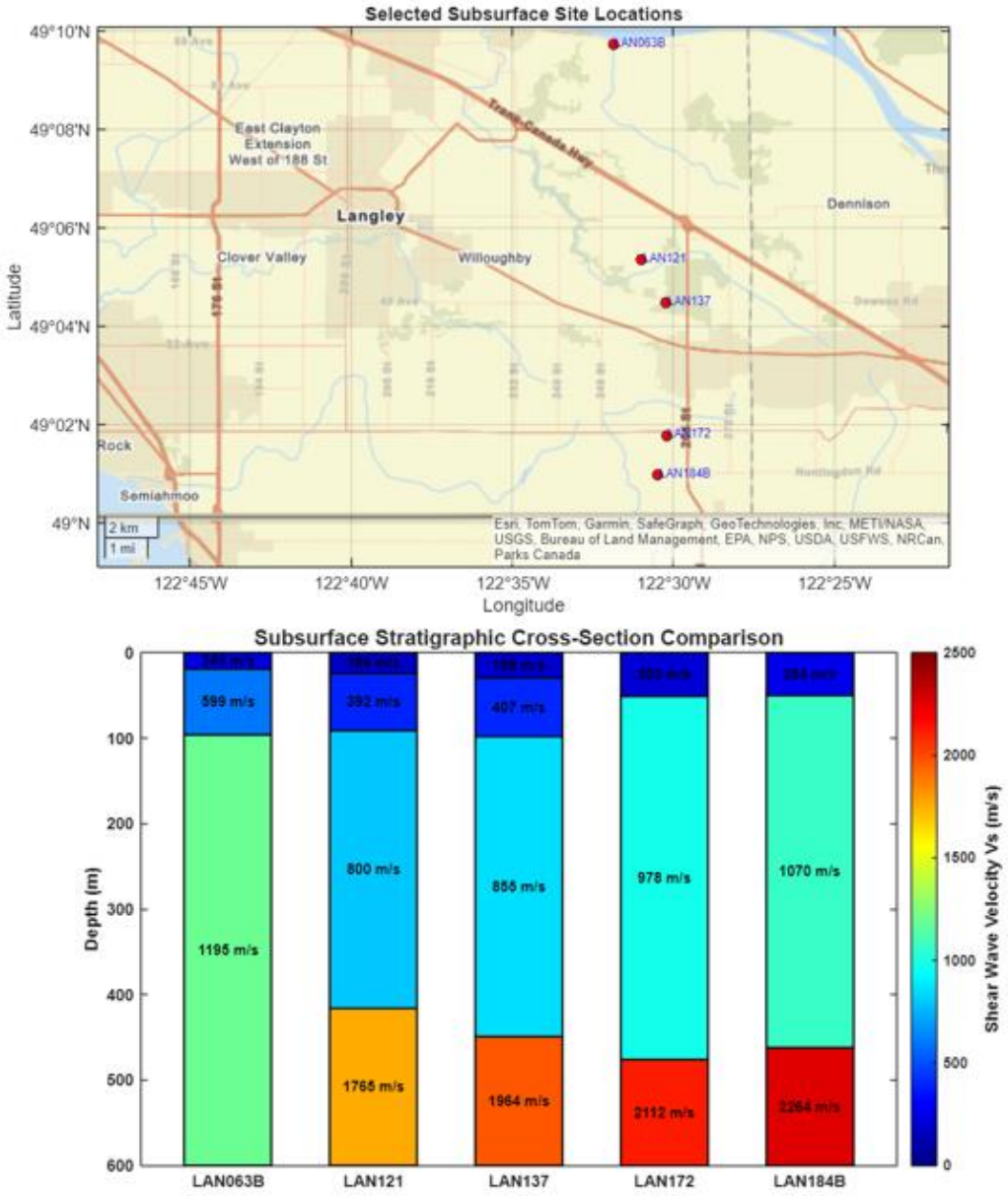


Figure 24: North-South transect showing subsurface layering for specified sites

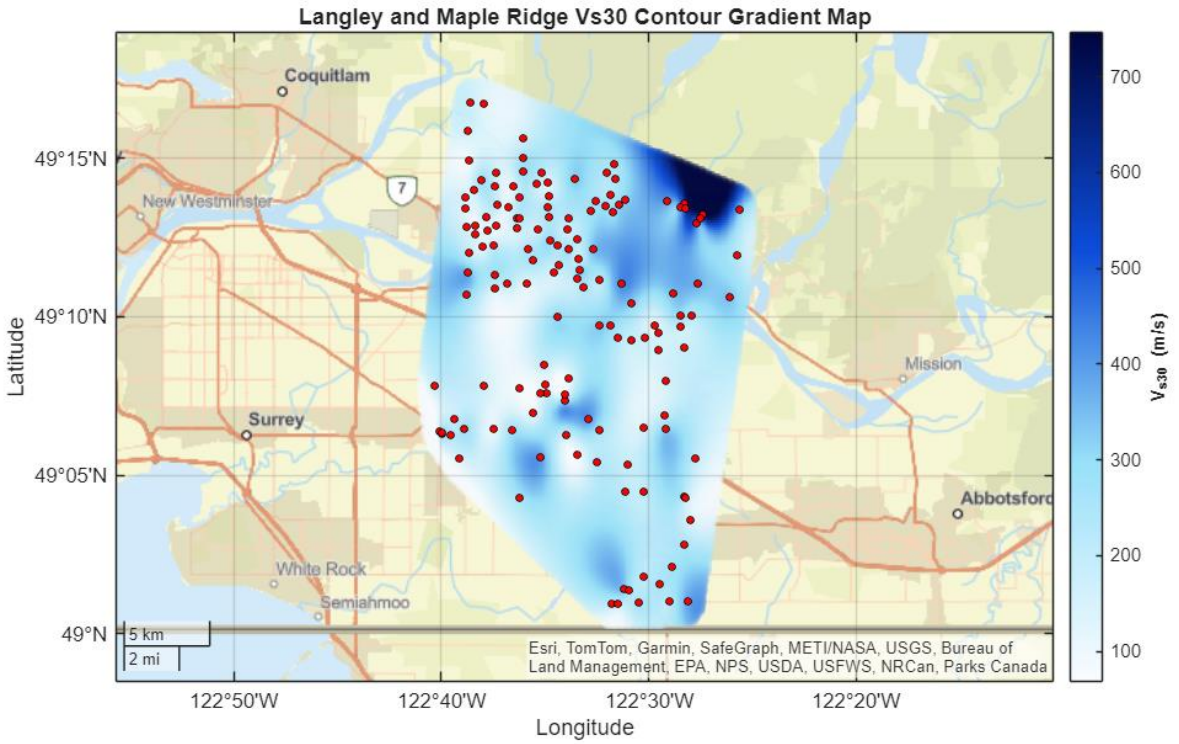
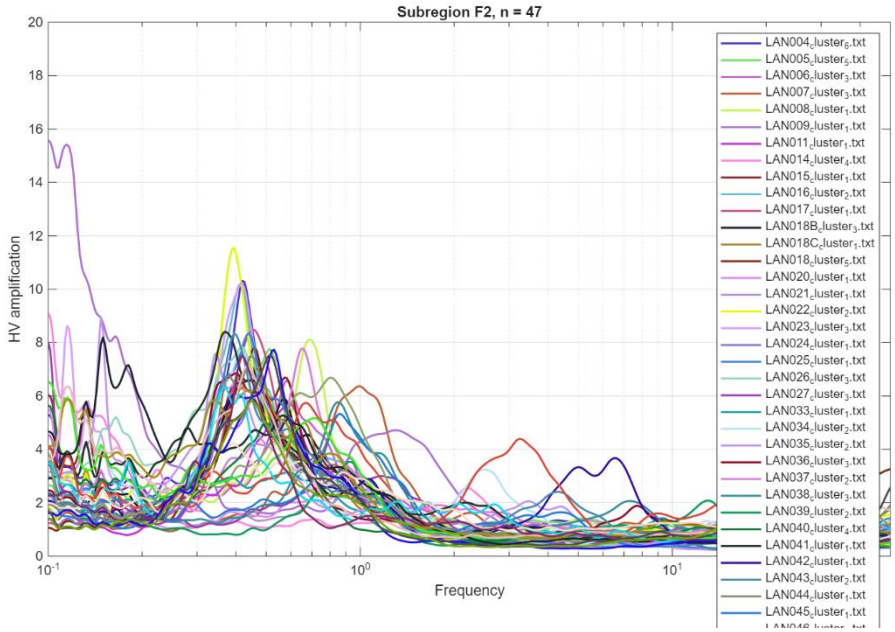


Figure 25: Vs30 map with additional Maple Ridge Data

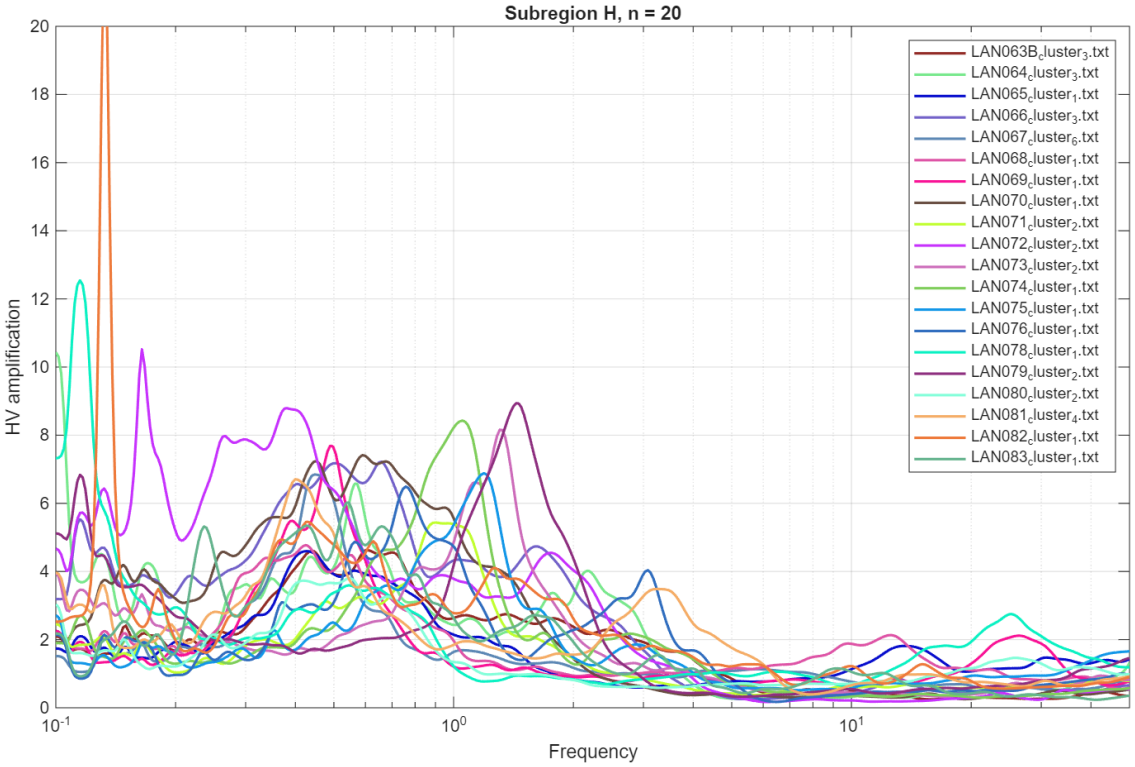
Appendices

Appendix A: Field Collected MHVSR Curves

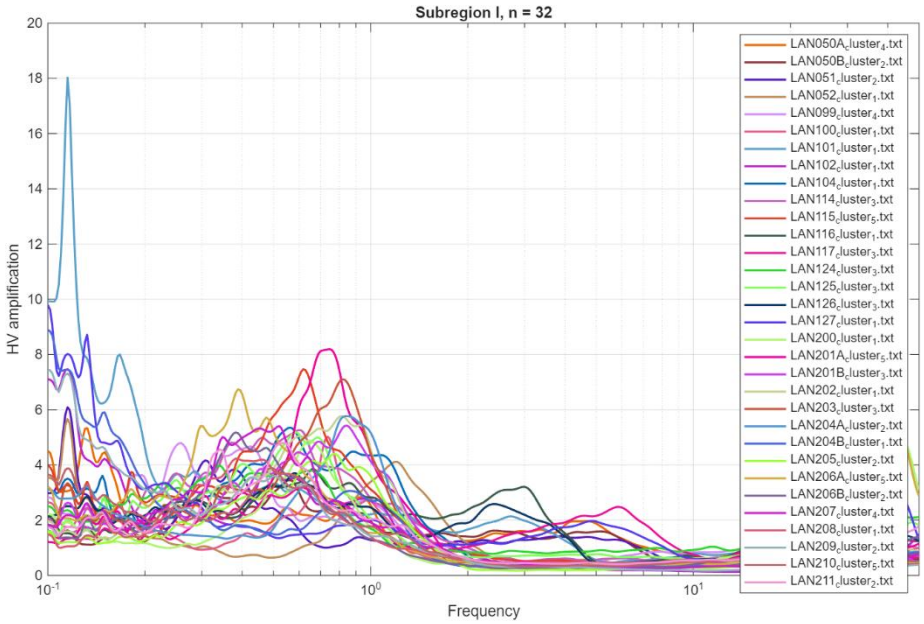
A1: MHVSR Curves for subregion F2



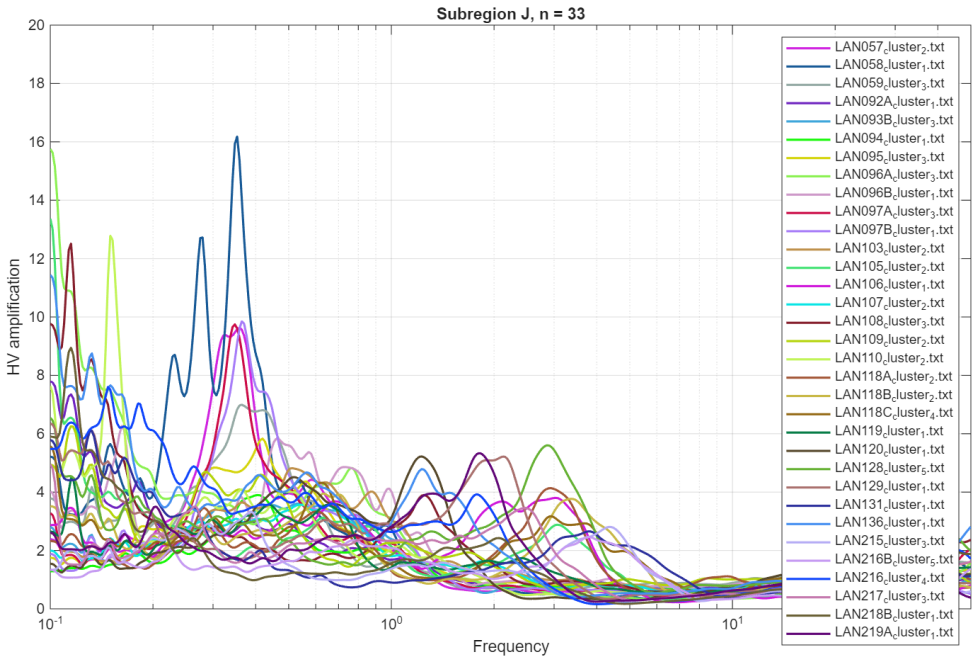
A2: MHVSR Curves for subregion H



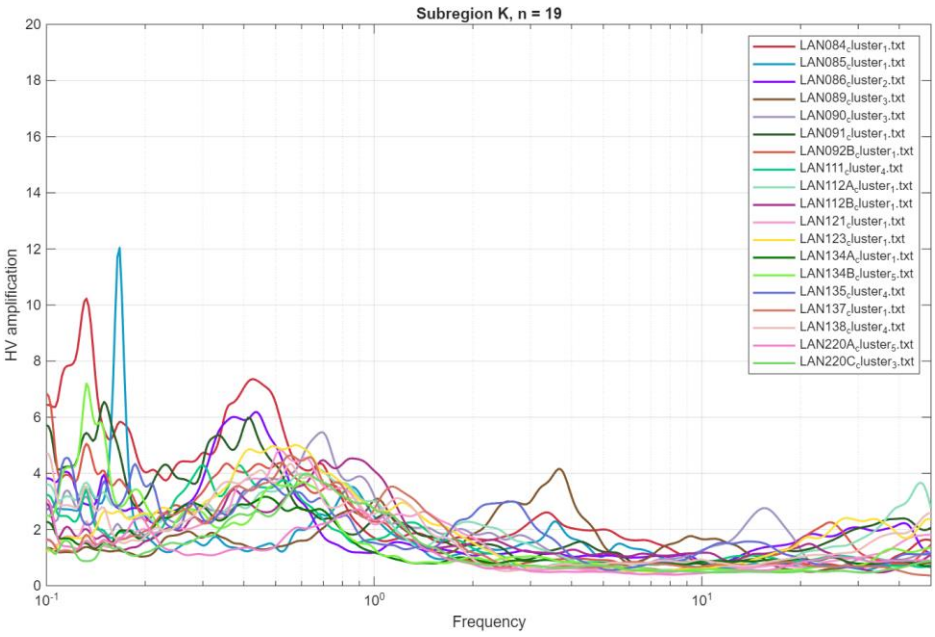
A3: MHVSR Curves for subregion I



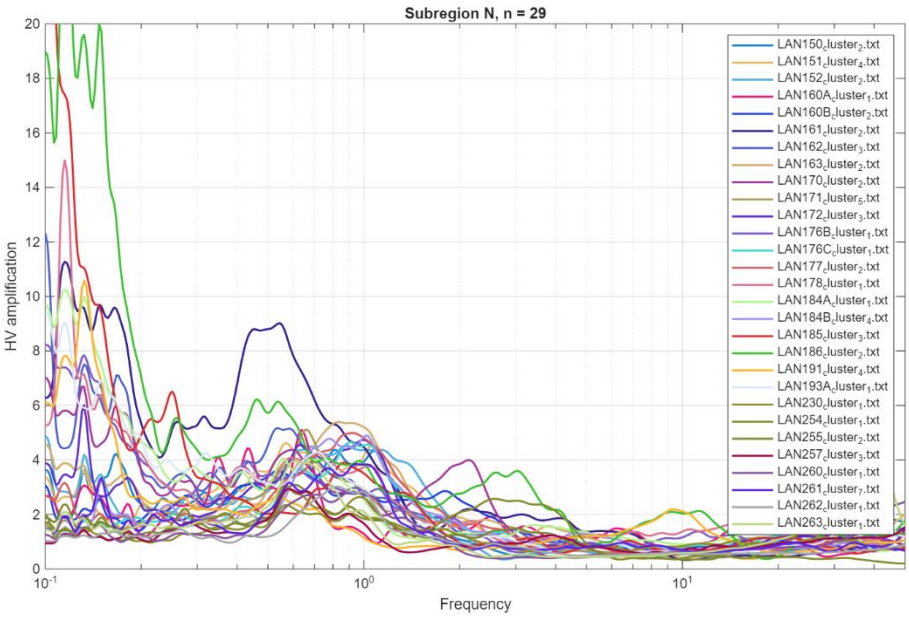
A4: MHVSR Curves for subregion J



A5: MHVSR Curves for subregion K

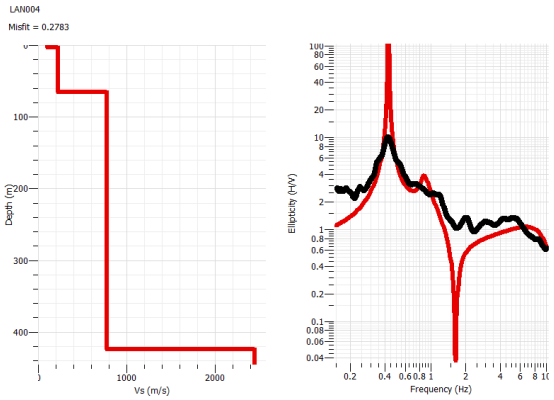


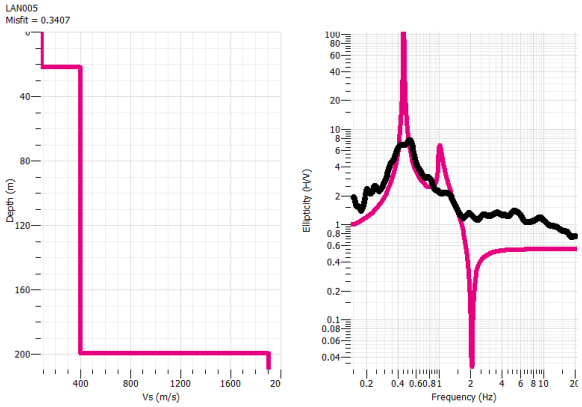
A6: MHVSR Curves for subregion N

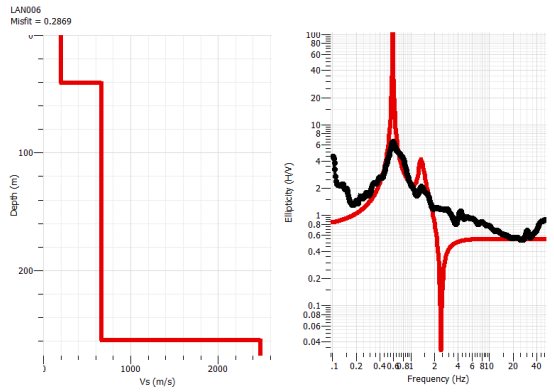


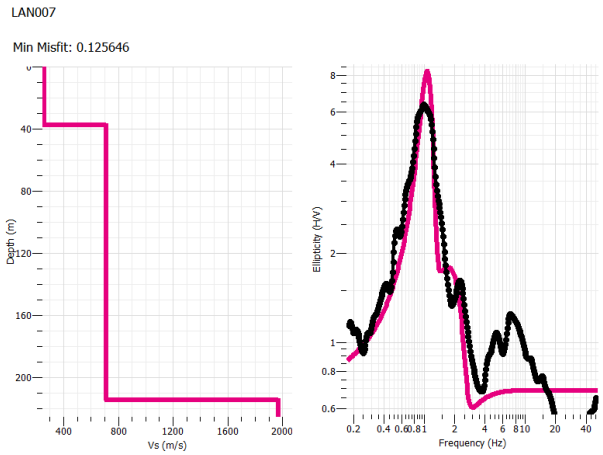
Appendix B: Initial Ground Profiles and Ellipticity Curves for each site

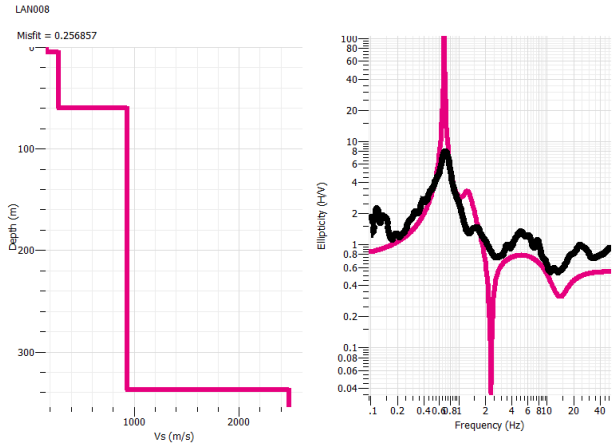
B1: F2 Region

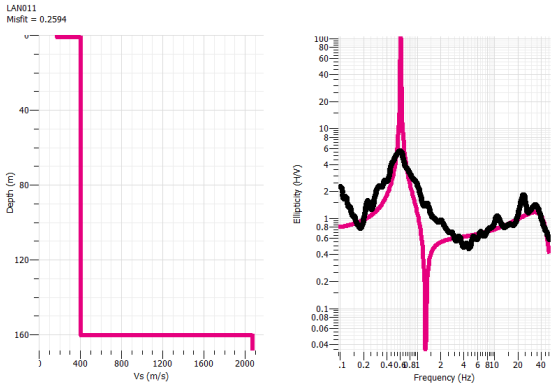

LAN004
Misfit = 0.2783
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)


LAN005
Misfit = 0.3407
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)


LAN006
Misfit = 0.2869
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

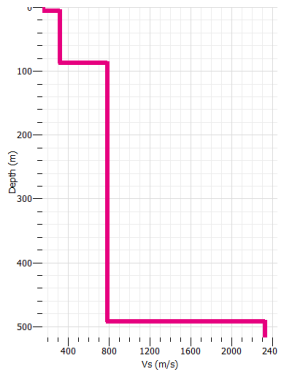

LAN007
Min Misfit: 0.125646
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

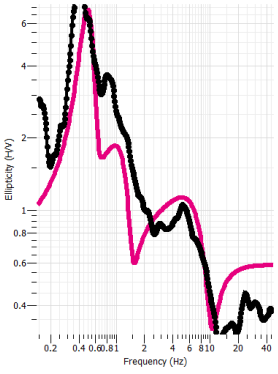

LAN008
Misfit = 0.256857
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

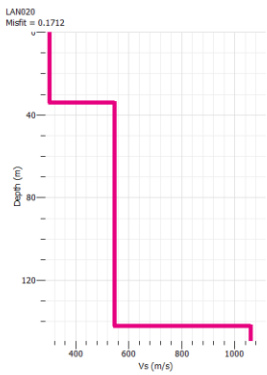

LAN011
Misfit = 0.2594
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

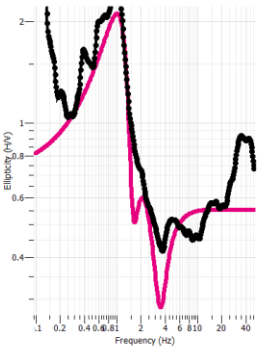
LAN016

Min Misfit: 0.187328



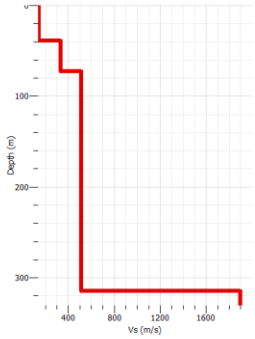


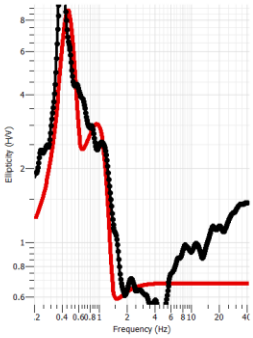




LAN022

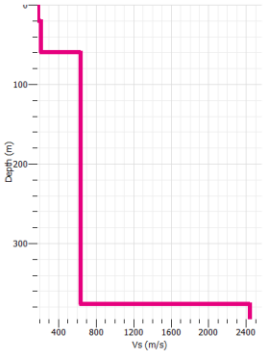
Min Misfit: 0.170722

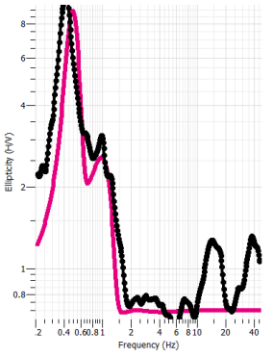




LAN023

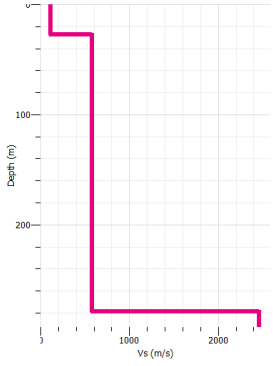
Min Misfit: 0.155723

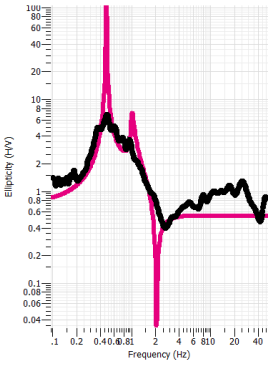




LAN024

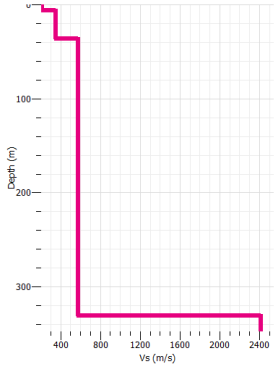
Misfit = 0.239

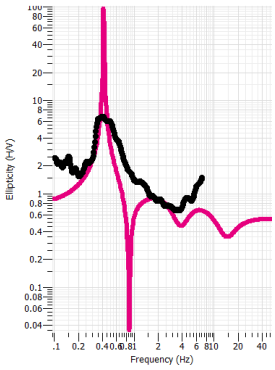


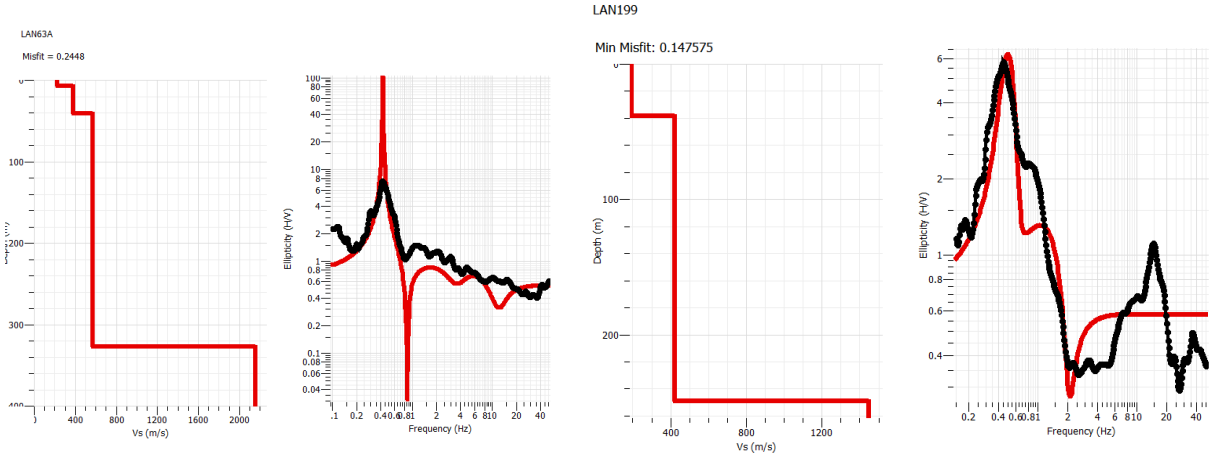


LAN036

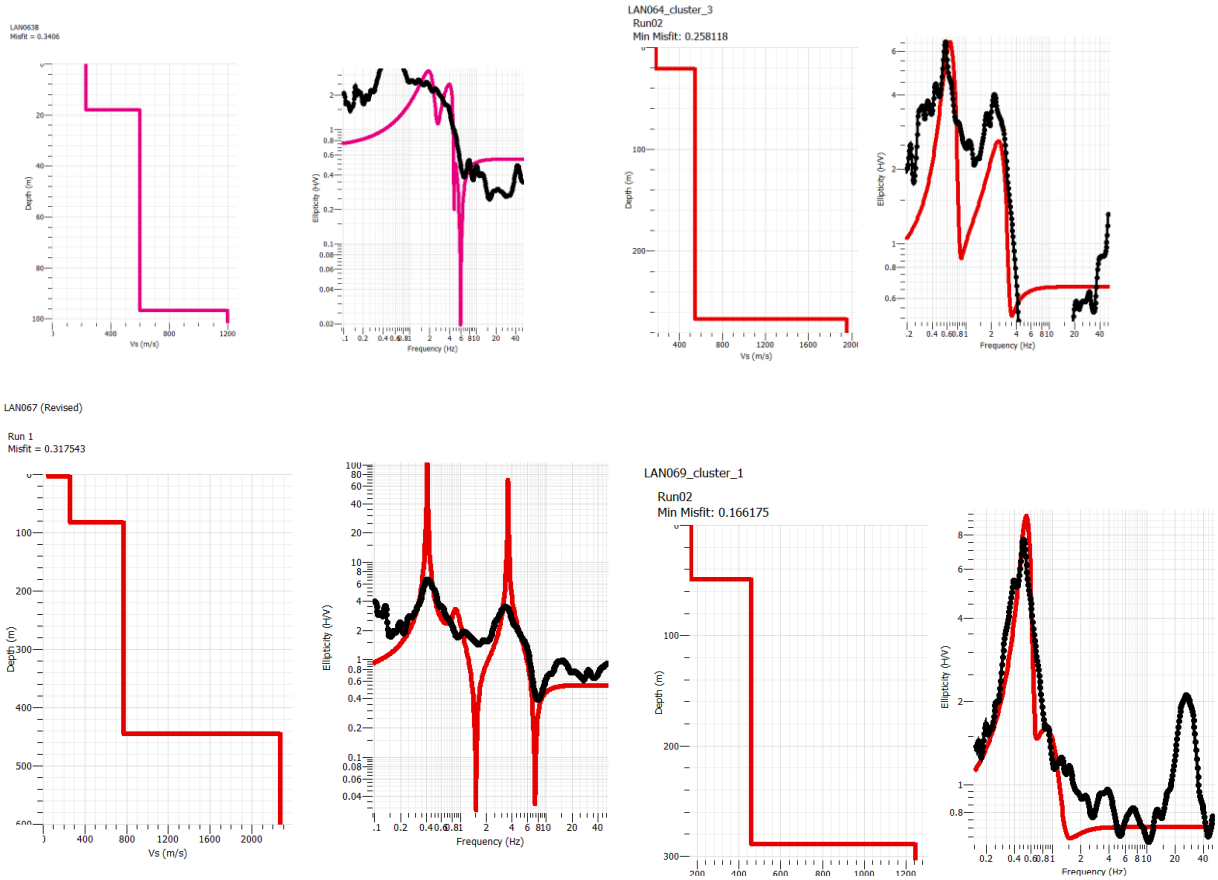
Misfit = 0.2409

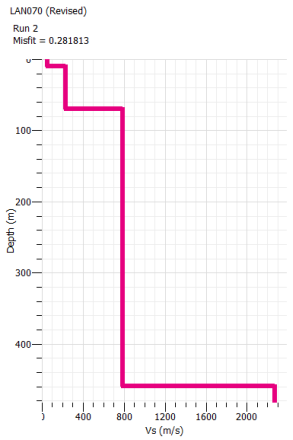


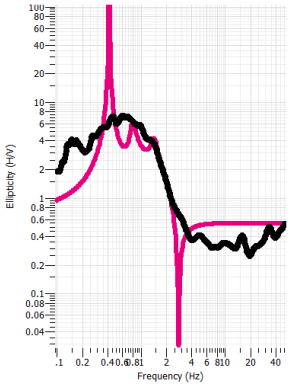


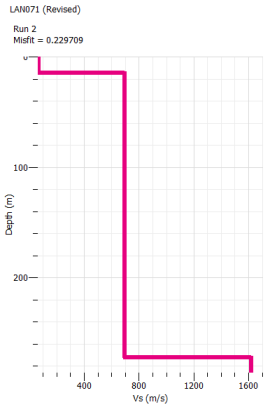

LAN63A
Misfit = 0.2448
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN199
Min Misfit: 0.147575
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

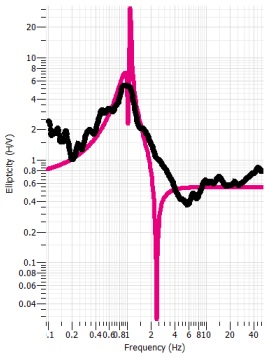
B2: H Region

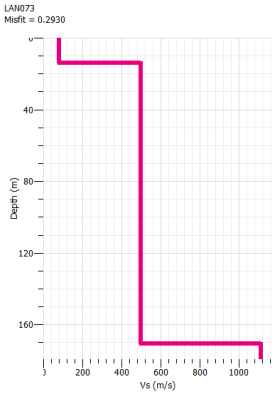

LAN063B
Misfit = 0.3406
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN064_cluster_3
Run02
Min Misfit: 0.258118
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN067 (Revised)
Run 1
Misfit = 0.317543
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN069_cluster_1
Run02
Min Misfit: 0.166175
Depth (m)
Ellipticity (H/V)
Frequency (Hz)

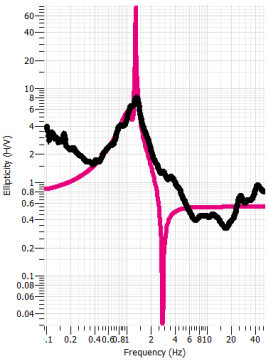

LAN070 (Revised)
Run 2
Misfit = 0.281813
Depth (m)
Vs (m/s)

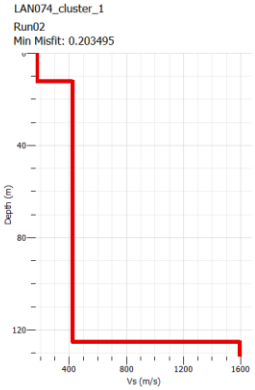

Ellipticity (H/V)
Frequency (Hz)

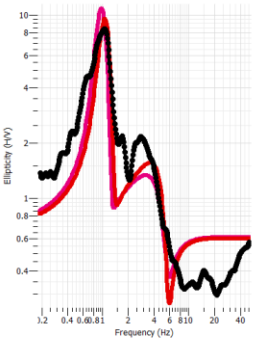

LAN071 (Revised)
Run 2
Misfit = 0.229709
Depth (m)
Vs (m/s)

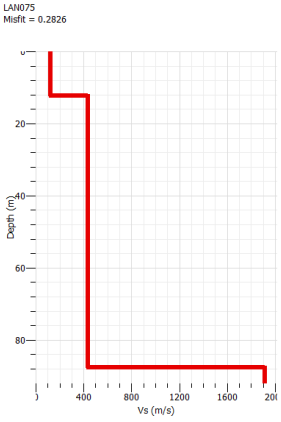

Ellipticity (H/V)
Frequency (Hz)

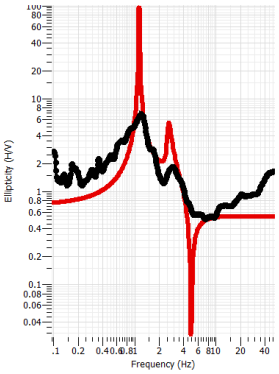

LAN073
Misfit = 0.2930
Depth (m)
Vs (m/s)

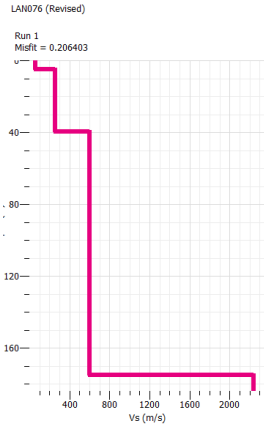

Ellipticity (H/V)
Frequency (Hz)

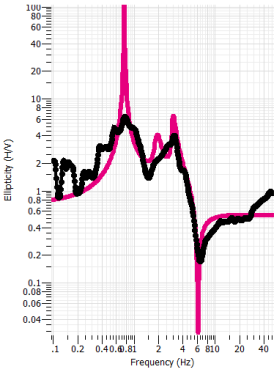

LAN074_cluster_1
Run02
Min Misfit: 0.203495
Depth (m)
Vs (m/s)

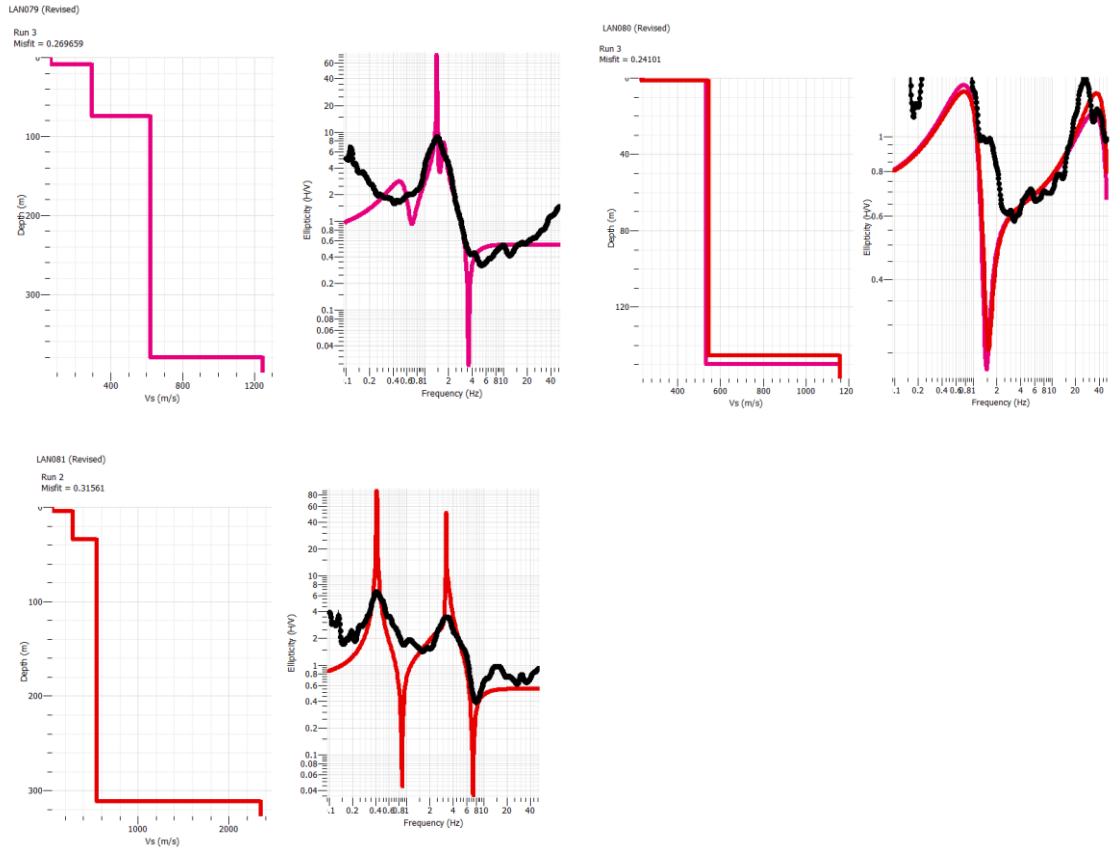

Ellipticity (H/V)
Frequency (Hz)


LAN075
Misfit = 0.2826
Depth (m)
Vs (m/s)

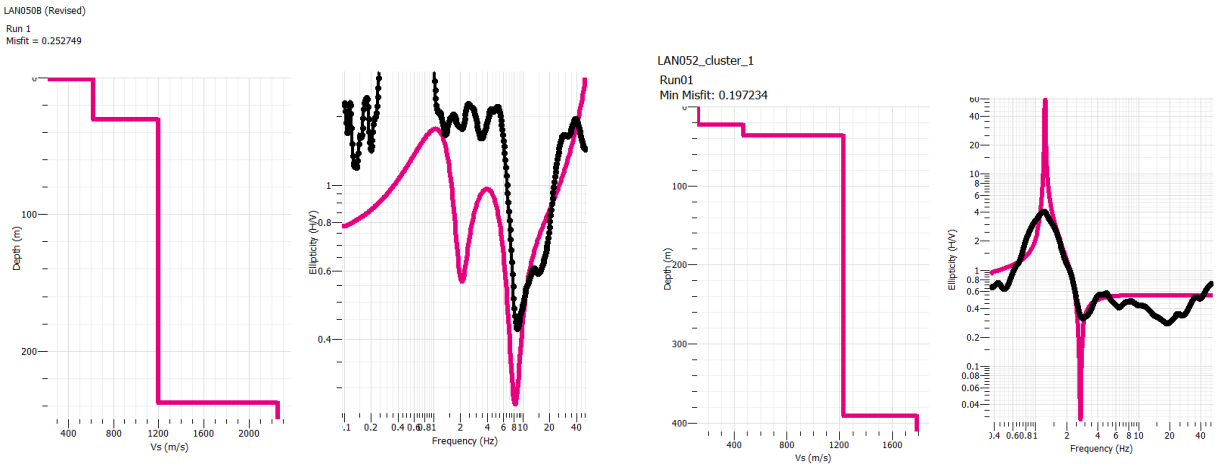

Ellipticity (H/V)
Frequency (Hz)

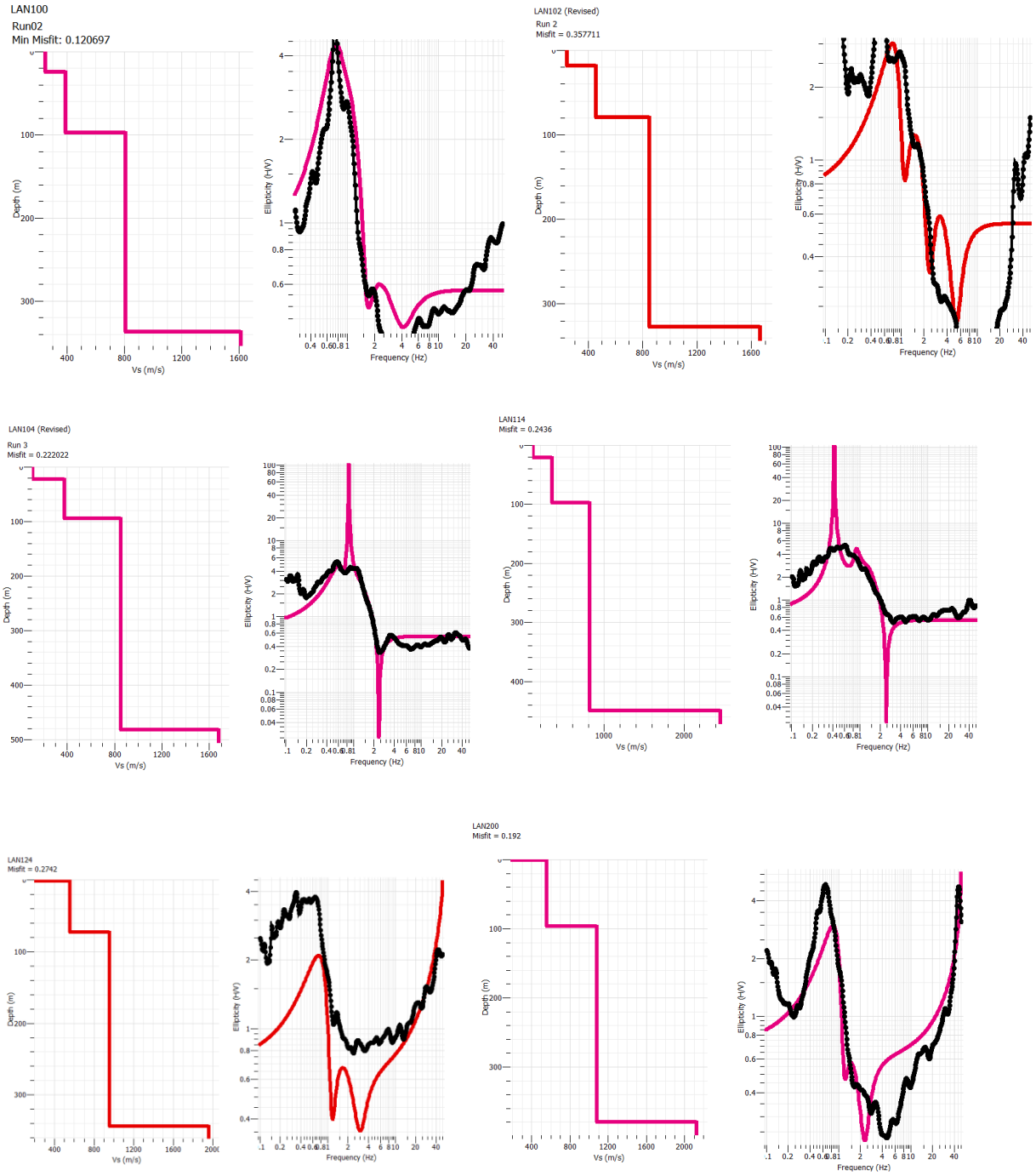

LAN076 (Revised)
Run 1
Misfit = 0.206403
Depth (m)
Vs (m/s)


Ellipticity (H/V)
Frequency (Hz)

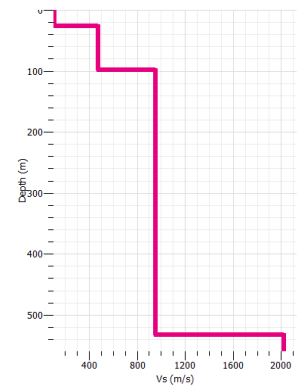

LAN079 (Revised)
Run 3
Misfit = 0.269659
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN080 (Revised)
Run 3
Misfit = 0.24101
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN081 (Revised)
Run 2
Misfit = 0.31561
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

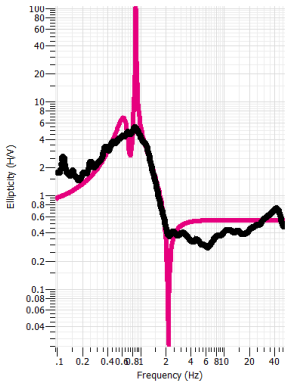
B3: I Region


LAN050B (Revised)
Run 1
Misfit = 0.252749
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN052_cluster_1
Run01
Min Misfit: 0.197234
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

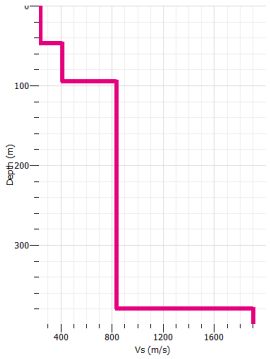

LAN100
Run02
Min Misfit: 0.120697
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN102 (Revised)
Run 2
Misfit = 0.357711
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN104 (Revised)
Run 3
Misfit = 0.222022
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN114
Misfit = 0.2436
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN124
Misfit = 0.2742
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN200
Misfit = 0.192
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

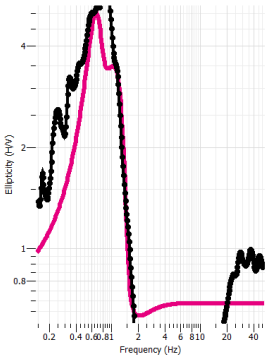
LAN201B
Misfit = 0.1953



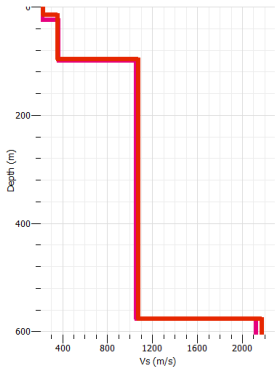


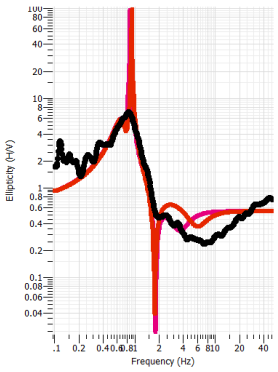
LAN202_cluster_1
Run01
Min Misfit: 0.182852



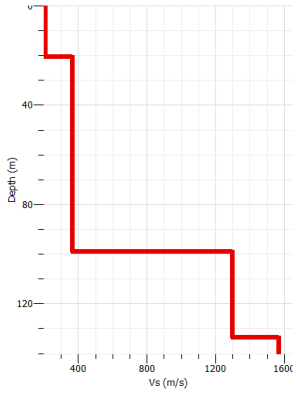


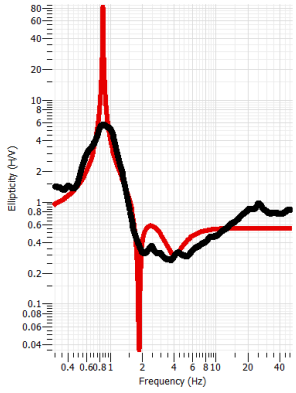
LAN203
Misfit = 0.2250



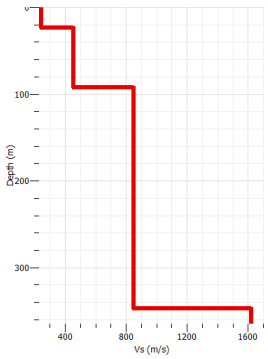


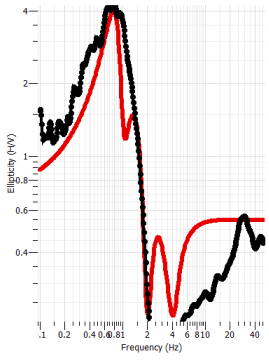
LAN204A_cluster_2
Run03
Min Misfit: 0.18879



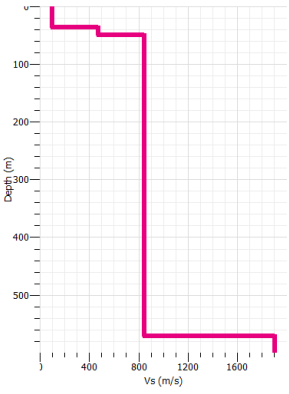


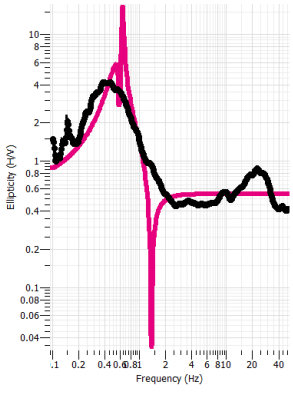
LAN205 (Revised)
Run 3
Misfit = 0.1963

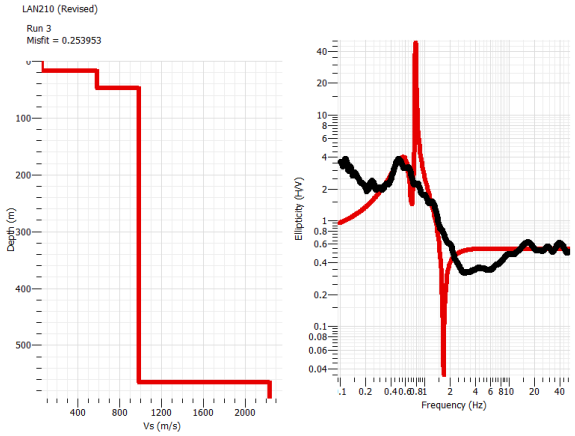




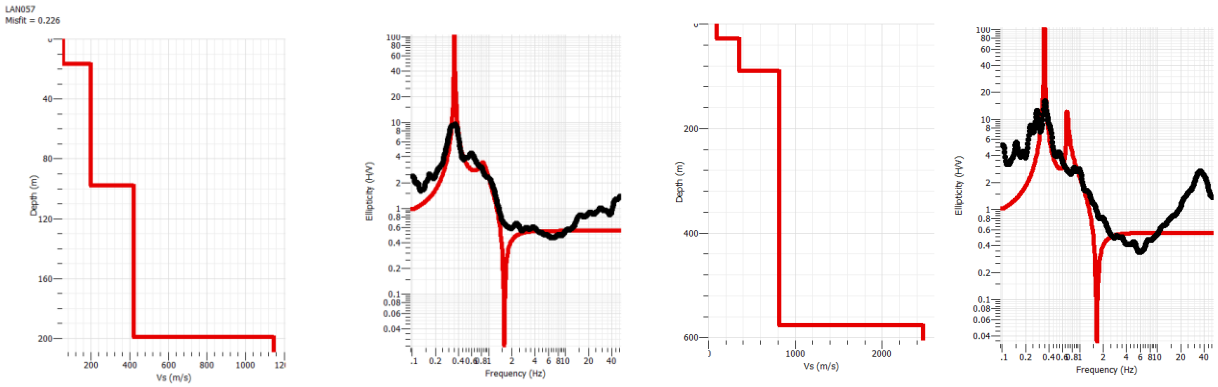
LAN208 (Revised)
Run 1
Misfit = 0.199949

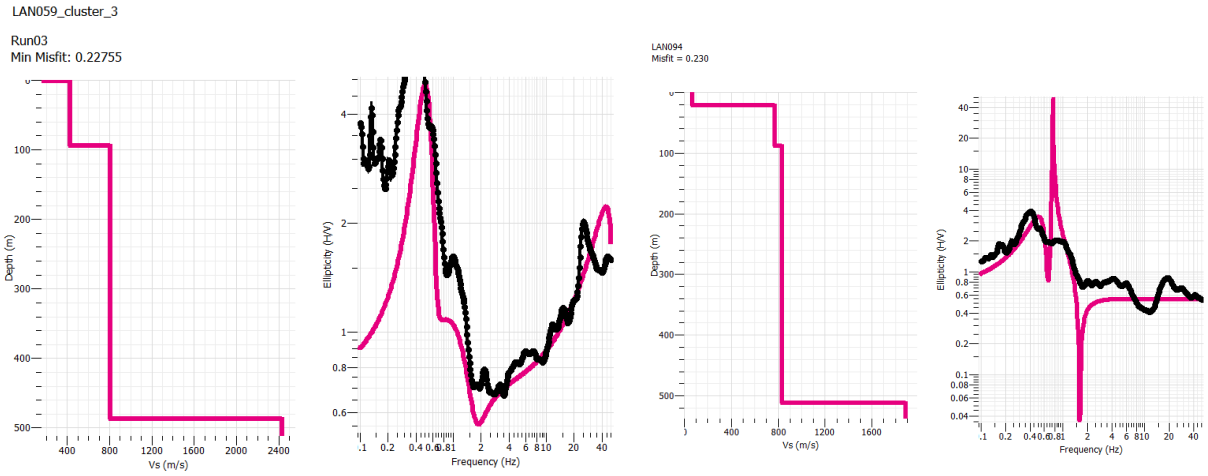





LAN210 (Revised)
Run 3
Misfit = 0.253953
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

B4: J Region

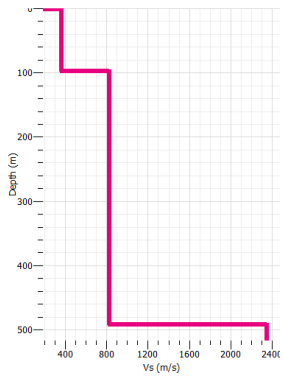

LAN057
Misfit = 0.226
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN058
Misfit 0.365
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

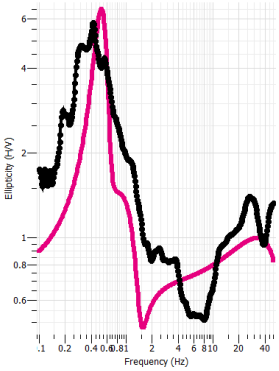

LAN059_cluster_3
Run03
Min Misfit: 0.22755
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN094
Misfit = 0.230
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

LAN095_cluster_3

Run03

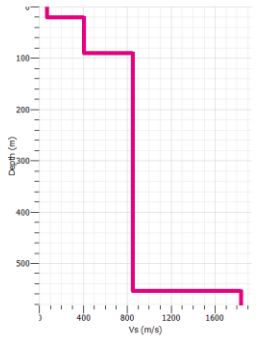
Min Misfit: 0.193422

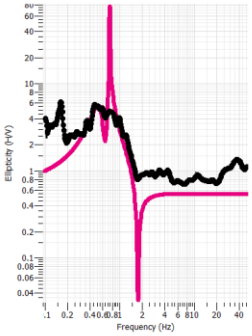




LAN096B

Misfit 0.307

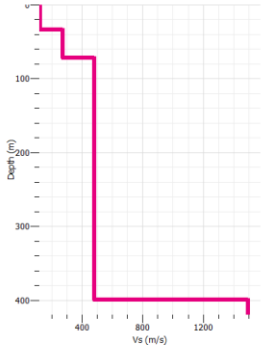


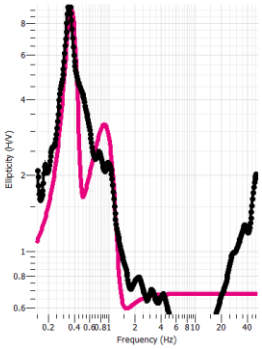


LAN097A_cluster_3

Run03

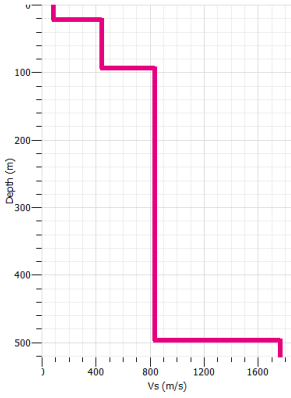
Min Misfit: 0.164304

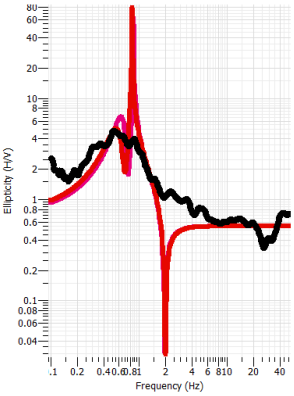




LAN103

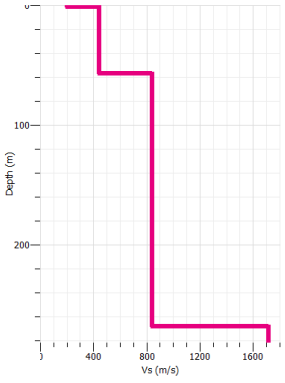
Misfit = 0.259

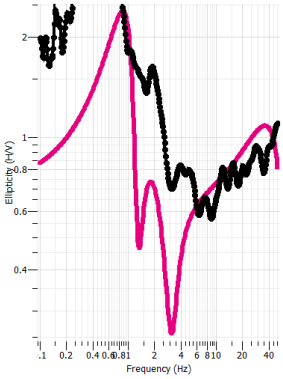




LAN107

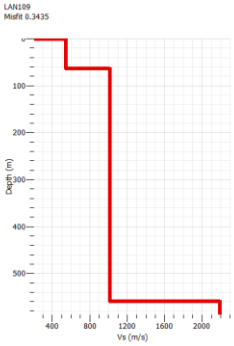
Misfit = 0.260

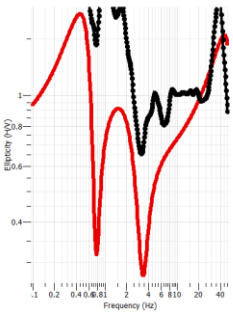


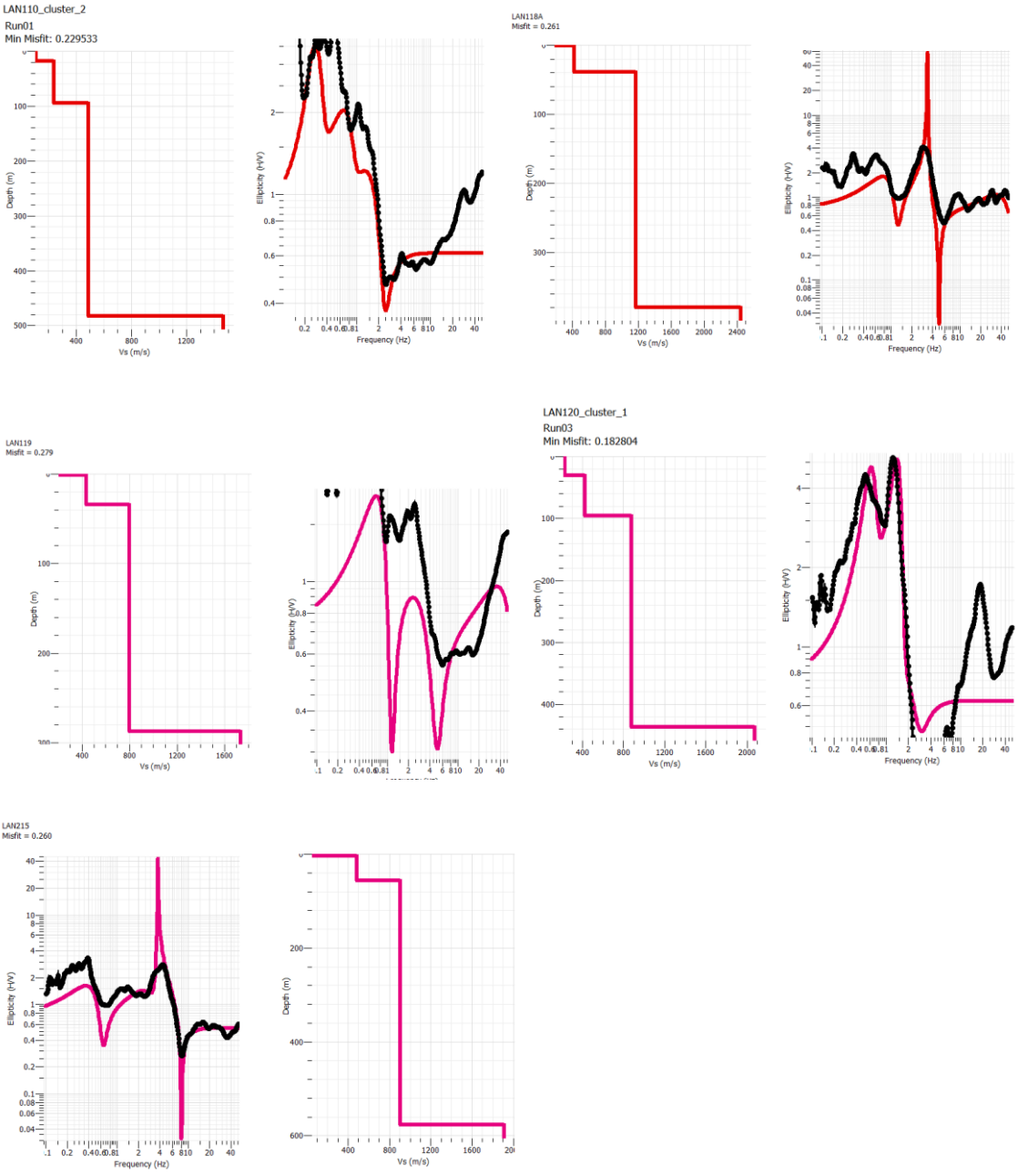


LAN109

Misfit 0.3435

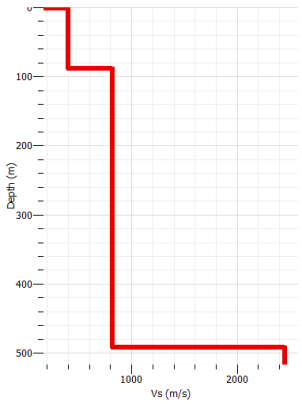


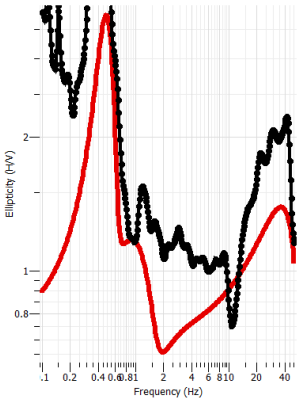



LAN110_cluster_2
Run01
Min Misfit: 0.229533
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN118A
Misfit = 0.261
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN119
Misfit = 0.279
Depth (m)
Vs (m/s)
Ellipticity (H/V)
LAN120_cluster_1
Run03
Min Misfit: 0.182804
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN215
Misfit = 0.260
Ellipticity (H/V)
Frequency (Hz)
Depth (m)
Vs (m/s)

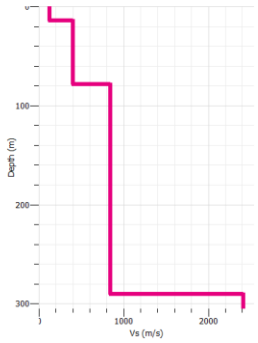
B5: K Region

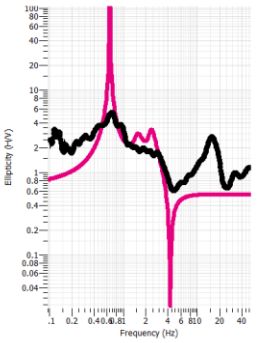
LAN086_cluster_2
Run02
Min Misfit: 0.24106



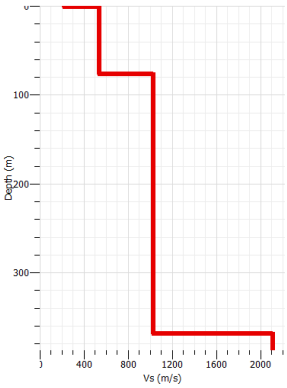


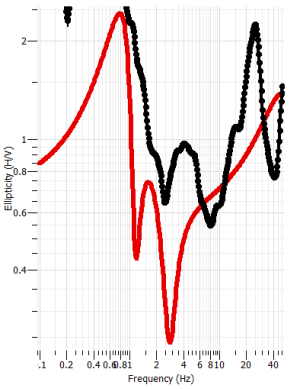
LAN090
Misfit = 0.347



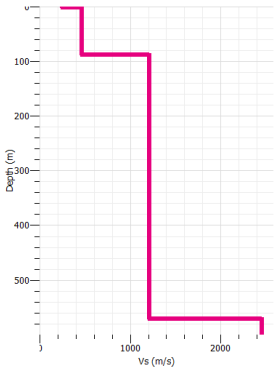


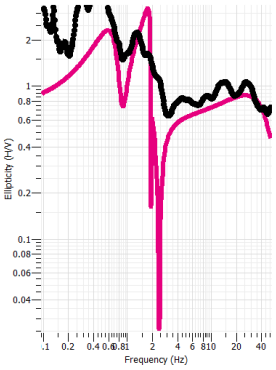
LAN092B
Misfit = 0.345



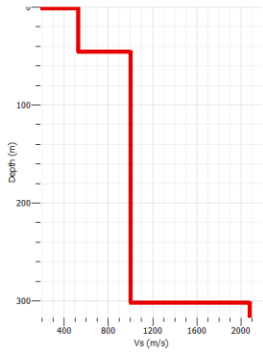


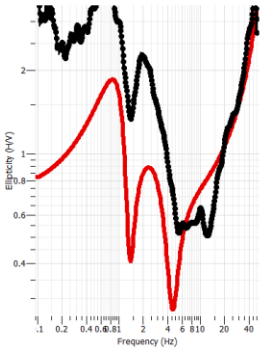
LAN111
Misfit = 0.2809



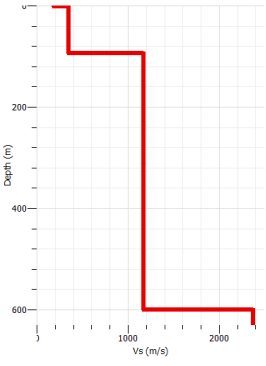


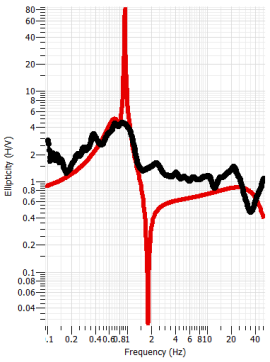
LAN112A
Misfit = 0.326

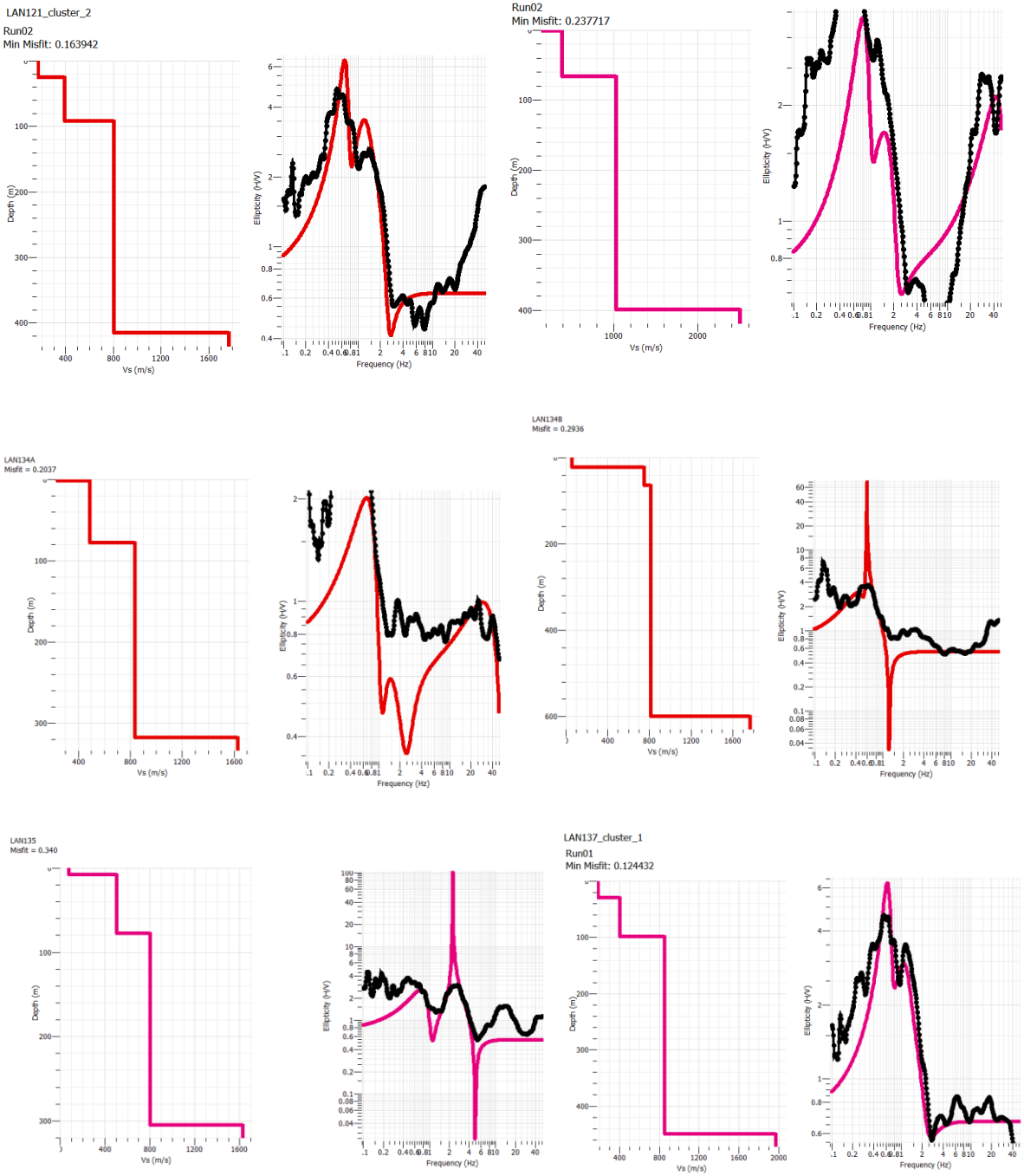


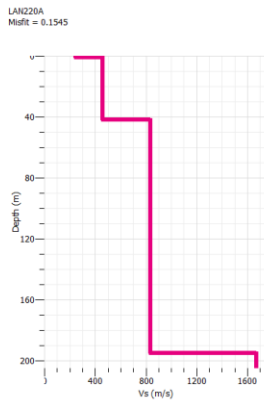


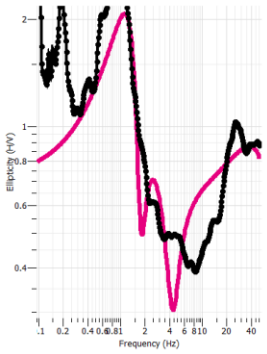
LAN112B
Misfit = 0.3059

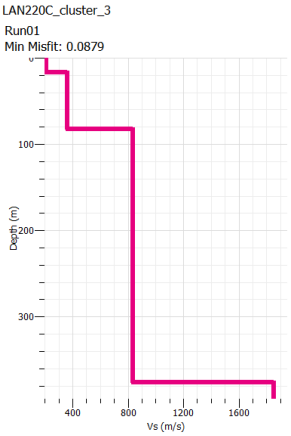


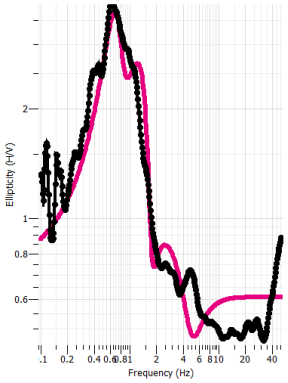



LAN121_cluster_2
Run02
Min Misfit: 0.163942
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN123_cluster_2
Run02
Min Misfit: 0.237717
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN134A
Misfit = 0.2037
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN134B
Misfit = 0.2936
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN135
Misfit = 0.340
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)
LAN137_cluster_1
Run01
Min Misfit: 0.124432
Depth (m)
Vs (m/s)
Ellipticity (H/V)
Frequency (Hz)

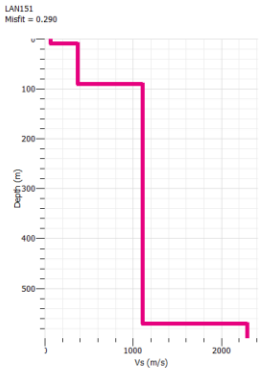

LAN220A
Misfit = 0.1545
Depth (m)
Vs (m/s)

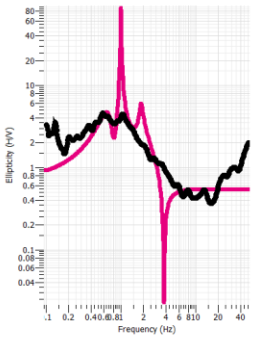

Ellipticity (H/V)
Frequency (Hz)

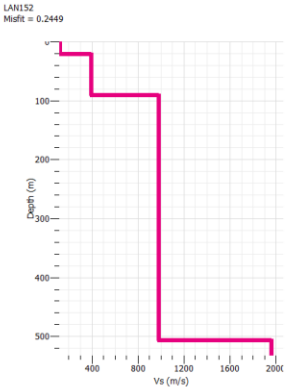

LAN220C_cluster_3
Run01
Min Misfit: 0.0879
Depth (m)
Vs (m/s)

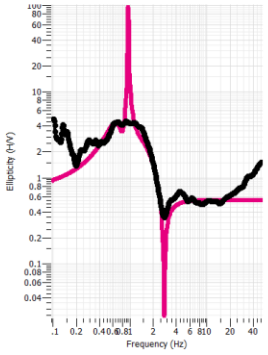

Ellipticity (H/V)
Frequency (Hz)

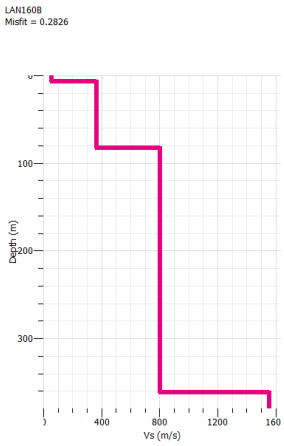
B6: N Region

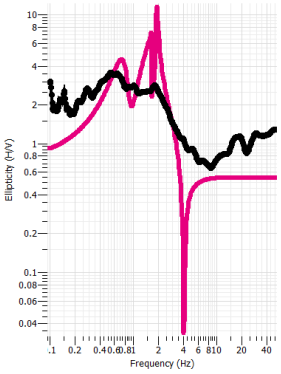

LAN151
Misfit = 0.290
Depth (m)
Vs (m/s)

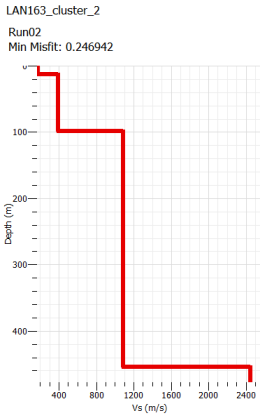

Ellipticity (H/V)
Frequency (Hz)

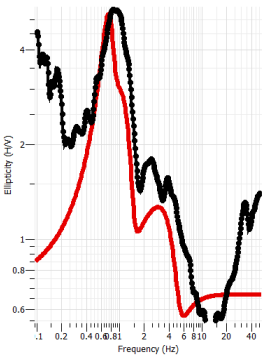

LAN152
Misfit = 0.2449
Depth (m)
Vs (m/s)


Ellipticity (H/V)
Frequency (Hz)

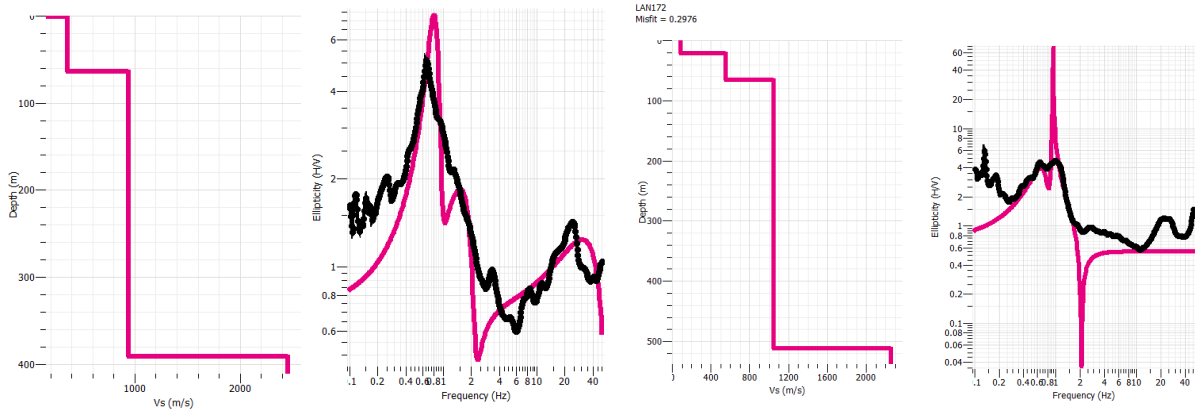

LAN160B
Misfit = 0.2826
Depth (m)
Vs (m/s)


Ellipticity (H/V)
Frequency (Hz)


LAN163_cluster_2
Run02
Min Misfit: 0.246942
Depth (m)
Vs (m/s)

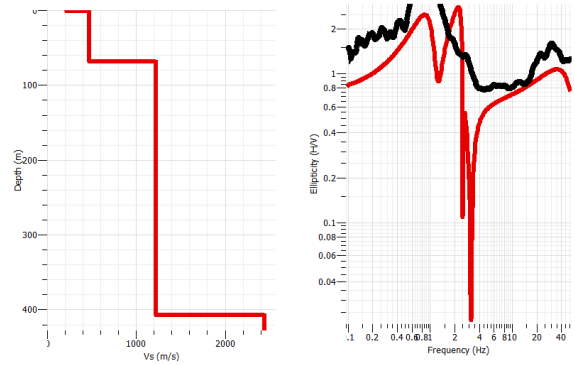

Ellipticity (H/V)
Frequency (Hz)

LAN171_cluster_5
Run03
Min Misfit: 0.143015

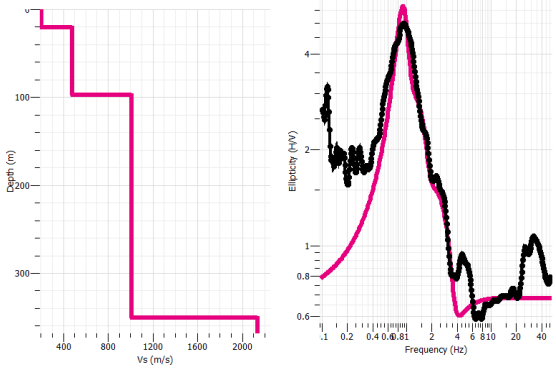


LAN172
Misfit = 0.2976

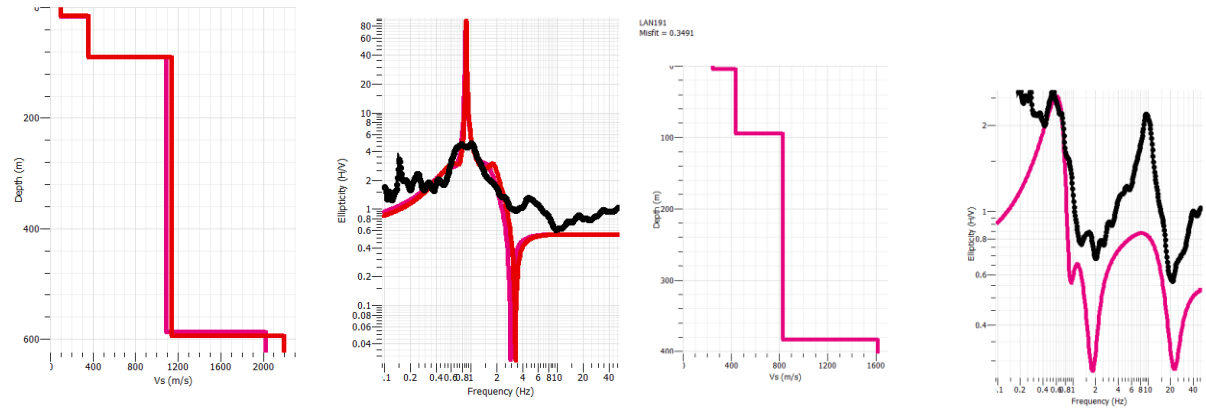
LAN176C
Misfit = 0.280



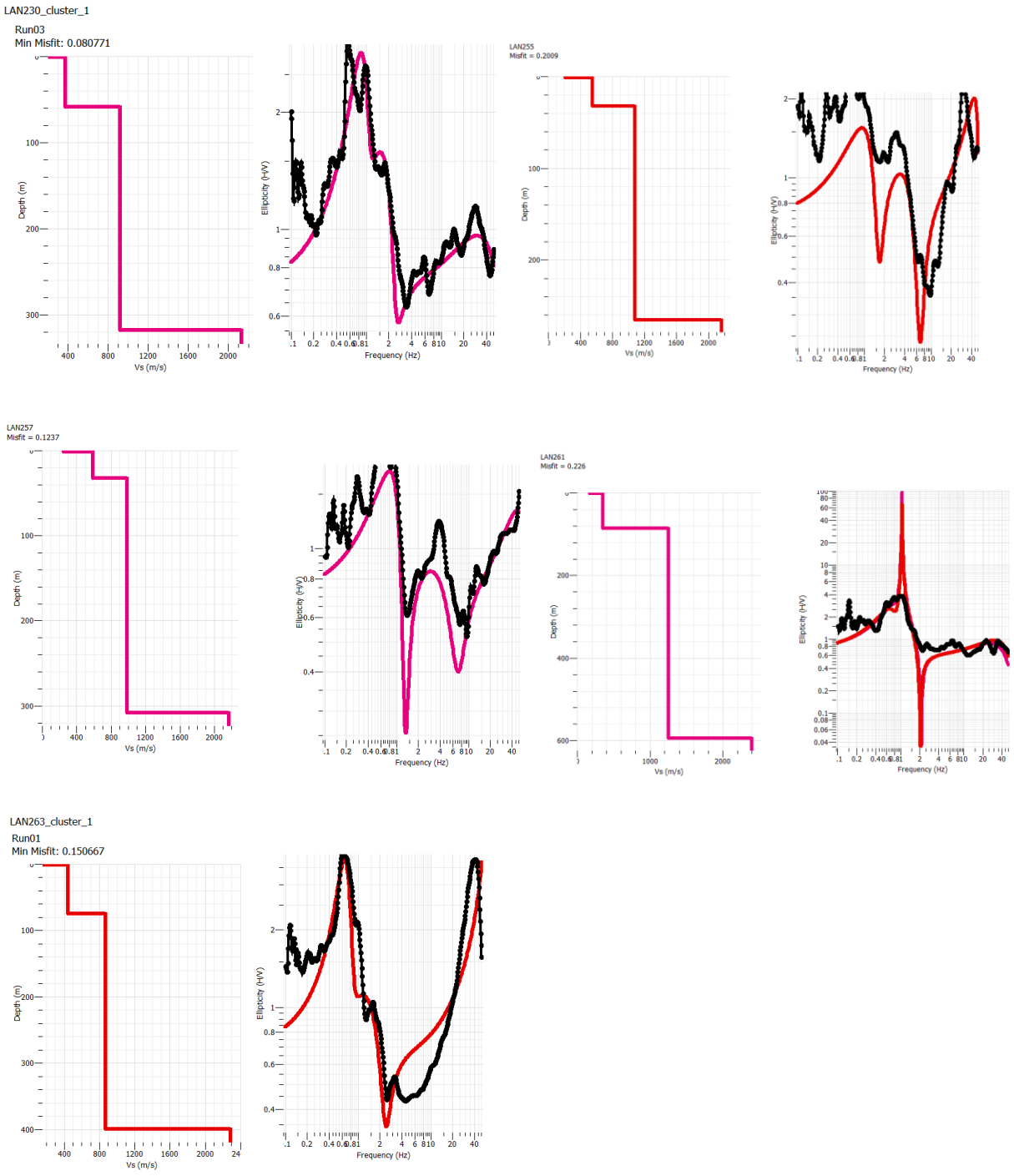
LAN177_cluster_5
Run02
Min Misfit: 0.16814



LAN184B
Misfit = 0.2810



LAN191
Misfit = 0.3491



Appendix C: Final inversion results Including Revised Results

FINAL Inversion Results

Geopsy *Figue* files included in this folder

Appendix D: Master spreadsheet

Also available at this link: Master_Spreadsheet.xlsx

Site Name	Successfully Inverted? (Y/N)	Final Parameters Used	Long	Lat	Layering (m):				Vs30
F2 Region									
LAN199	Y	Option 2	-122.583699	49.141635	h	ztop	zbot	vs	vs30
						33	0	33	121
						191	33	224	429
							224		1956
LAN023	Y	Option 1	-122.613753	49.184164	h	ztop	zbot	vs	vs30
						30	0	30	140
						33	20	53	200
						293	53	346	421
							346		1271
LAN022	Y	Option 1	-122.623492	49.181818	h	ztop	zbot	vs	vs30
						41	0	41	176
						17	41	58	252
						361	58	419	570
							419		2140
LAN016	Y	Option 1	-122.623398	49.188872	h	ztop	zbot	vs	vs30
						20	0	20	120
						68	20	88	250

						262	88	350	519		
							350		1625		
LAN007	Y		Option 2	-122.596925	49.201902	h	ztop	zbot	vs	vs30	
						37	0	37	258		258
						178	37	215	716		
							215		1962		
LAN004	Y		Option 1	-122.644205	49.200461	h	ztop	zbot	vs	vs30	
						3	0	3	100	192.8251121	
						61	3	64	215		
						359	64	423	775		
							423		2438		
LAN006	Y		Option 2	-122.624745	49.204452	h (m)	ztop (m)	zbot (m)	Vs (m/s)	vs30	
						40	0	40	203		203
						218	40	258	666		
							258		2485		
LAN005	Y		Option 2	-122.633371	49.203617	h (m)	ztop (m)	zbot (m)	Vs (m/s)	vs30	
						21	0	21	84	110.0917431	
						177	21	198	400		
							198		1907		
LAN008	Y		Option 1	-122.592784	49.196599	h (m)	ztop (m)	zbot (m)	Vs (m/s)	vs30	
						15	0	15	71	110.5919003	
						66	15	81	250		
						389	81	470	690		

												470	1225
LAN011	Y	Option 2	-122.6451	49.19008	h	ztop	zbot	vs	vs30				
						1.16	0	1.16	173	385.903287			
						157.84	1.16	159	406				
								159	2074				
LAN020	Y	Option 2	-122.646098	49.178437	h	ztop	zbot	vs	vs30				
						33	0	33	302	302			
						108	33	141	548				
								141	1062				
LAN024	Y	Option 2	-122.59763	49.18408	h	ztop	zbot	vs	vs30				
						27	0	27	112	121.8260379			
						251	27	278	579				
								278	2460				
LAN036	Y	Option 1	-122.573609	49.16677	h	ztop	zbot	vs	vs30				
						6	0	6	213	303.1360202			
						30	6	36	339				
						294	36	330	577				
								330	2411				
LAN063A	Y	Option 1	-122.539895	49.162388	h	ztop	zbot	vs	vs30				
						7	0	7	225	327.3771538			
						33	7	40	380				
						285	40	325	570				
								325	2156				

H Region

LAN064	Y	Option 2	-122.525204	49.155428	h	ztop	zbot	vs	vs30		
						21	0	21	187	232.5634518	
						245	21	266	539		
						266			1955		
LAN067	Y	Option 1	-122.514147	49.154443	h	ztop	zbot	vs	vs30		
						3.4	0	3.4	50	176.1517615	
						77.6	3.4	81	260		
						363	81	444	772		
						444			2266		
LAN069	Y	Option 2	-122.503516	49.155761	h	ztop	zbot	vs	vs30		
						48	0	48	178	178	
						239	48	287	463		
						287			1246		
LAN074	Y	Option 2	-122.474771	49.167089	h	ztop	zbot	vs	vs30		
						11	0	11	178	281.958354	
						114	11	125	426		
						125			1594		
LAN081	Y	Option 1	-122.471345	49.15063	h	ztop	zbot	vs	vs30		
						4	0	4	63	193.4967556	
						29	4	33	284		
						278	33	311	554		
						311			2406		
LAN063B	N	Option 2	-122.531077	49.162167							

LAN070	N	Option 2	-122.494949	49.162123							
LAN071	Y	Option 4	-122.492604	49.164599	h	ztop	zbot	vs	vs30		
						46	0	46	207		207
							46		977		
LAN076	Y	Option 5	-122.473984	49.161778	h	ztop	zbot	vs	vs30		
						9	0	9	114	237.8340366	
						63	9	72	445		
						178	72	250	931		
							250		2337		
LAN079	Y	Option 2	-122.492484	49.149202	h	ztop	zbot	vs	vs30		
						13	0	13	86	157.3170732	
						107	13	120	430		
							120		876		
LAN080	N		-122.462444	49.154345							

LAN073	Y	Option 2	-122.492489	49.158017	h	ztop	zbot	vs	vs30	
						14	0	14	80	144.7924253
						156	14	170	497	
							170		1108	
LAN075	Y	Option 2	-122.465739	49.167403	h	ztop	zbot	vs	vs30	
						12	0	12	122	214.9758065
						75	12	87	437	
							87		1908	
I Region										
LAN052	Y	Option 3	-122.632326	49.130439	h	ztop	zbot	vs	vs30	
						30	0	30	98	98
						21	30	51	477	
						491	51	542	998	
							542		1577	
LAN100	Y	Option 3	-122.647977	49.107589	h	ztop	zbot	vs	vs30	
						22	0	22	250	276.598755
						75	22	97	391	
						240	97	337	810	
							337		1620	
LAN104	Y	Option 3	-122.609273	49.107416	h	ztop	zbot	vs	vs30	
						17	0	17	92	136.5901552
						79	17	96	373	

						394	96	490	844		
							490		1703		
LAN202	N		-122.667241	49.106257							
LAN204A	Y	Option 3	-122.666176	49.104999	h	ztop	zbot	vs	vs30		
						20	0	20	209	243.2733119	
						78	20	98	362		
						35	98	133	1294		
							133		1569		
LAN050B	Y	Option 6	-122.672368	49.130218	h	ztop	zbot	vs	vs30		
						50	0	50	162		162
						548	50	598	973		
							598		2151		
LAN102	Y	Option 6	-122.624893	49.107549	h	ztop	zbot	vs	vs30		
						10	0	10	41	98.45985401	
						179	10	189	329		
							189		909		
LAN114	Y	Option 3	-122.652462	49.092235	h	ztop	zbot	vs	vs30		
						19	0	19	96	130.8071632	
						78	19	97	350		
						352	97	449	820		
							449		2454		

LAN124	N		-122.678608	49.074683							
LAN200	Y	Option 6	-122.65658	49.11276	h	ztop	zbot	vs	vs30		
						57	0	57	170		170
						524	57	581	1000		
							581		1931		
LAN201B	Y	Option 3	-122.6677248	49.10648274	h	ztop	zbot	vs	vs30		
						24	0	24	112	132.2222222	
						72	24	96	476		
						436	96	532	952		
							532		2024		
LAN203	Y	Option	-122.6662483	49.10606883	h	ztop	zbot	vs	vs30		
						14	0	14	213	270.7627119	
						82	14	96	355		
						480	96	576	1074		
							576		2177		
LAN205	Y	Option 3	-122.6590133	49.10460059	h	ztop	zbot	vs	vs30		
						22	0	22	193	219.3276495	
						69	22	91	351		
						485	91	576	1103		
							576		2178		

LAN208	N			-122.6527207	49.09736934						
LAN210	N			-122.6548701	49.09553278						
J Region											
LAN059	Y		Option 10	-122.564452	49.134055	h	ztop	zbot	vs	vs30	
							35	0	35	94	94
							285	35	320	474	
								320		1442	
LAN095	N		Option 3	-122.567576	49.122383						
LAN097A	Y		Option 3	-122.582433	49.126264	h	ztop	zbot	vs	vs30	
							33	0	33	124	124
							39	33	72	264	
							328	72	400	490	

								400		1495	
LAN110	Y		Option 6	-122.53933	49.10698	h	ztop	zbot	vs	vs30	
							60	0	60	256	256
							518	60	578	1116	
								578		2462	
LAN120	Y		Option 3	-122.541467	49.090075	h	ztop	zbot	vs	vs30	
							29	0	29	222	225.5249514
							66	29	95	418	
							340	95	435	876	
								435		2073	
LAN057	Y		Option 3	-122.603611	49.129412	h	ztop	zbot	vs	vs30	
							15	0	15	52	82.45418327
							83	15	98	199	
							100	98	198	427	
								198		1145	
LAN094	Y		Option 3	-122.56771	49.126043	h	ztop	zbot	vs	vs30	
							20	0	20	61	87.99182904
							69	20	89	765	
							422	89	511	833	
								511		1883	
LAN103	Y		Option 3	-122.592878	49.116112	h	ztop	zbot	vs	vs30	
							22	0	22	86	109.4575746
							70	22	92	438	
							404	92	496	836	
								496		1765	
LAN107	Y		Option 6	-122.566488	49.10461	h	ztop	zbot	vs	vs30	

| | | |

LAN096B	Y		Option 3	-122.587241	49.126362	h	ztop	zbot	vs	vs30
							19	0	19	78 110.8850174
							70	19	89	408
							463	89	552	853
								552		1827
LAN058	Y		Option 3	-122.582975	49.131123	h	ztop	zbot	vs	vs30
							25	0	25	97 110.2869518
							65	25	90	350
							485	90	575	816
								575		2466
K Region										
LAN086	Y		Option 6	-122.486615	49.132748	h	ztop	zbot	vs	vs30
							76	0	76	147 147
							524	76	600	843
								600		1878
LAN121	Y		Option 3	-122.516467	49.089282	h	ztop	zbot	vs	vs30
							24	0	24	169 190.6966033
							67	24	91	392
							325	91	416	800
								416		1765
LAN123	Y		Option 3	-122.462204	49.091969	h	ztop	zbot	vs	vs30

						25	0	25	182	199.3540968	
						70	25	95	381		
						353	95	448	823		
							448		2325		
LAN137	Y		Option 3	-122.504006	49.074696	h	ztop	zbot	vs	vs30	
						29	0	29	188	191.4335752	
						69	29	98	407		
							98	449	855		
							449		1964		
LAN220C	Y		Option 3	-122.4714	49.0721	h	ztop	zbot	vs	vs30	
						28	0	28	231	237.2521957	
						67	28	95	382		
						311	95	406	870		
							406		2040		
LAN090	Y		Option 3	-122.487061	49.114746	h	ztop	zbot	vs	vs30	
						14	0	14	126	198.7912088	
						64	14	78	402		
						220	78	298	836		
							298		2405		
LAN111	N			-122.525781	49.111295						

LAN112B	Y	Option 3	-122.486542	49.107555	h	ztop	zbot	vs	vs30		
						1.4	0	1.4	162	330.3023201	
						90.6	1.4	92	348		
						506	92	598	1171		
							598		2373		
LAN134A	N		-122.525858	49.072488							
LAN135	Y	Option 3	-122.518842	49.074633	h	ztop	zbot	vs	vs30		
						7	0	7	74	213.3794694	
						69	7	76	500		
						229	76	305	804		
							305		1629		
LAN220A	Y	Option 6	-122.470718	49.07179532	h	ztop	zbot	vs	vs30		
						48	0	48	197		197
						516	48	564	1206		
							564		2230		
LAN134B	N		-122.52589	49.074013							

LAN112A	Y		Option 6	-122.504109	49.108287	h		ztop	zbot	vs	vs30
								53	0	53	145
								505	53	558	839
									558		1783
LAN092B	N			-122.524149	49.125907						
N Region											
LAN163	Y		Option 3	-122.47173	49.046787	h		ztop	zbot	vs	vs30
								12	0	12	167
								85	12	97	383
								356	97	453	1095
									453		2440
LAN171	Y		Option 3	-122.481255	49.034907	h		ztop	zbot	vs	vs30
								23	0	23	227
								72	23	95	466
								261	95	356	874
									356		1831
LAN177	Y		Option 3	-122.490892	49.025828	h		ztop	zbot	vs	vs30

252.4112076

257.8581295

						19	0	19	198	252.2617253	
						77	19	96	479		
						302	96	398	1010		
							398		2130		
LAN230	Y	Option 3	-122.5202	49.0235	h	ztop	zbot	vs	vs30		
						1.2	0	1.2	197	342.161976	
						46.8	1.2	48	353		
						229	48	277	830		
							277		2344		
LAN263	Y	Option 3	-122.4686	49.0171	h	ztop	zbot	vs	vs30		
						1	0	1	138	383.0204082	
						46	1	47	408		
						281	47	328	802		
							328		2235		
LAN151	N		-122.4817	49.066387							
LAN152	Y	Option 3	-122.46605	49.059589	h	ztop	zbot	vs	vs30		
						20	0	20	115	150.6765677	
						70	20	90	397		
						416	90	506	981		
							506		1965		

LAN160B	N			-122.50806	49.049048						
LAN172	Y	Option 6		-122.503592	49.0297	h	ztop	zbot	vs	vs30	
							51	0	51	203	203
							425	51	476	978	
								476		2112	
LAN176C	Y	Option 6		-122.51577	49.02317	h	ztop	zbot	vs	vs30	
							49	0	49	249	249
							434	49	483	1129	
								483		2434	
LAN184B	Y	Option 6		-122.50826	49.016294	h	ztop	zbot	vs	vs30	
							50	0	50	254	254
							412	50	462	1070	
								462		2264	
LAN191	N			-122.534029	49.008988						
LAN255	Y	Option 6		-122.5298848	49.01603936	h	ztop	zbot	vs	vs30	

						48	0	48	195		195
						537	48	585	1146		
							585		2410		
LAN257	Y		Option 6	122.5245087	-49.01583436	h	ztop	zbot	vs	vs30	
						53	0	53	160		160
						545	53	598	1116		
							598		1799		
LAN261	Y		Option 6	122.4835587	-49.01692097	h	ztop	zbot	vs	vs30	
						49	0	49	215		215
						471	49	520	1121		
							520		2287		

Appendix E: Matlab code for producing Vs_{30} Map

```
%VS30 Map - Asha Punia

clear all;
clc;

% Database [
data.LAN199   = [49.141635, -122.583699, 180.0];
data.LAN023   = [49.184164, -122.613753, 194.0244];
data.LAN022   = [49.181818, -122.623492, 144.0];
data.LAN016   = [49.188872, -122.623398, 144.0];
data.LAN007   = [49.201902, -122.596925, 258.0];
data.LAN004   = [49.200461, -122.644205, 192.8251];
data.LAN006   = [49.204452, -122.624745, 203.0];
data.LAN005   = [49.203617, -122.633371, 110.0917];
data.LAN008   = [49.196599, -122.592784, 256.5121];
data.LAN011   = [49.190080, -122.645100, 385.9033];
data.LAN020   = [49.178437, -122.646098, 302.0];
data.LAN024   = [49.184080, -122.597630, 120.9363];
data.LAN036   = [49.166770, -122.573609, 303.136];
data.LAN063A  = [49.162388, -122.539895, 327.3772];
data.LAN064   = [49.155428, -122.525204, 232.5635];
```

```
data.LAN067   = [49.154443, -122.514147, 176.1518];
data.LAN069   = [49.155761, -122.503516, 178.0];
data.LAN074   = [49.167089, -122.474771, 281.9584];
data.LAN081   = [49.150630, -122.471345, 186.9077];
data.LAN063B  = [49.162167, -122.531077, 307.5957];
data.LAN070   = [49.162123, -122.494949, 101.4028];
data.LAN076   = [49.161778, -122.473984, 178.0488];
data.LAN079   = [49.149202, -122.492484, 157.3171];
data.LAN073   = [49.158017, -122.492489, 144.7924];
data.LAN075   = [49.167403, -122.465739, 214.9758];
data.LAN052   = [49.130439, -122.632326, 165.493];
data.LAN100   = [49.107589, -122.647977, 276.5988];
data.LAN104   = [49.107416, -122.609273, 276.5988];
data.LAN204A  = [49.104999, -122.666176, 243.2733];
data.LAN050B  = [49.130218, -122.672368, 162.0];
data.LAN102   = [49.107549, -122.624893, 98.45985];
data.LAN114   = [49.092235, -122.652462, 134.5794];
data.LAN200   = [49.112760, -122.656580, 170.0];
data.LAN201B  = [49.106482, -122.667725, 132.2222];
data.LAN203   = [49.106068, -122.666248, 270.7627];
data.LAN205   = [49.104600, -122.659013, 219.3276];
data.LAN059   = [49.134055, -122.564452, 94.0];
data.LAN095   = [49.122383, -122.567576, 352.1506];
data.LAN097A  = [49.126264, -122.582433, 182.3044];
data.LAN110   = [49.106980, -122.539330, 110.31];
data.LAN120   = [49.090075, -122.541467, 225.525];
data.LAN057   = [49.129412, -122.603611, 82.45418];
data.LAN094   = [49.126043, -122.567710, 87.99183];
data.LAN103   = [49.116112, -122.592878, 109.4576];
data.LAN107   = [49.104610, -122.566488, 155.0];
data.LAN215   = [49.071426, -122.603382, 258.3244];
data.LAN119   = [49.094220, -122.557376, 69.34843];
data.LAN118A  = [49.093043, -122.586641, 406.2595];
data.LAN109   = [49.113179, -122.548322, 406.2595];
data.LAN096B  = [49.126362, -122.587241, 110.885];
data.LAN058   = [49.131123, -122.582975, 110.287];
data.LAN086   = [49.132748, -122.486615, 147.0];
data.LAN121   = [49.089282, -122.516467, 190.6966];
data.LAN123   = [49.091969, -122.462204, 199.3541];
data.LAN137   = [49.074696, -122.504006, 191.4336];
data.LAN220C  = [49.072100, -122.471400, 220.4639];
data.LAN090   = [49.114746, -122.487061, 198.7912];
data.LAN112B  = [49.107555, -122.486542, 330.3023];
data.LAN135   = [49.074633, -122.518842, 213.3795];
data.LAN200A  = [49.071795, -122.470718, 197.0];
data.LAN112A  = [49.108287, -122.504109, 145.0];
data.LAN163   = [49.046787, -122.471730, 252.4112];
data.LAN171   = [49.034907, -122.481255, 257.8581];
data.LAN177   = [49.025828, -122.490892, 258.394];
data.LAN230   = [49.023500, -122.520200, 342.162];
data.LAN263   = [49.017100, -122.468600, 383.0204];
data.LAN152   = [49.059589, -122.466050, 150.6766];
data.LAN172   = [49.029700, -122.503592, 203.0];
data.LAN176C  = [49.023170, -122.515770, 249.0];
data.LAN184B  = [49.016294, -122.508260, 254.0];
```

```
data.LAN255   = [49.016039, -122.529885, 195.0];
data.LAN257   = [49.015834, -122.524509, 160.0];
data.LAN261   = [49.01692097, -122.483559, 215.0];

% Setting up the geometries of the map
siteNames = fieldnames(data);
numSites = length(siteNames);
lats = zeros(numSites, 1);
lons = zeros(numSites, 1);
vs30 = zeros(numSites, 1);
for i = 1:numSites
    lats(i) = data.(siteNames{i})(1);
    lons(i) = data.(siteNames{i})(2);
    vs30(i) = data.(siteNames{i})(3);
end

% Generating the Map
figure('Color', 'w', 'Name', 'Vs30 Heatmap');

% Setting up the margins
margin = 0.025;
minLat = min(lats) - margin;
maxLat = max(lats) + margin;
minLon = min(lons) - margin;
maxLon = max(lons) + margin;

[latGrid, lonGrid] = meshgrid(linspace(minLat, maxLat, 250), linspace(minLon, maxLon,
250));

% Setting up the Vs30 grid
Vs30_grid = griddata(lats, lons, vs30, latGrid, lonGrid, 'v4');

% Creating polygons for the boundaries of map
k = convhull(lats, lons);
hullPoly = polyshape(lats(k), lons(k));

bufferDistance = 0.012;
bufferedPoly = polybuffer(hullPoly, bufferDistance);

[polyLats, polyLons] = boundary(bufferedPoly);

inBoundary = inpolygon(latGrid, lonGrid, polyLats, polyLons);

Vs30_grid(~inBoundary) = NaN;

% Clipping
Vs30_grid(Vs30_grid < min(vs30)) = min(vs30);
Vs30_grid(Vs30_grid > max(vs30)) = max(vs30);

validIdx = ~isnan(Vs30_grid(:));

% Plot background interpolations
geoscatter(latGrid(validIdx), lonGrid(validIdx), 18, Vs30_grid(validIdx), 'filled',
'MarkerFaceAlpha', 0.22, 'HandleVisibility', 'off');
hold on;
```

```
% Plotting sites
geoscatter(lats, lons, 40, 'red', 'filled', 'MarkerEdgeColor', 'k');

% Applying map limits & map style
geolimits([minLat, maxLat], [minLon, maxLon]);
geobasemap streets;

% Selecting the gradient colours
numColors = 256;
darkBlue  = [0.02, 0.08, 0.35]; % Deep Blue (High Vs30)
midBlue   = [0.15, 0.45, 0.85]; % Vibrant Mid Blue
lightCyan = [0.85, 0.98, 1.00]; % Bright Cyan/White (Low Vs30)
half = numColors / 2;

customMap = [
    linspace(darkBlue(1), midBlue(1), half)', linspace(darkBlue(2), midBlue(2),
half)', linspace(darkBlue(3), midBlue(3), half)';
    linspace(midBlue(1), lightCyan(1), half)', linspace(midBlue(2), lightCyan(2),
half)', linspace(midBlue(3), lightCyan(3), half)'
];

colormap(flipud(customMap));
clim([min(vs30), max(vs30)]);

% Setup Colourbar
c = colorbar;
c.Label.String = 'V_{s30} (m/s)';
c.Label.FontWeight = 'bold';

% Labelling of the sites
for i = 1:numSites
    labelText = sprintf('%.1f', vs30(i));
    text(lats(i) + 0.0005, lons(i) + 0.0005, labelText, 'FontSize', 7.5,
'FontWeight', 'bold', 'Color', [0.2, 0.2, 0.2]);
end

title('Vs30 Contour Gradient Map');
```

Appendix H: Maple Ridge Coordinates and Vs30

Site	lon	lat	VS30
MR002	-122.600855	49.250333	260
MR004	-122.6225077	49.2424976	171.3074205
MR007	-122.624456	49.235405	209.1493939
MR036	-122.6011452	49.2429416	284.1176471
MR037	-122.586235	49.24202	210.8206221

MR039	-122.5587723	49.2389299	326.787736
MR040	-122.581249	49.237237	122.2671377
MR041	-122.590447	49.2365	179.1966313
MR042	-122.6093906	49.2352244	106.1194446
MR043	-122.5275373	49.24688987	250.6442111
MR044	-122.5338352	49.24219711	232.9868775
MR045	-122.5273315	49.23885632	234.9450469
MR131	-122.63354	49.238217	286.6990616
PMW010	-122.643257	49.278823	179.5998496
PMW018	-122.64488	49.26427	164.792887
PMW019	-122.60095	49.26021111	168.2224146
PMW101	-122.6325	49.27871	128.2
PMW126	-122.644322	49.248501	84.2718301
MR008	-122.6403092	49.23328727	139.1720541
MR011	-122.6469796	49.22927409	155.5887097
MR016	-122.6471777	49.22380057	126.7265696
MR019	-122.621701	49.225176	194.0927703
MR023	-122.6296191	49.21913901	161.6894792
MR026	-122.6457441	49.21389411	146.3669314
MR033	-122.6387185	49.21018966	157.1934088
MR035	-122.6228178	49.2145107	86.7756315
MR048	-122.5802531	49.2298514	175.3114542
MR051	-122.6043712	49.2294862	229.6986836

MR053	-122.613057	49.224108	108.242514
MR054B	-122.60585	49.21854	229
MR056	-122.604461	49.218154	223.1415548
MR058	-122.5805556	49.22388889	247.3002412
MR060	-122.605522	49.213018	142.5580757
MR063	-122.580031	49.219125	241.3323945
MR065	-122.543247	49.227721	252.3650867
MR070	-122.5639305	49.21862279	213.9469844
MR072	-122.588633	49.212262	172.604971
MR075	-122.579328	49.206622	307.7060761
MR077	-122.56532	49.212526	158.9223432
MR079	-122.572886	49.203995	220.4365741
MR081	-122.556904	49.20737	190.8540055
MR089	-122.54542	49.202087	342.9578556
MR091	-122.564101	49.202205	291.5247844
MR093	-122.571529	49.193574	190.2352995
MR095	-122.556074	49.196881	150.0014433
MR097	-122.576455	49.189938	143.6921979
MR099	-122.555413	49.191521	327.1079174
MR103	-122.556799	49.186432	208.9951509
MR105	-122.539569	49.185878	187.5042294
MR107	-122.551963	49.181903	231.0665049
MR132	-122.638574	49.214382	92.00805751

MR134	-122.629326	49.212172	116.4
MR147	-122.5460737	49.222268	271.926408
MR109	-122.5223395	49.1839727	426.9052827
MR112	-122.514326	49.17394	319.4945142
MR116	-122.4795865	49.1791692	402.4024648
MR118	-122.4795865	49.1791692	290.8644032
MR120	-122.461308	49.183872	362.961516
MR124	-122.428978	49.198692	230.2279946
MR128	-122.435193	49.176926	337.4201186
MR139	-122.530606	49.230516	221.1278015
MR141	-122.484975	49.227321	388.5148393
MR144	-122.535058	49.224523	294.5738793
MR146	-122.519441	49.227804	320.3034061
MR150	-122.529306	49.221327	286.9989236
MR158	-122.523634	49.225435	316.4693417
MR160	-122.456819	49.220533	745.8089441
MR162	-122.42699	49.223041	349.5260356
MR154B	-122.4710124	49.225873	561.2907716
MR152B	-122.47411	49.22437	374.6040309
MR157	-122.461663	49.216057	478.3043599
MR159	-122.45935	49.218055	344.5444616
MR155B	-122.470354	49.223787	607.716678